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# ST67W611 Wi-Fi 6 + Bluetooth® LE + Thread

Santa Clara Wireless Team

# Agenda

# X-Cube-ST67W61 Package

# Device Setup and Flashing

# BLE Commissioning App

# Generating BLE Commissioning app using NUCLEO-U575ZI-Q as host processor (Hands-on)

# Porting to a new host mcu using STM32CubeMX

# Generating the BLE Commissioning app using NUCLEO-G0B1RE processor (Hands-on)

# Documentation

# PLEASE DOWNLOAD ALL THE TOOLS LISTED BELOW Prerequisites

- X\_CUBE\_ST67W61\_V1.1.0 (To be installed under C drive)  
[https://github.com/STMicroDEMO/ST67\\_Experts\\_Training/releases/tag/v1.1.0](https://github.com/STMicroDEMO/ST67_Experts_Training/releases/tag/v1.1.0)
- STMicroelectronics.X-CUBE-ST67W61.1.1.0.pack  
[https://github.com/STMicroDEMO/ST67\\_Experts\\_Training/releases/tag/v1.1.0\\_CubeMX](https://github.com/STMicroDEMO/ST67_Experts_Training/releases/tag/v1.1.0_CubeMX)
- [STM32CubeIDE v1.16.1+](#)
- [STM32CubeProgrammer v2.18.0+](#) To be installed in default directory  
(C:\Program Files\STMicroelectronics\STM32Cube\STM32CubeProgrammer)
- [STM32CubeMX v6.15.0](#)
- [HyperTerminal – TeraTerm v.5.4.0](#)
- [ST BLE Toolbox](#) (To be downloaded on mobile)



***NOTE: If your host platform is Ubuntu, check the next slide.***

# Prerequisites - Ubuntu

**PLEASE DOWNLOAD ALL THE TOOLS LISTED IN THE PREVIOUS SLIDE AS A ROOT USER.**

- You will need elevated permissions to run ST Tools – STM32CubeProgrammer, STM32CubeIDE, STM32CubeMX.
- If root user access is not available to you, ensure that you have all permissions enabled for installed tools manually. Commands to do this:
  - `Chmod +x <folder_path>`
  - `Chmod -R 777 <folder_path>`
- Instead of TeraTerm, you can use Minicom or Cútecom.



# Testing Tools on Ubuntu

- You can check that STM32CubeProgrammer CLI is working with `STM32_Programmer_CLI _version`. This should return the version.
- Always open STM32CubeMX and STM32CubeIDE as root user.
- If any of the tools do not work, you might need to add them to path using `export PATH="$PATH:$HOME/<Path_to_bin_folder_of_the_tool>"`
- Check that STM32CubeMX is working by following the next steps.
- To check that STM32CubeIDE is working, you can run an existing example and flash.

# X\_CUBE\_ST67W61\_V1.1.0

[https://github.com/STMicroDEMO/ST67\\_Experts\\_Training/releases/tag/v1.1.0](https://github.com/STMicroDEMO/ST67_Experts_Training/releases/tag/v1.1.0)

← ↻ 🔒 [https://github.com/STMicroDEMO/ST67\\_Experts\\_Training/releases/tag/v1.1.0](https://github.com/STMicroDEMO/ST67_Experts_Training/releases/tag/v1.1.0) 📄 🗨️ ⭐ ⚙️

☰ STMicroDEMO / ST67\_Experts\_Training 🔍 Type / to search 🗑️ + 🕒

<> Code 🕒 Issues 🔗 Pull requests 🔄 Actions 📁 Projects 📖 Wiki 🛡️ Security 📈 Insights ⚙️ Settings

[Releases](#) / v1.1.0

## X-CUBE-ST67W61\_V1.1.0.RC1\_Setup.exe

🛡️ STMicroDEMO released this 24 minutes ago 📦 v1.1.0 🔑 4649534 ✅

This is X-CUBE-ST67W61\_V1.1.0.RC1\_Setup.exe. The package needs to be downloaded and placed under C drive

▼ **Assets** 3

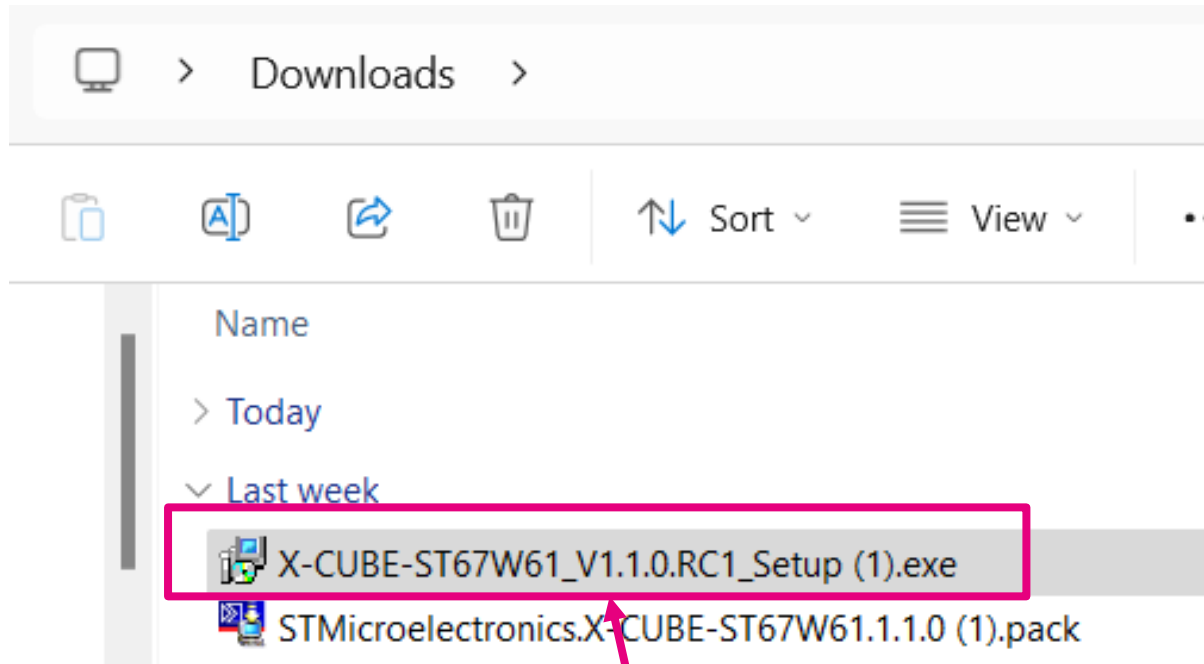
|                                                         |                                         |   |        |                |
|---------------------------------------------------------|-----------------------------------------|---|--------|----------------|
| 📦 <a href="#">X-CUBE-ST67W61_V1.1.0.RC1_Setup.1.exe</a> | sha256:9428890914a826891034034ac3b7c... | 📄 | 318 MB | 19 minutes ago |
| 📄 <a href="#">Source code (zip)</a>                     |                                         |   |        | 1 hour ago     |
| 📄 <a href="#">Source code (tar.gz)</a>                  |                                         |   |        | 1 hour ago     |

😊

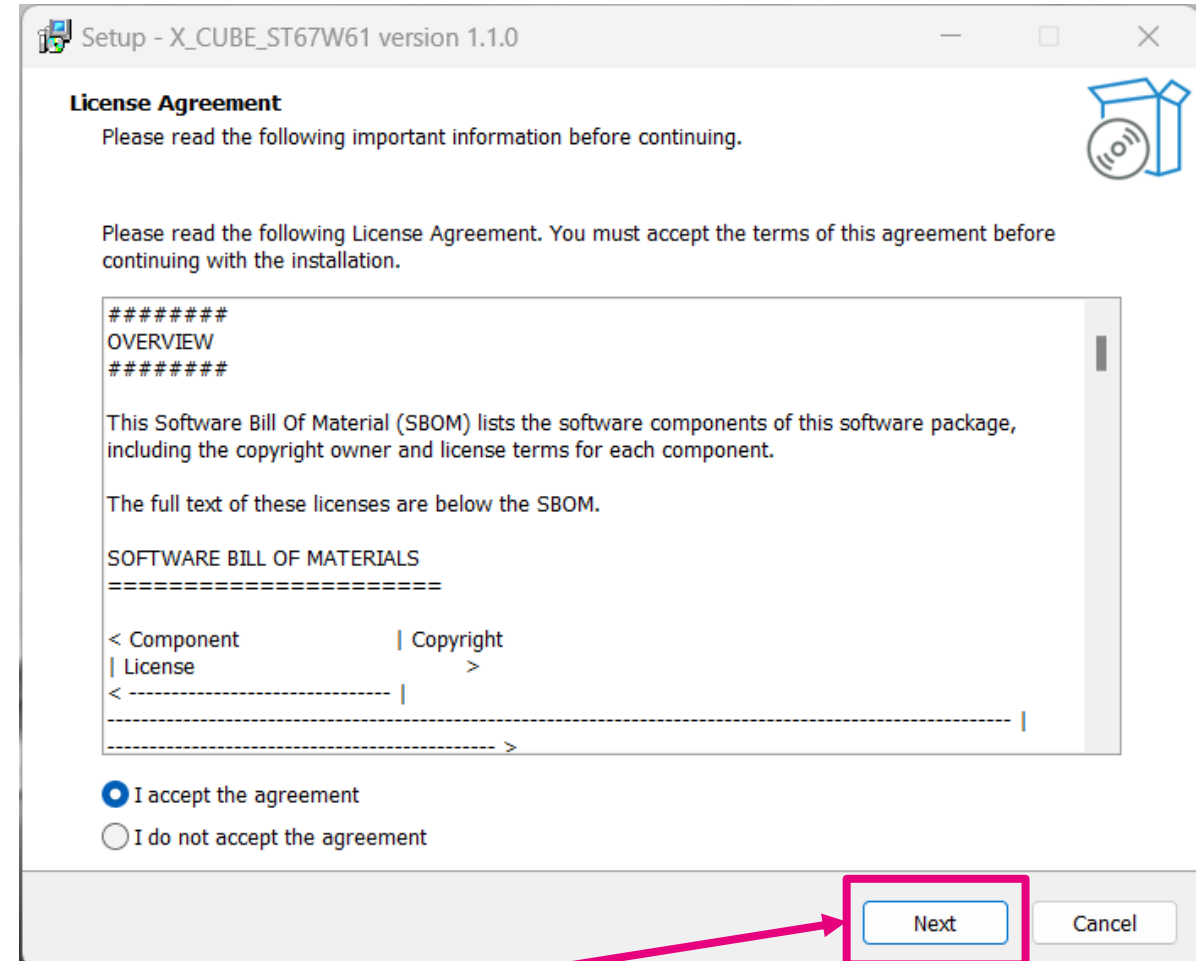
Download the  
package



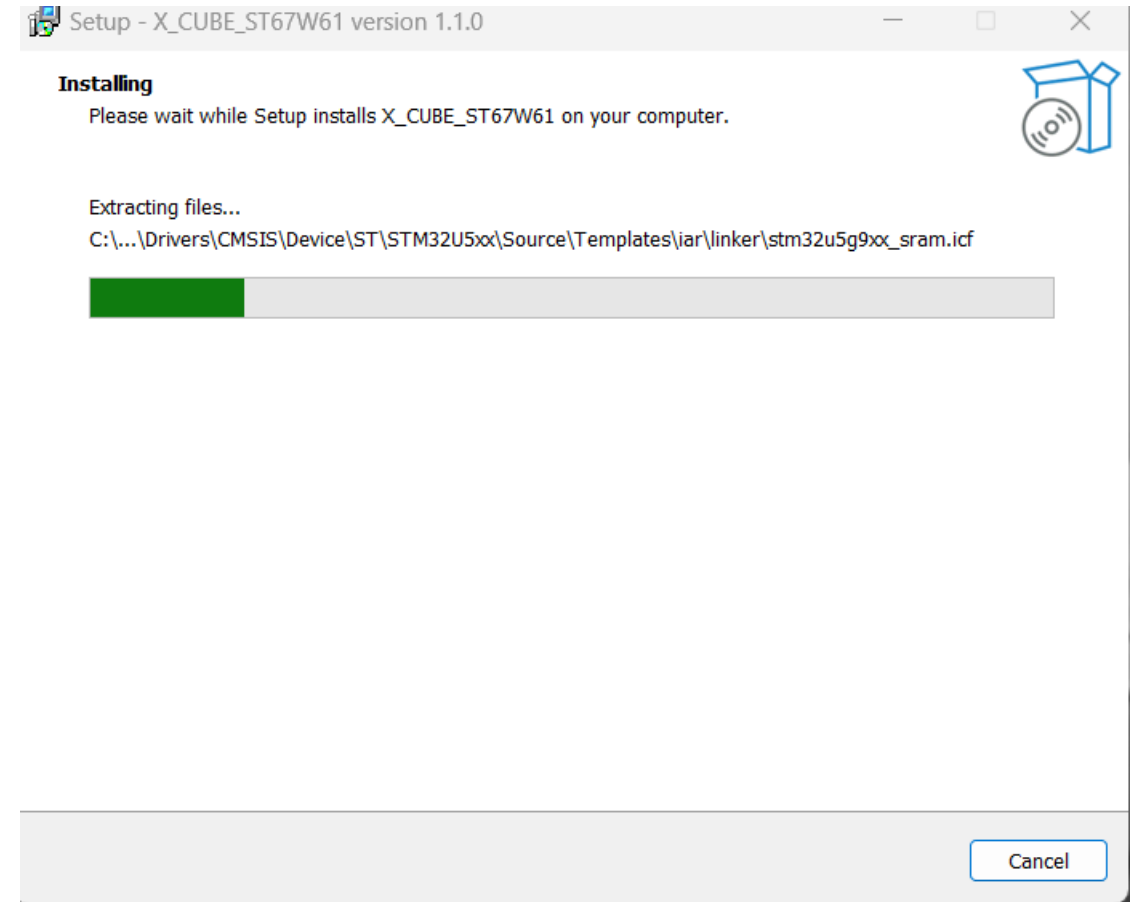
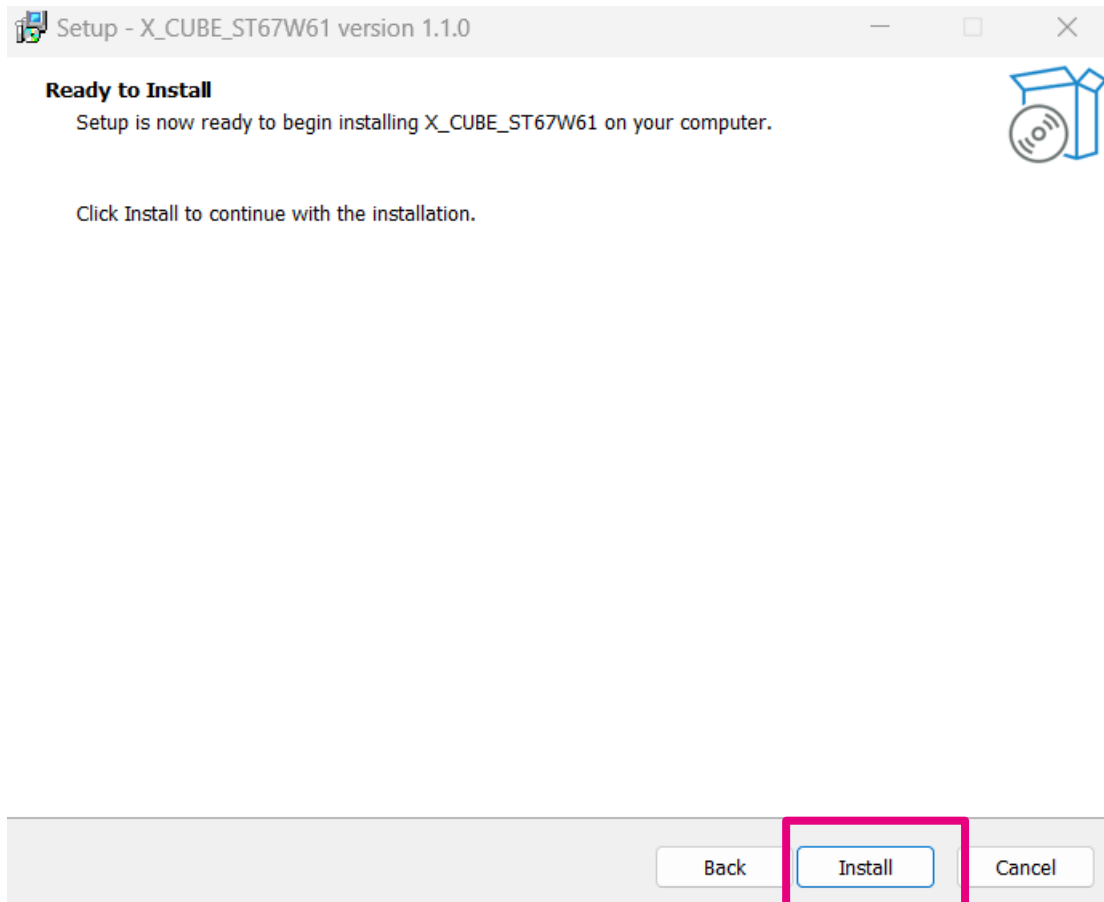
# X\_CUBE\_ST67W61\_V1.1.0



Under Download-> Double-click on the package



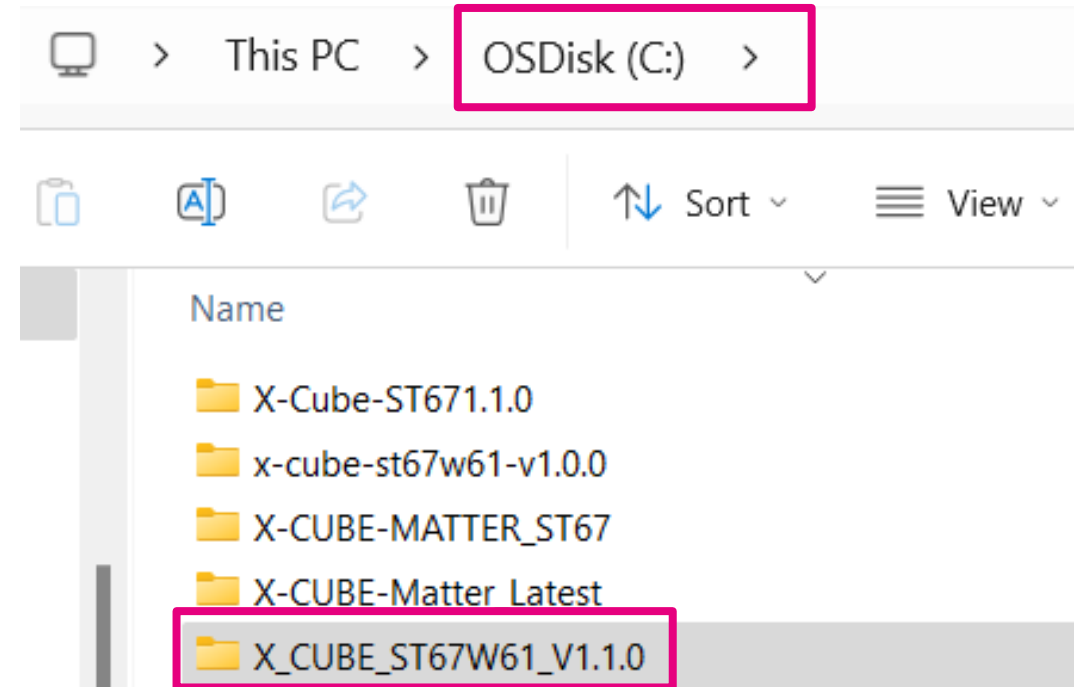
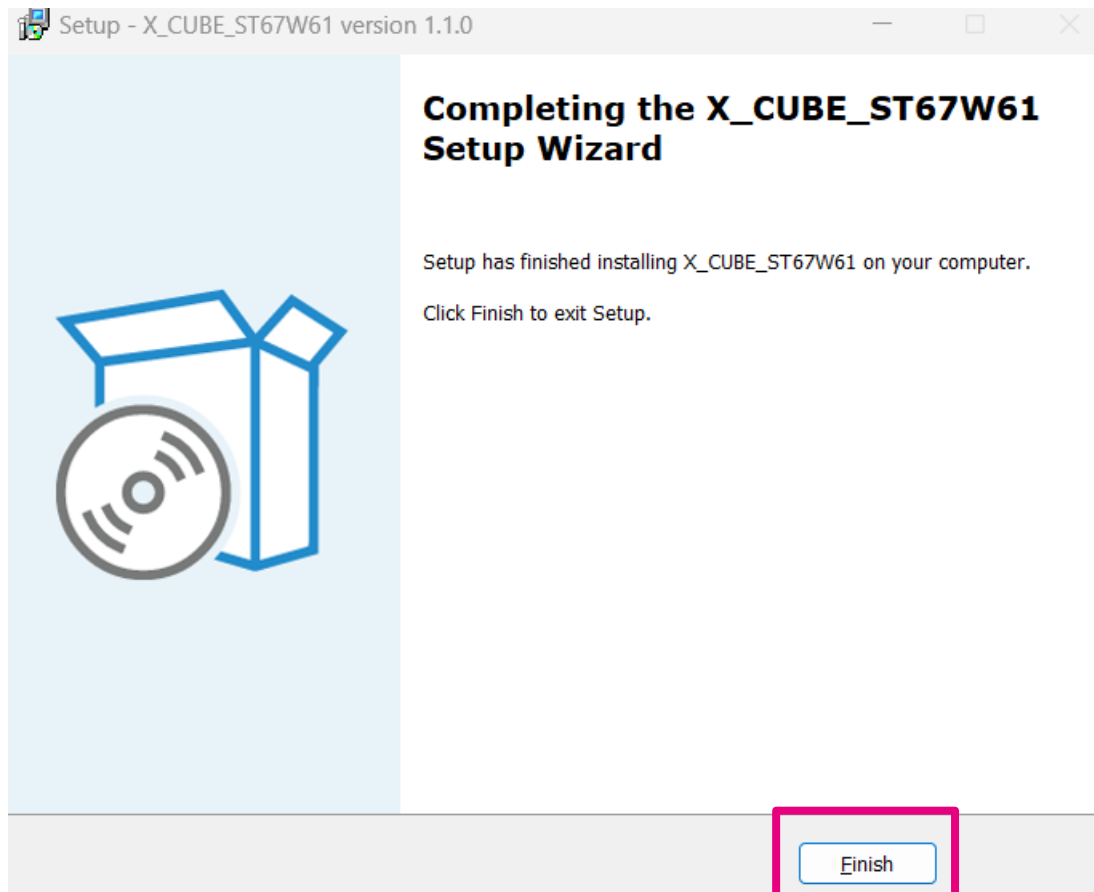
# X\_CUBE\_ST67W61\_V1.1.0



Package is installing



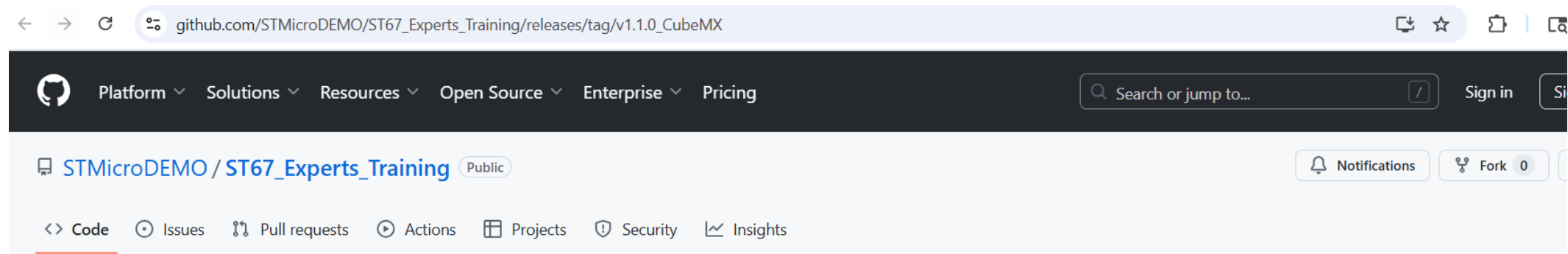
# X\_CUBE\_ST67W61\_V1.1.0



Package is installed  
under C drive

# STMicroelectronics.X-CUBE-ST67W61.1.1.0.pack

[https://github.com/STMicroDEMO/ST67\\_Experts\\_Training/releases/tag/v1.1.0\\_CubeMX](https://github.com/STMicroDEMO/ST67_Experts_Training/releases/tag/v1.1.0_CubeMX)



Releases / v1.1.0\_CubeMX

## STMicroelectronics.X-CUBE-ST67W61.1.1.0.pack

Latest

Compare

STMicroDEMO released this 41 minutes ago v1.1.0\_CubeMX 4649534

This is STMicroelectronics.X-CUBE-ST67W61.1.1.0.pack. This package needs to be loaded in STM32CubeMX software for the hands-on session

### Assets 3

STMicroelectronics.X-CUBE-ST67W61.1.1.0.1.pack

sha256: 211a3841d6c2273128443cc7beb15...



94.3 MB

42 minutes ago

Source code (zip)

1 hour ago

Source code (tar.gz)

1 hour ago

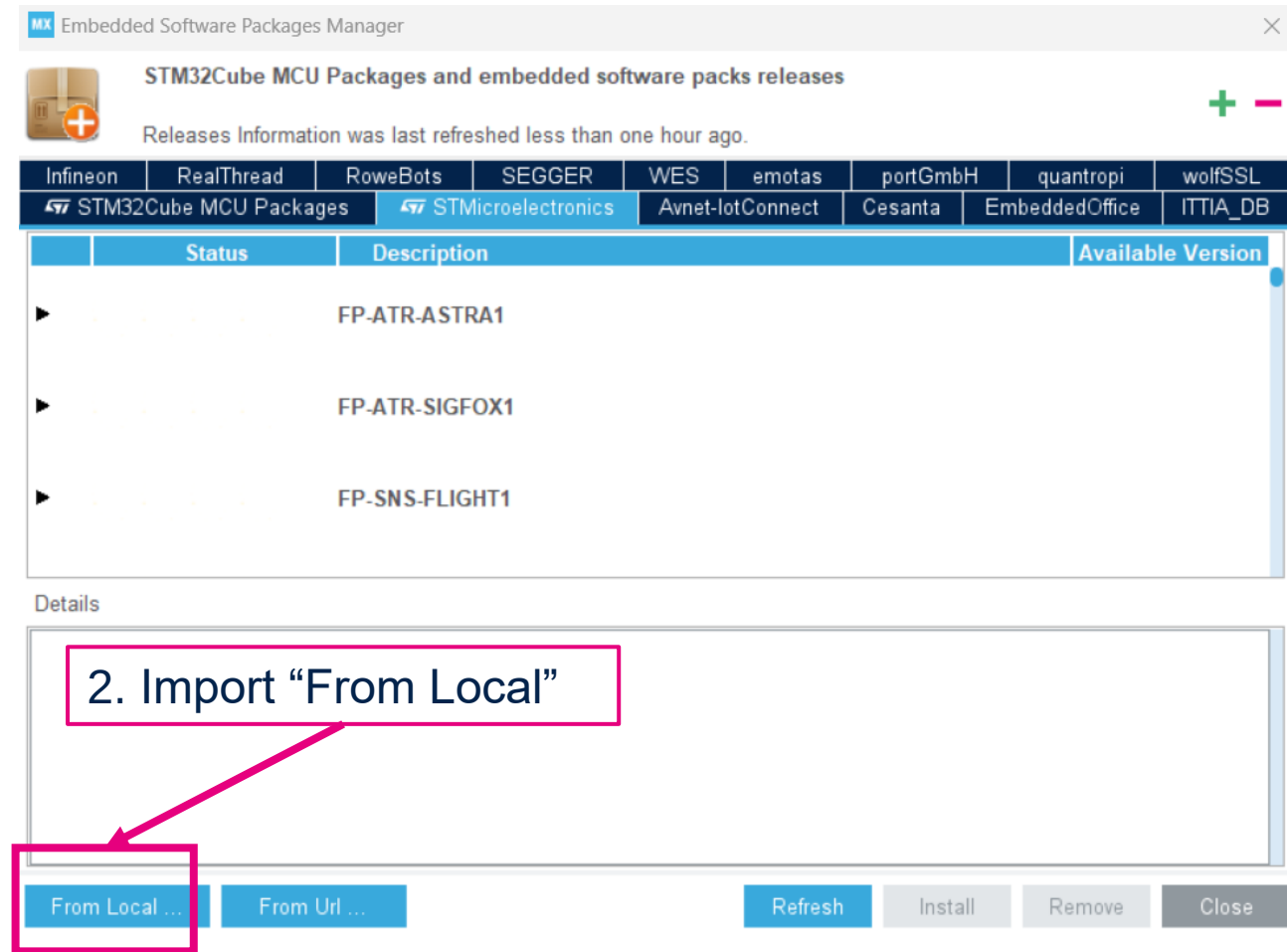
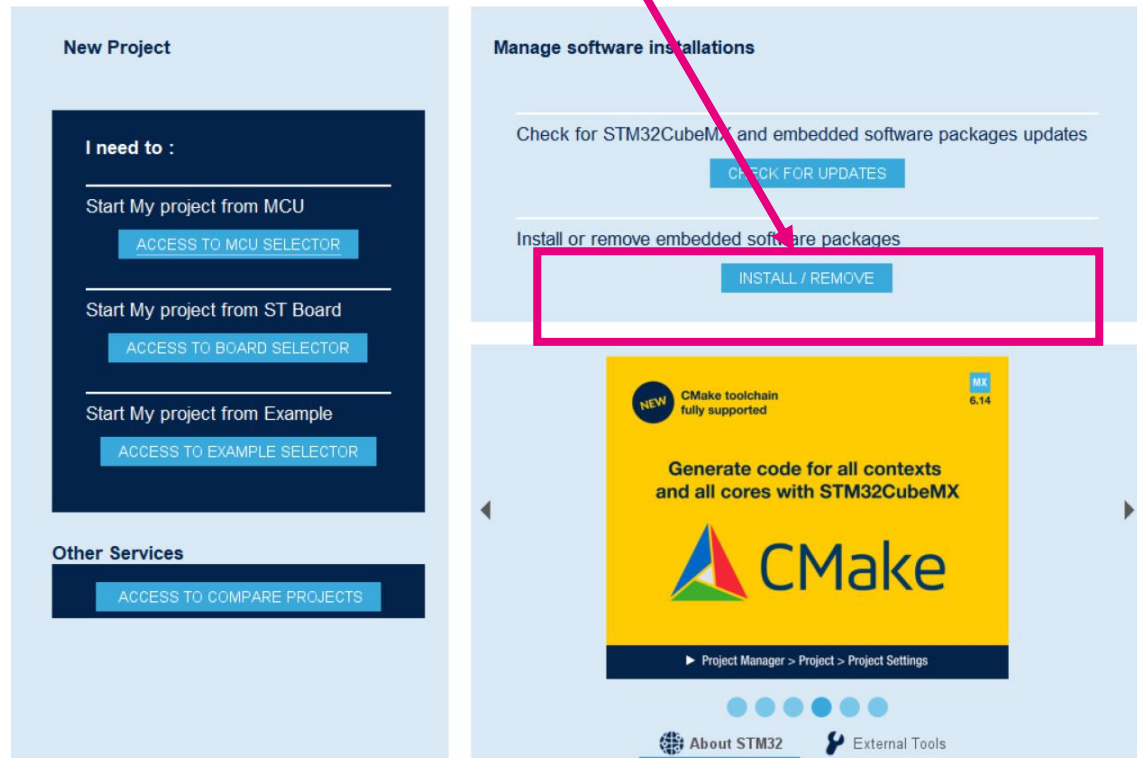
Download the package



# STMMicroelectronics.X-CUBE-ST67W61.1.1.0.pack

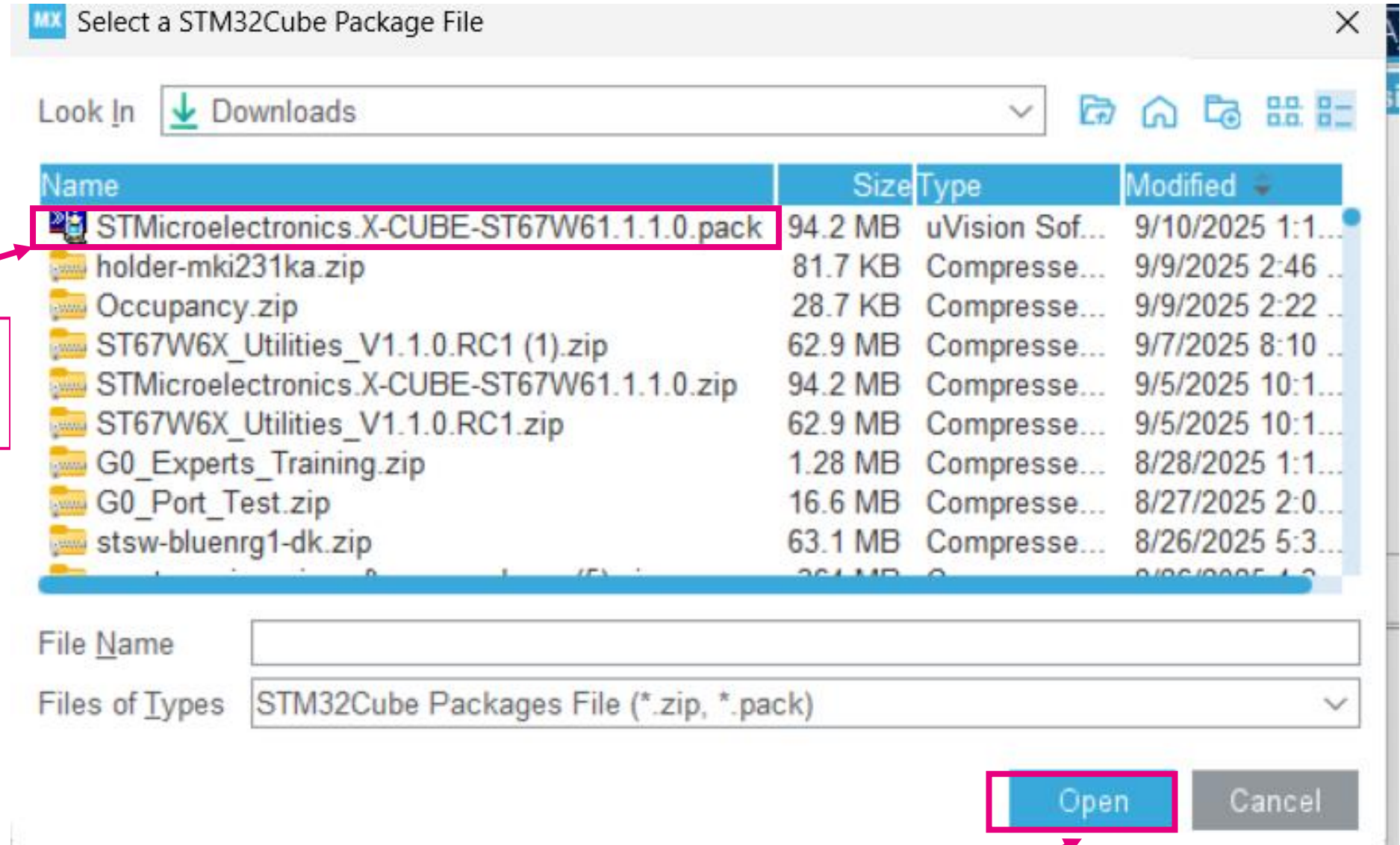
## Open STM32CubeMX with admin rights

1. Go to “Install/Remove” software packages



2. Import “From Local”

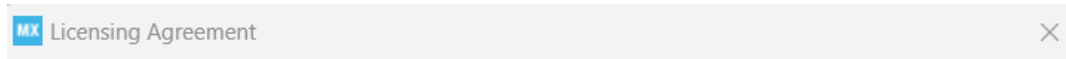
# STMicroelectronics.X-CUBE-ST67W61.1.1.0.pack



3. Select  
STMicroelectronics.X-CUBE-  
ST67W61.1.1.0.pack

4. Click Open

# STMicroelectronics.X-CUBE-ST67W61.1.1.0.pack



## STMicroelectronics X-CUBE-ST67W61 1.1.0 License Agreement

Please read and accept the following agreement carefully to finish the installation:

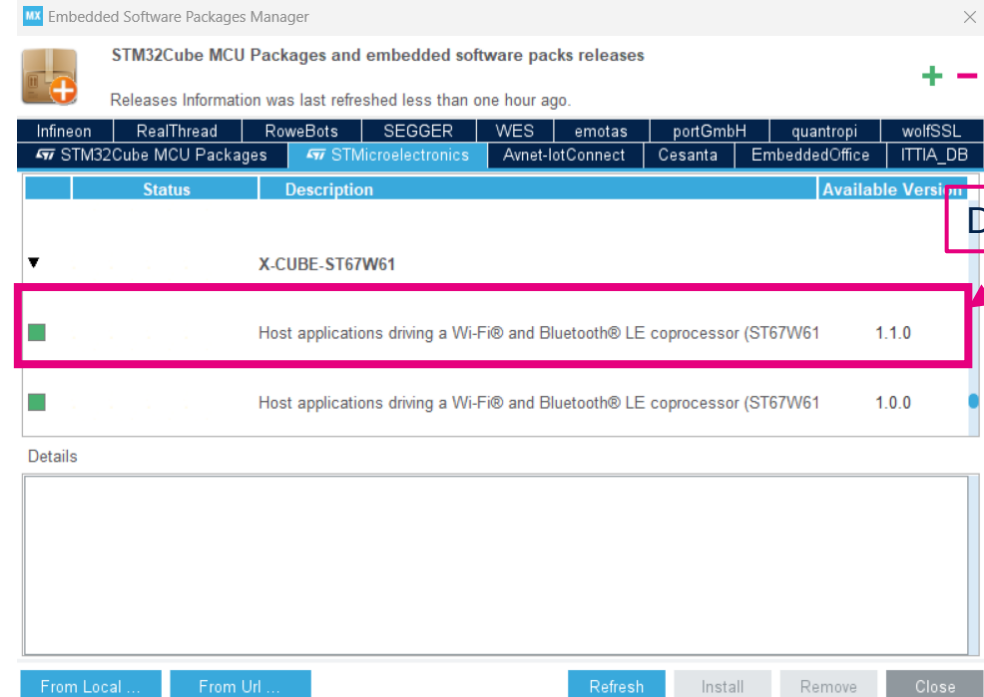
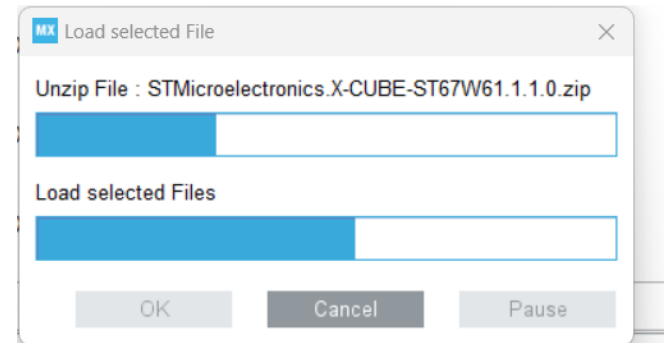
[Click here to open the license agreement](#)

☒ I have read, and I agree to the terms of this license agreement

☐ I do not accept the terms of this license agreement

Finish

Cancel



Downloaded

# X-CUBE-FREERTOS

Download latest X-CUBE-FREERTOS

MX Embedded Software Packages Manager

STM32Cube MCU Packages and embedded software packs releases

Releases Information was last refreshed 4 hours ago.

| Infinion               | RealThread         | RoweBots         | SEGGER  | WES            | emotas   | portGmbH | quantropi | wolfSSL |
|------------------------|--------------------|------------------|---------|----------------|----------|----------|-----------|---------|
| STM32Cube MCU Packages | STMicroelectronics | Avnet-lotConnect | Cesanta | EmbeddedOffice | ITTIA_DB |          |           |         |

| Status | Description                                                                     | Available Version |
|--------|---------------------------------------------------------------------------------|-------------------|
| ▼      | X-CUBE-FREERTOS                                                                 |                   |
| ■      | FreeRTOS STM32Cube expansion package for STM32H5/U5/WBA/C0/U0/N6/U3 series      | 1.3.1             |
| □      | FreeRTOS STM32Cube expansion package for STM32H5/U5/WBA/C0/U0/N6/U3 series (Siz | 1.3.0             |
| □      | FreeRTOS STM32Cube expansion package for STM32H5/U5/WBA/C0/U0/N6/U3 series (Siz | 1.2.0             |
| □      | FreeRTOS STM32Cube expansion package for STM32H5/U5/WBA/C0/U0/N6/U3 series (Siz | 1.1.0             |

# STM32U5

MX Embedded Software Packages Manager

 STM32Cube MCU Packages and embedded software packs releases

Releases Information was last refreshed 4 hours ago.

| Infinion                                                                                                                   | RealThread                                                                                           | RoweBots          | SEGGER  | WES            | emotas   | portGmbH | quantropi | wolfSSL |
|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------|---------|----------------|----------|----------|-----------|---------|
|  STM32Cube MCU Packages                   |  STMicroelectronics | Avnet-IotConnect  | Cesanta | EmbeddedOffice | ITTIA_DB |          |           |         |
| Description                                                                                                                | Installed Version                                                                                    | Available Version |         |                |          |          |           |         |
| ▼ STM32U5                                                                                                                  |                                                                                                      |                   |         |                |          |          |           |         |
|  STM32Cube MCU Package for STM32U5 Series | 1.8.0                                                                                                | 1.8.0             |         |                |          |          |           |         |
| <input type="checkbox"/> STM32Cube MCU Package for STM32U5 Series (Size : 390 MB)                                          |                                                                                                      | 1.7.0             |         |                |          |          |           |         |
| <input type="checkbox"/> STM32Cube MCU Package for STM32U5 Series (Size : 351 MB)                                          |                                                                                                      | 1.6.0             |         |                |          |          |           |         |
| <input type="checkbox"/> STM32Cube MCU Package for STM32U5 Series (Size : 306 MB)                                          |                                                                                                      | 1.5.0             |         |                |          |          |           |         |

Download STM32U5  
(v.1.7.0)

# STM32G0

Embedded Software Packages Manager

STM32Cube MCU Packages and embedded software packs releases

Releases Information was last refreshed 20320 days ago.

| Infinion                                                      | RealThread         | RoweBots         | SEGGER            | WES            | emotas   | portGmbH | quantropi | wolfSSL |
|---------------------------------------------------------------|--------------------|------------------|-------------------|----------------|----------|----------|-----------|---------|
| STM32Cube MCU Packages                                        | STMicroelectronics | Avnet-IotConnect | Cesanta           | EmbeddedOffice | ITTIA_DB |          |           |         |
| Description                                                   | Installed Version  |                  | Available Version |                |          |          |           |         |
| ▶ STM32F4                                                     |                    |                  |                   |                |          |          |           |         |
| ▶ STM32F7                                                     |                    |                  |                   |                |          |          |           |         |
| ▼ STM32G0                                                     |                    |                  |                   |                |          |          |           |         |
| ■ STM32Cube MCU Package for STM32G0 Series                    | 1.6.2              |                  | 1.6.2             |                |          |          |           |         |
| □ STM32Cube MCU Package for STM32G0 Series (Size : 204.68 MB) |                    |                  | 1.6.1             |                |          |          |           |         |
| □ STM32Cube MCU Package for STM32G0 Series (Size : 203 MB)    |                    |                  | 1.6.0             |                |          |          |           |         |

Download latest STM32G0

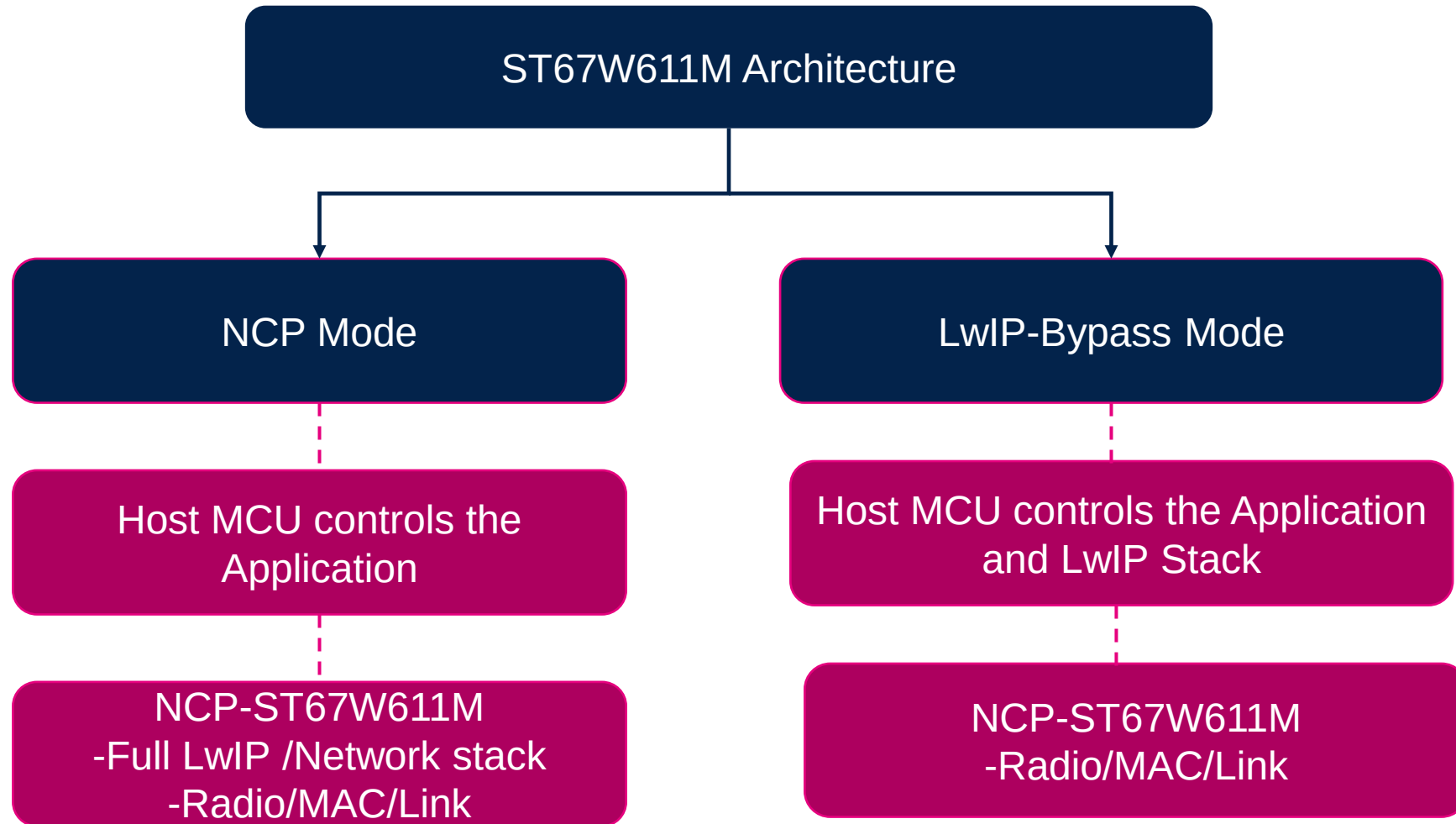




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# X-Cube-ST67W61

# ST67W611M Architecture

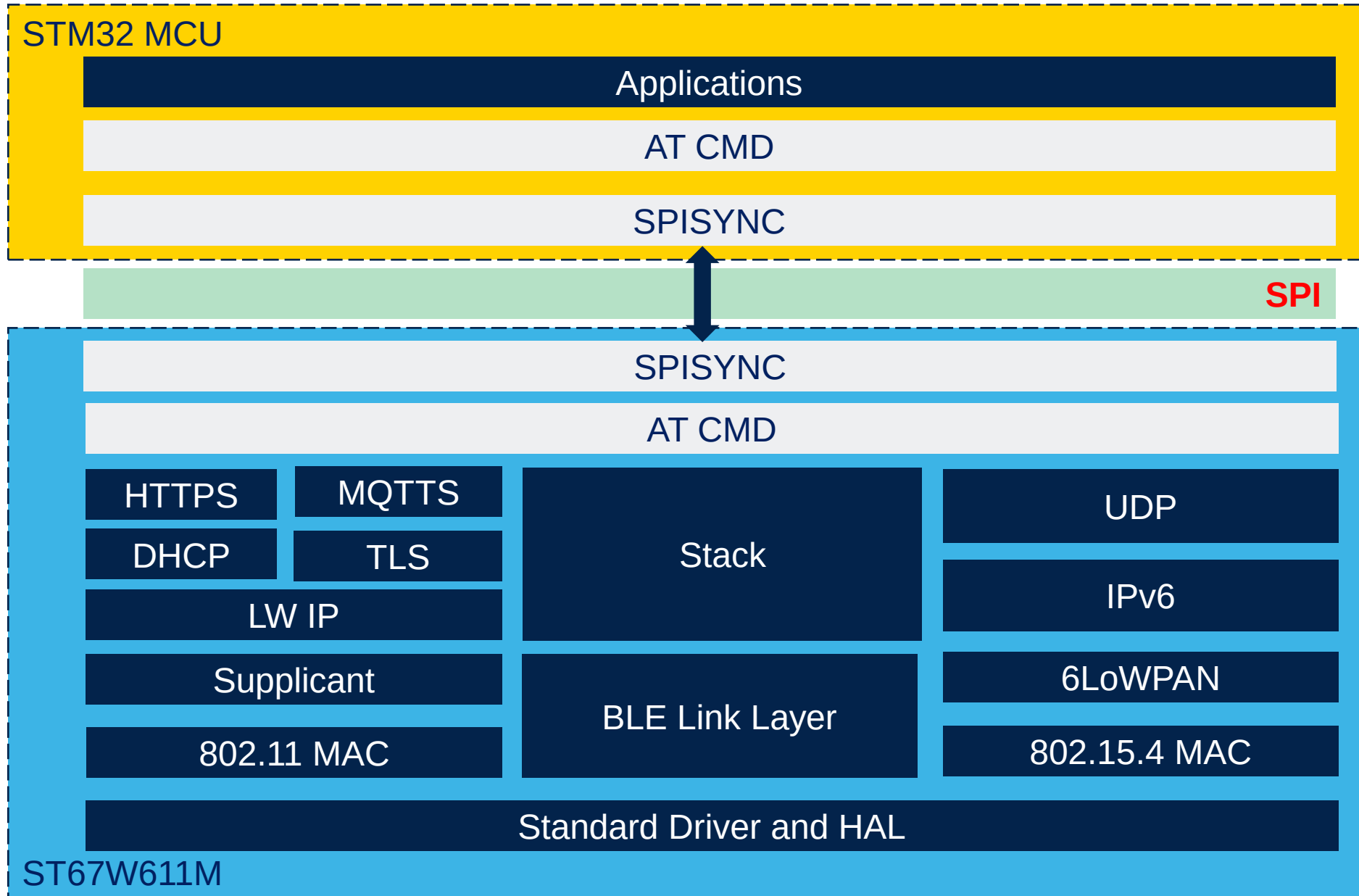


# T01:NCP Mode

- **LwIP (TCP/IP stack) and all networking protocols** run on the **NCP (ST67W611M)**, not on the host (STM32 MCU)
- The NCP (ST67W611M) is **responsible for the entire network stack**, including application protocols (HTTPS, MQTT), security (TLS), and wireless operations (radio, MAC, link layer)
- The **host (STM32 MCU) runs only the application logic** and interacts with the NCP using simple AT commands
- Communication between the host and NCP is done via SPI using AT commands and the SPISYNC protocol
- This mode is ideal when the host MCU has limited resources or when the user wants to simplify host software development and integration.



# T01:NCP Mode

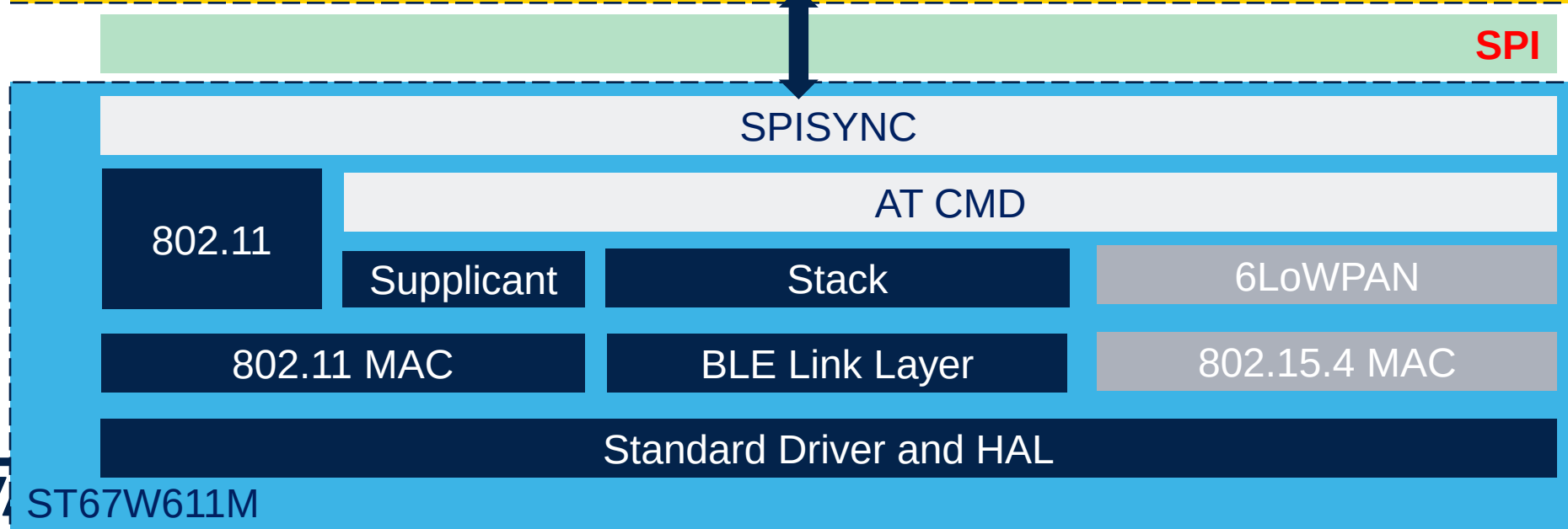
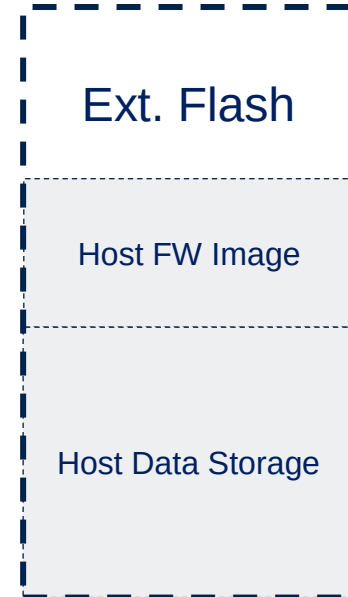
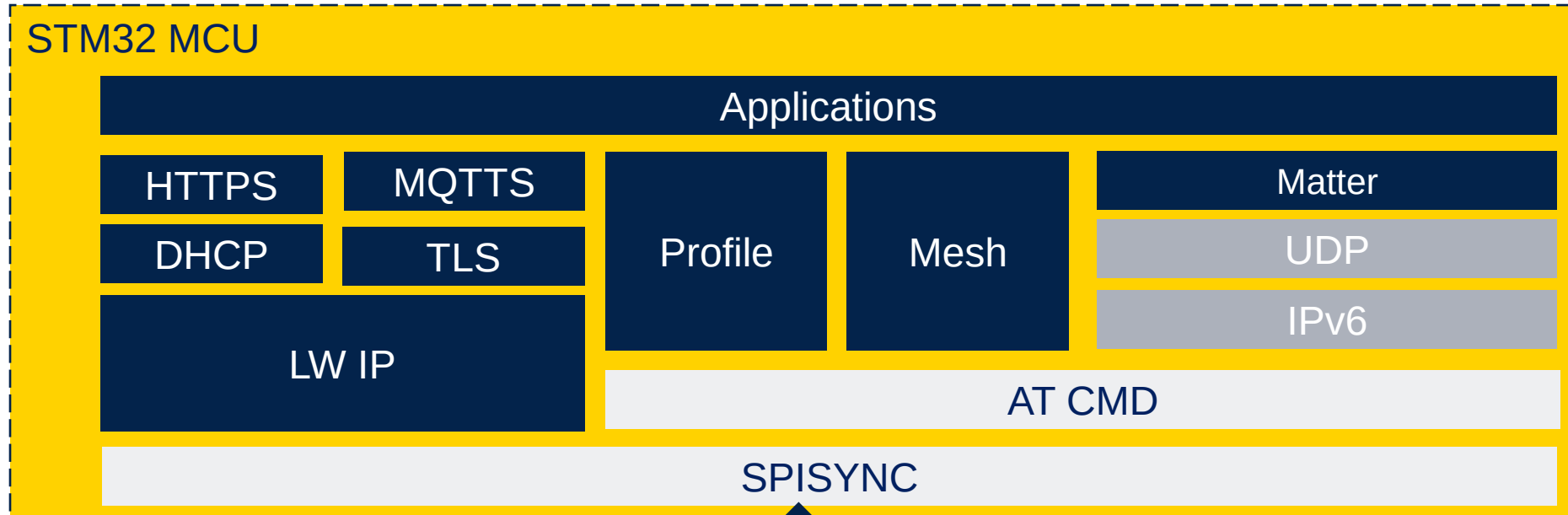


## T02: LwIP bypass

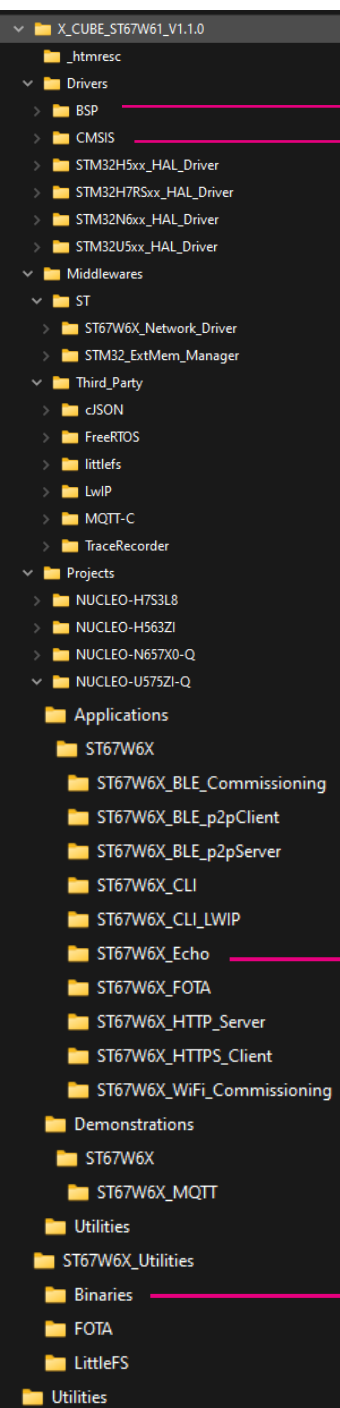
- **LwIP (TCP/IP stack) runs on the host (STM32 MCU), not on the NCP**
- The NCP (ST67W611M) is responsible only for the radio, MAC, and link layer operations.
- **The host, allowing for more flexibility** and has full control over the network stack advanced networking features
- Communication between the host and NCP is done via SPI using AT commands and the **SPISYNC protocol**
- This mode is ideal **when the host MCU is powerful enough to handle the network stack** and/ or the **user needs advanced networking features or customizations.**



# T02: LwIP bypass

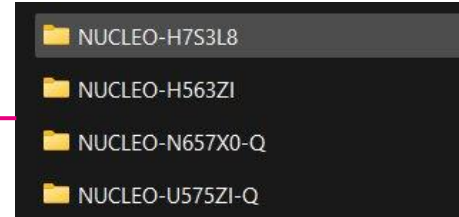


# X-CUBE-ST67W61 package overview

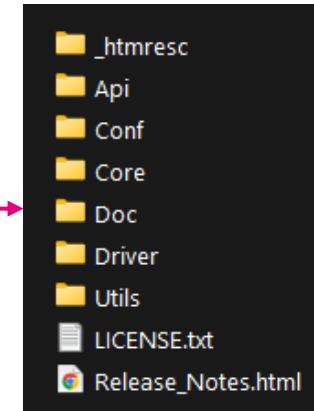


Board support package  
CMSIS libraries

Device HAL drivers



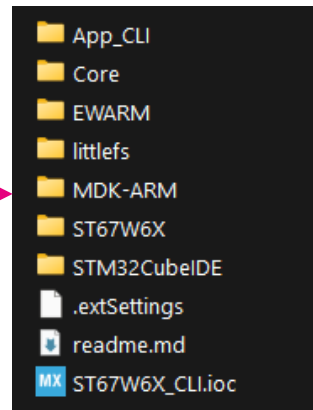
Variety of  
**Nucleo** hosts



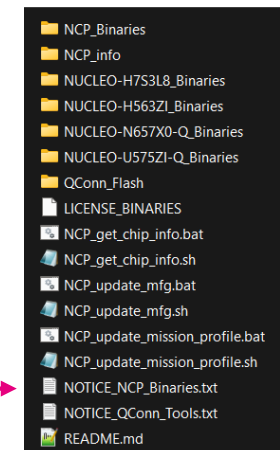
APIs include files  
Templates for middleware and driver config files  
Integration between generic API and driver

W61 driver  
AT commands / responses  
SPI driver

UART Output logs  
Memory usage statistics  
CLI Console



**STM32CubeIDE** and **EWARM** Project Files



**NCP and  
Host binaries**



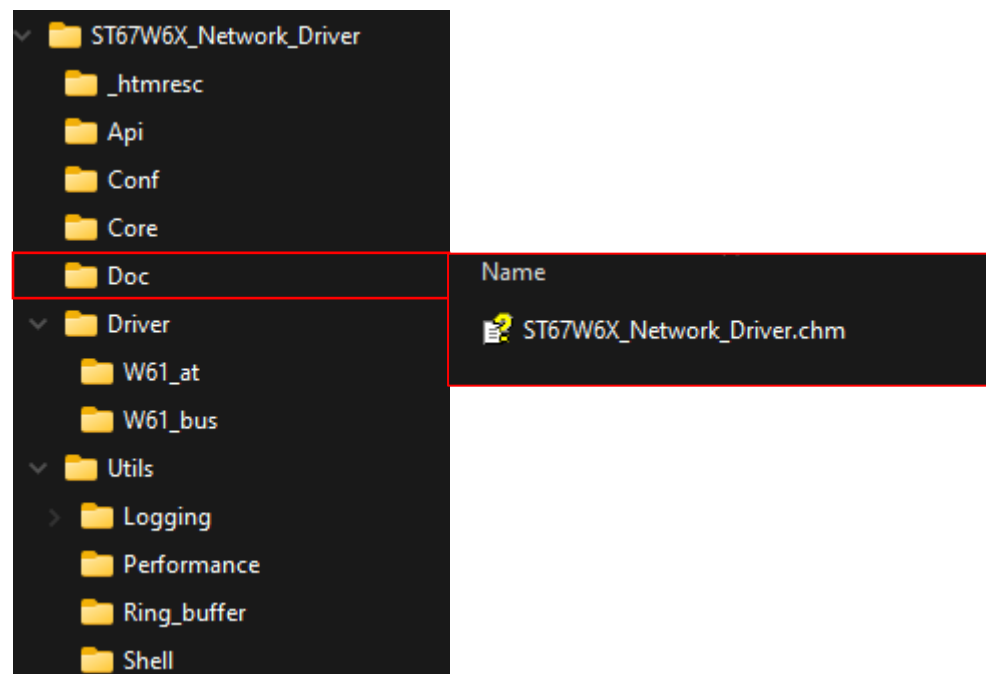
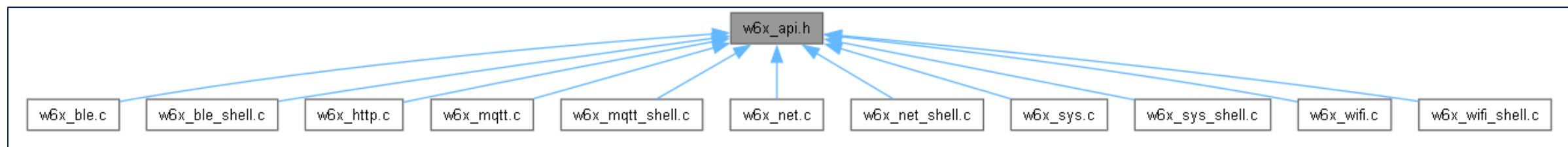
# X-CUBE-ST67W61 middleware

|                 | Scope / description                                                                                                                                                                                       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| W61_bus         | This is the <b>SPI driver for communication</b> between the host and the ST67W61.                                                                                                                         |
| W61_at          | It does the <b>formatting of the AT commands</b> to be sent, and it implements a receive tasks <b>to decode the received messages</b> from the W61 and dispatch them to the correspondent module.         |
| W6x service API | The Core directory implements an adaptation layer between the W61 AT driver and the API proposed to the application. The APIs are "Zephyr-like" for the Wi-Fi® and socket API for the network operations. |
| Utils           | Console handling, debug, trace, performance measurements.                                                                                                                                                 |

All the above submodules make use of FreeRTOS™



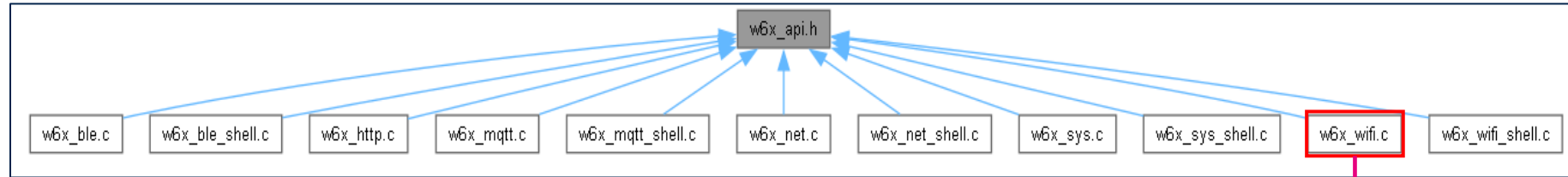
# API Documentation



Found in \Middlewares\ST\ST67W6X\_Network\_Driver\Doc

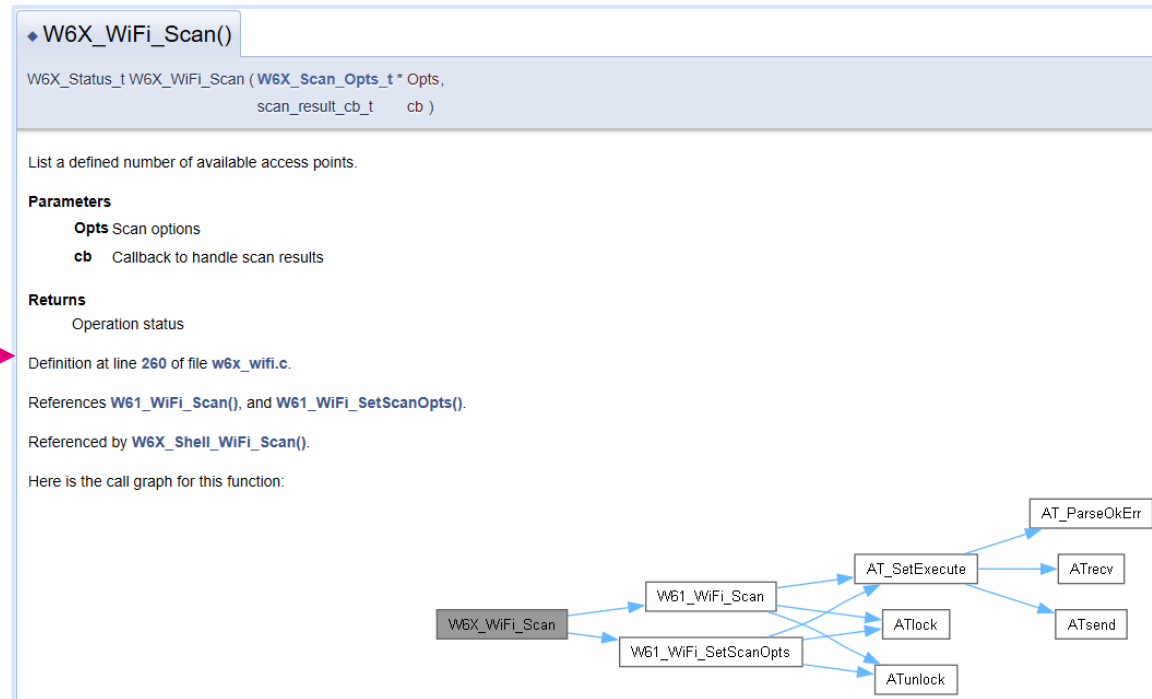
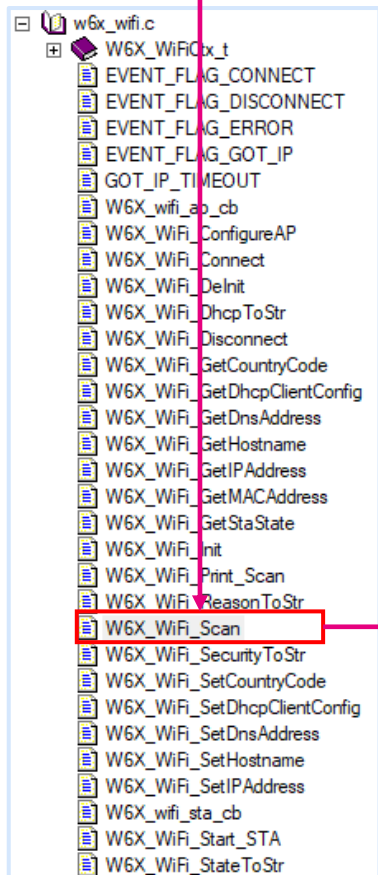
- API Definitions
- Overview of API calls
- Correlated AT Commands

# API Documentation cont...



Selecting **w6x\_wifi.c** will provide the different API categories

Selecting the API, **W6X\_WiFi\_Scan()**, will provide information and the call graph



# X-CUBE-ST67W61 applications

- Applications
  - ST67W6X
    - ST67W6X\_BLE\_Commissioning
    - ST67W6X\_BLE\_p2pClient
    - ST67W6X\_BLE\_p2pServer
    - ST67W6X\_CLI
    - ST67W6X\_CLI\_LWIP
    - ST67W6X\_Echo
    - ST67W6X\_FOTA
    - ST67W6X\_HTTP\_Server
    - ST67W6X\_HTTPS\_Client
    - ST67W6X\_WiFi\_Commissioning
- Demonstrations
  - ST67W6X
    - ST67W6X\_MQTT

|                             | Scope / description                                                                                                                                                                                                                                |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bluetooth® LE commissioning | Uses Bluetooth® LE to provide Wi-Fi® SSID and password after scanning available AP                                                                                                                                                                 |
| Bluetooth® LE p2p client    | Point-to-Point communication demo using Bluetooth® LE (GATT Client)<br>Scan and connect to a p2pServer<br>Write messages & receive notifications from the p2pServer                                                                                |
| Bluetooth® LE p2p server    | Point-to-Point communication demo using Bluetooth® LE (GATT server)<br>Advertises and wait for a connection from either: <ul style="list-style-type: none"> <li>- p2pClient app</li> <li>- <i>ST BLE Toolbox</i> smartphone application</li> </ul> |
| Echo                        | TCP Echo feature over Wi-Fi<br>- Good starting point for generic user development                                                                                                                                                                  |
| CLI                         | ST67 evaluation via CLI and for WTS tests the Wi-Fi solution.<br>- Wi-Fi API: scan, connect, disconnect, etc, NET socket API and iperf, Bluetooth® LE API                                                                                          |
| CLI_LWIP                    | Evaluate and to test the X-NUCLEO-67W61M1 Wi-Fi, Bluetooth LE and Network interface with LwIP solutions via command line interface (CLI).                                                                                                          |
| Echo FOTA                   | FOTA (firmware update over-the-air) feature demo over Wi-Fi<br>- Using HTTP                                                                                                                                                                        |
| HTTP server                 | HTTP server over Network API<br>The server, running on the host board, has been validated with 1 client, using HTTP requests to send and receive data to and from the server.                                                                      |
| HTTPS client                | HTTPS Client over Network API demo<br>- Basic GET requests to retrieve worldwide weather reports                                                                                                                                                   |
| Wi-Fi® commissioning        | Wi-Fi® commissioning project using an HTTP server, hosted by a device configured as a soft access point (SoftAP) and station (STA).                                                                                                                |
| MQTT Demo                   | Starting point for MQTT user development<br>- Sends IKS sensor data to an MQTT broker                                                                                                                                                              |

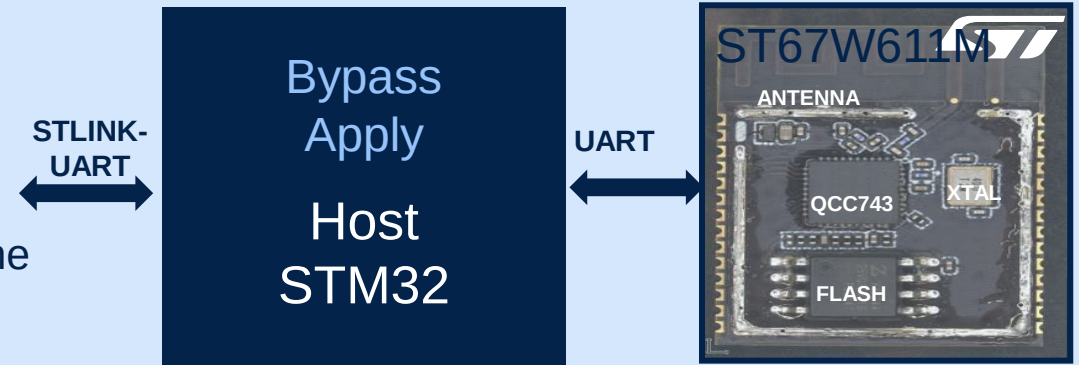


# ST67W611M Modes of Operation

## Manufacturing and Mission Modes

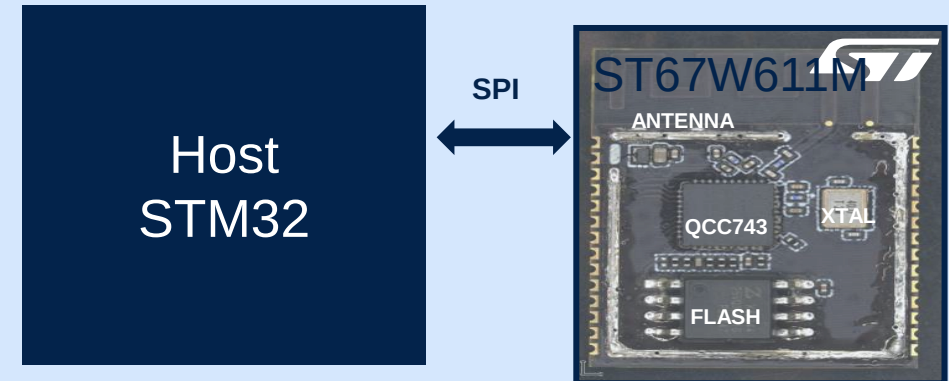
- **Manufacturing mode**

- Host-less Application
- Dedicated to RF performances monitoring and calibration.
- QC provides graphical tools to control the device through the **UART** interface



- **Mission mode (Functional)**

- Implementing the AT command for the different supported protocol (WIFI / BLE / MQTT /..) through the **SPI** interface.
- The ROM boot loader loads and executes the program from SPI Flash to boot the system.

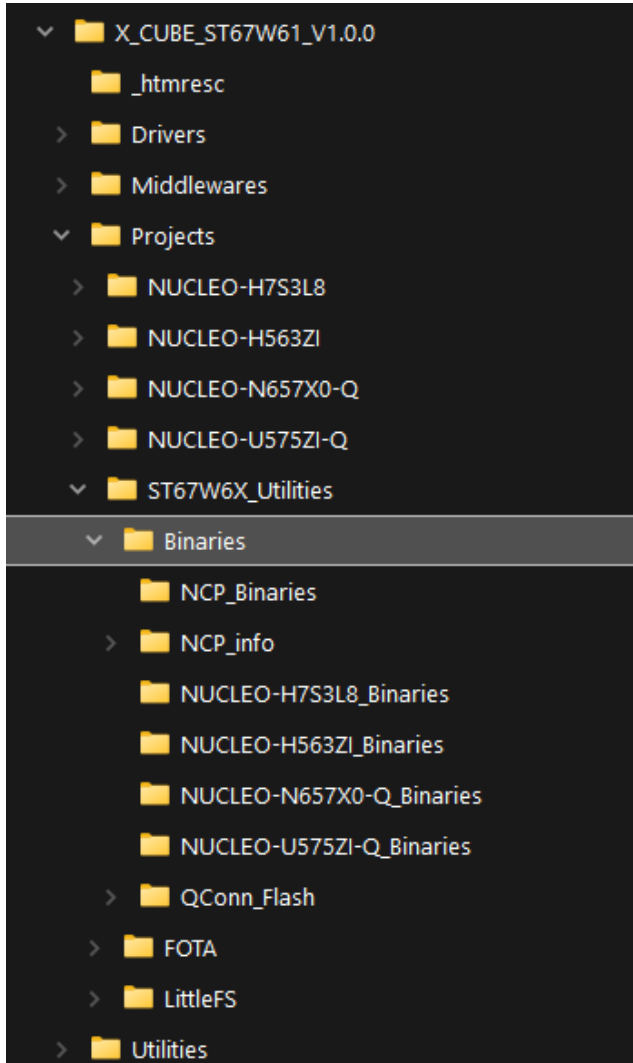


# X-CUBE-ST67W1 NCP tools

Scripts for flashing the latest mission or manufacturing firmware:

**`..\x-cube-st67w61-v1.0.0\Projects\ST67W6X_Uilities\Binaries`**

| Tool                                  | Description                                                                                                                         |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>NCP_get_chip_info.bat</b>          | Script to retrieve ST67 MAC address and locking information even if no firmware is loaded onto the ST67 device.                     |
| <b>NCP_update_mfg.bat</b>             | Script to flash the ST67 module with the manufacturing application, which is used for RF evaluation.                                |
| <b>NCP_update_mission_profile.bat</b> | Script to flash the ST67 module with the mission profile application, which is used for normal operations and development purposes. |



# ST67W611M Security Overview

**Two modes, each with a header containing a digital signature and versioning info**

## A. Mission mode:

- **Bootloader binary**
  - Authenticated by the ROM code during initial boot process
  - Versioning info to prevent rollback
- **Mission profile binary**
  - Wi-Fi® & Bluetooth® LE binary used for operational tasks
  - Authenticated by bootloader to ensure integrity & authenticity before execution

## B. Manufacturing mode:

- **Bootloader binary**
  - Authenticated by the ROM code during initial boot process
  - Versioning info to prevent rollback
- **MFG firmware binary**
  - Used for test and production configuration
  - Authenticated by bootloader to ensure integrity & authenticity before execution





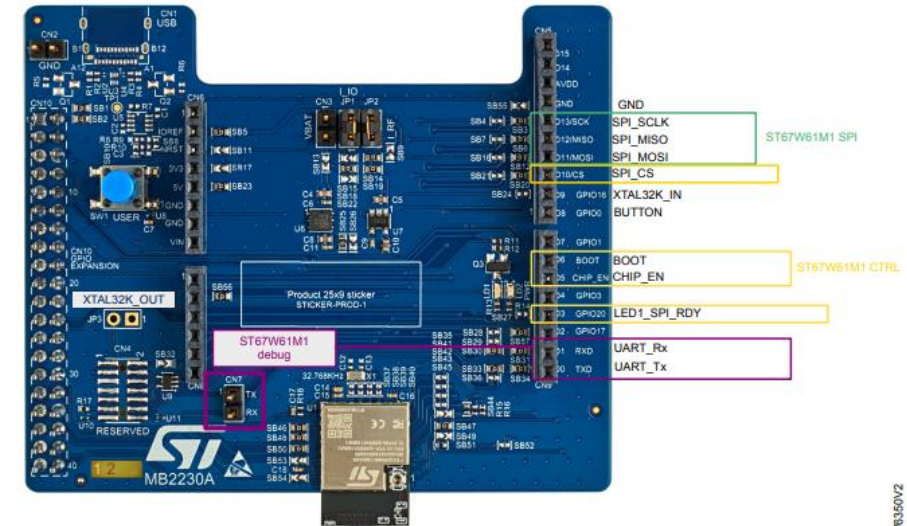


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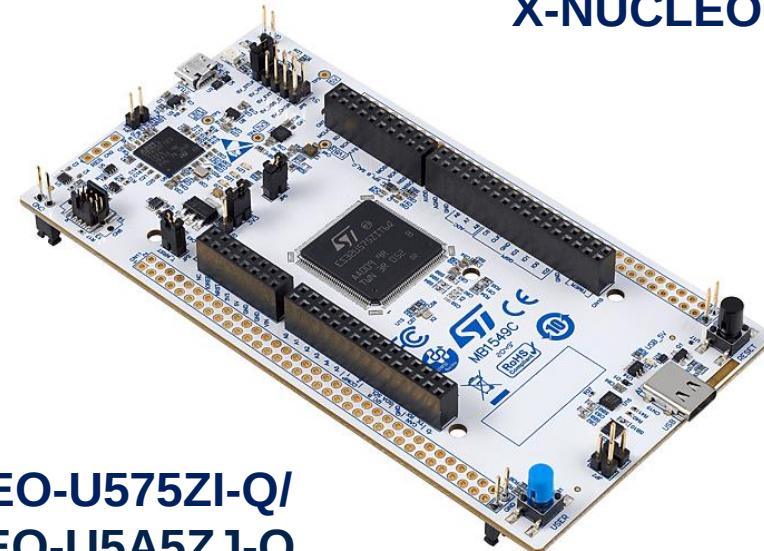
# Flashing ST67 & Device Setup

# STM32U5 and ST67W611M configuration

- 1x NUCLEO-U575ZI-Q/ NUCLEO-U5A5ZJ-Q board
- 1x X-NUCLEO-67W61M1 board
- Boards connected through Arduino® connectors
  - Key signals: SPI (SCK, MISO, MOSI, CS), SPI\_RDY, CHIP\_EN, BOOT, UART\_TX/RX
- Power: MicroUSB connection to NUCLEO-U575ZI-Q



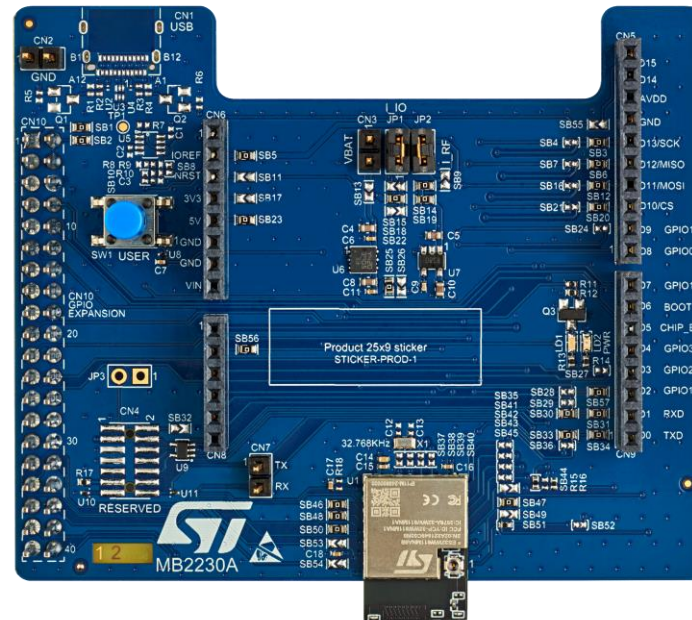
X-NUCLEO-67W61M1



NUCLEO-U575ZI-Q/  
NUCLEO-U5A5ZJ-Q



# Nucleo-U5 and X-Nucleo-67W61M1 Configuration



When stacking the Nucleo and X-Nucleo boards the Arduino connectors must be aligned.

The MicroUSB port is used when programming the STM32U575 or ST67W611M1.

The MicroUSB port is also used to read the UART messages from the STM32U575 in a serial terminal.

# Flashing ST67W611M1 on NUCLEO-U575ZI-Q

## Mission Mode (Functional Mode) Setup

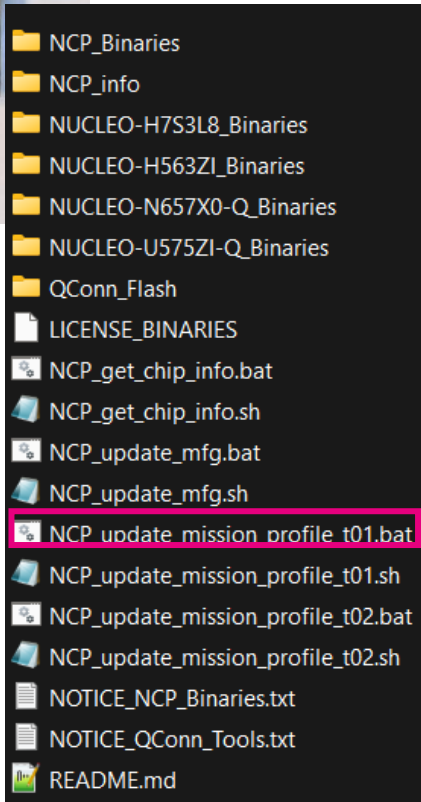
### Prerequisites:

- Installed STM32CubeProgrammer
  - Installed in default directory (C:\Program Files\STMicroelectronics\STM32Cube\STM32CubeProgrammer)
- Installed X-CUBE-ST67W61
  - Installed on C: drive (C:\X\_CUBE\_ST67W61\_V1.1.0)\*\*

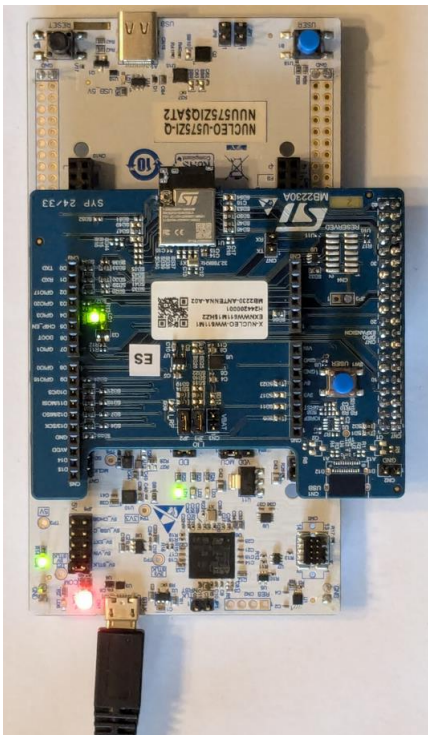
### Using batch scripts located:

C:\X\_CUBE\_ST67W61\_V1.1.0\Projects\ST67W6X\_Uilities\Binaries

1. Nucleo-U575ZI-Q and X-Nucleo-67W61M1 stacked together
2. USB cable connected to microUSB port on Nucleo-U575ZI-Q and Windows PC.
3. Double-click the batch files to execute
  1. NCP\_update\_mission\_profile\_t01.bat will flash the ST67W61M with the spi\_wifi binary and the host processor with CLI application.



*\*\*Note: Path length limits and paths with spaces in them for the x-cube-st67w61-v1.0.0 may cause issues. Saving on C: drive will prevent these issues.*



# Flashing ST67W611M1 with a Shell Script - Ubuntu

## Mission Mode (Functional Mode) Setup

### Prerequisites:

- Installed STM32CubeProgrammer
  - Installed in default directory (home\STMicroelectronics\STM32Cube\STM32CubeProgrammer)
- Installed X-CUBE-ST67W61
  - Installed in home (home\X\_CUBE\_ST67W61\_V1.1.0)\*\*

### Using shell scripts located:

~\X\_CUBE\_ST67W61\_V1.1.0\Projects\ST67W6X\_Utilityies\Binaries

1. Nucleo-U575ZI-Q and X-Nucleo-67W61M1 stacked together
2. USB cable connected to microUSB port on Nucleo-U575ZI-Q.
3. First run this command to correct the format of the shell script
  - `dos2unix NCP_update_mission_profile_t01.sh`
    1. NCP\_update\_mission\_profile\_t01.sh will flash the ST67W61M with the spi\_wifi binary and the host processor with CLI application.
    2. Run `./NCP_update_mission_profile_t01.sh`



*\*\*Note: Path length limits and paths with spaces in them for the x-cube-st67w61-v1.0.0 may cause issues. Saving on home will prevent these issues.*

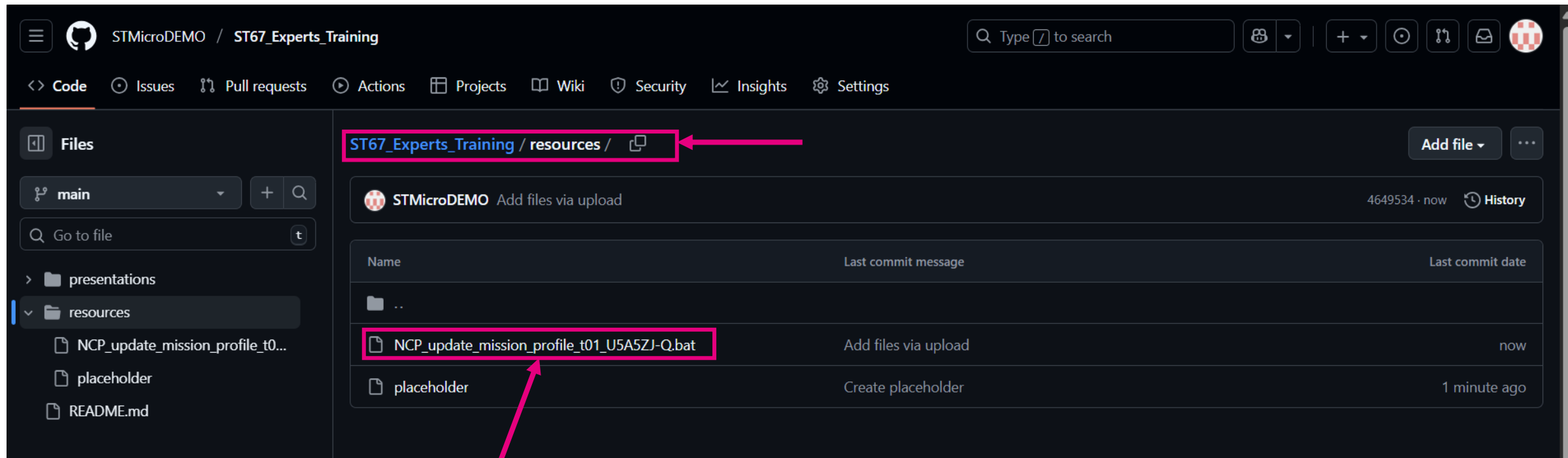
ST Restricted

# Flashing ST67W611M1 on NUCLEO-U5A5ZJ-Q

## Mission Mode (Functional Mode) Setup

- If you are using **NUCLEO-U5A5ZJ-Q**, use the modified .bat script to flash the ST67 with the mission mode.

[ST67\\_Experts\\_Training/resources at main · STMicroDEMO/ST67\\_Experts\\_Training](#)



The screenshot shows the GitHub interface for the repository `STMicroDEMO / ST67_Experts_Training`. The left sidebar displays the file tree with the `resources` folder selected. The main content area shows the `resources` directory contents, including the file `NCP_update_mission_profile_t01_U5A5ZJ-Q.bat`, which is highlighted with a pink box. A pink arrow points from this file to the download instruction at the bottom of the slide. Another pink arrow points to the breadcrumb path `ST67_Experts_Training / resources /` at the top of the file list.

| Name                                        | Last commit message  | Last commit date |
|---------------------------------------------|----------------------|------------------|
| ..                                          |                      |                  |
| NCP_update_mission_profile_t01_U5A5ZJ-Q.bat | Add files via upload | now              |
| placeholder                                 | Create placeholder   | 1 minute ago     |



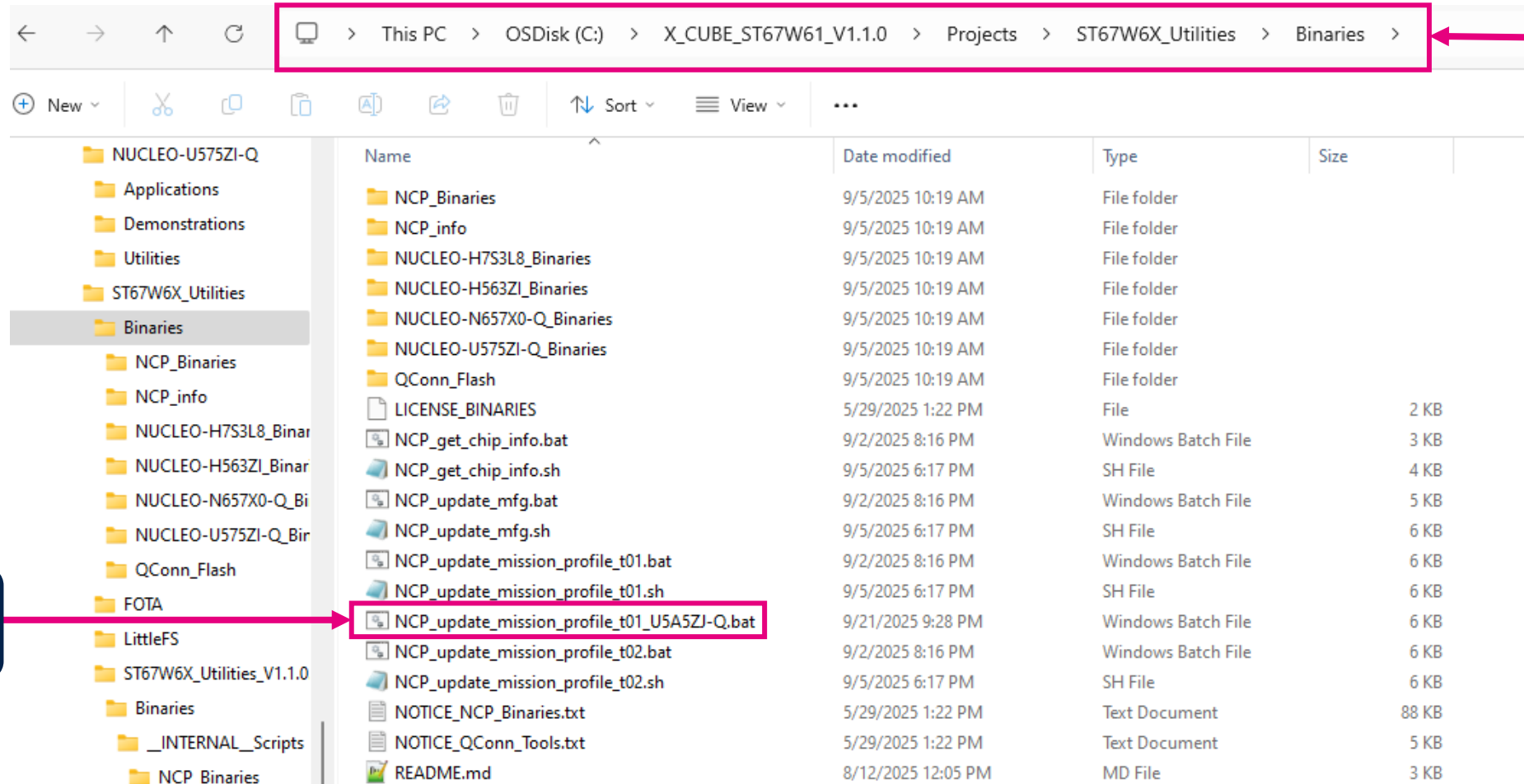
Download: `NCP_update_mission_profile_t01_U5A5ZJ-Q.bat`



# Flashing ST67W611M1 on NUCLEO-U5A5ZJ-Q

## Mission Mode (Functional Mode) Setup

- If you are using **NUCLEO-U5A5ZJ-Q**, use the modified .bat script to flash the ST67 with the mission mode.



```
#####  
"# You are about to load a signed binary to the NCP."  
"# This will lock the ST67W61M if not yet locked."  
#####  
Are you sure to proceed? (Y/N)  
Press Y to continue or N to abort: |
```

- ```
Opening and parsing file: Bootloader.bin  
  
Memory Programming ...  
File      : Bootloader.bin  
Size       : 8.96 KB  
Address    : 0x08000000  
  
Erasing memory corresponding to segment 0:  
Erasing internal memory sectors [0 1]  
Download in Progress:  
██████████████████████████████████████ 100%  
  
File download complete  
Time elapsed during download operation: 00:00:00.168
```

```
Opening and parsing file: ST67W6X_CLI.bin

Memory Programming ...
File           : ST67W6X_CLI.bin
Size          : 186.91 KB
Address       : 0x08000000

Erasing memory corresponding to segment 0:
Erasing internal memory sectors [0 23]
Download in Progress:
██████████████████████████████████████████████████████████████████████████ 100%

File download complete
Time elapsed during download operation: 00:00:02.186
```

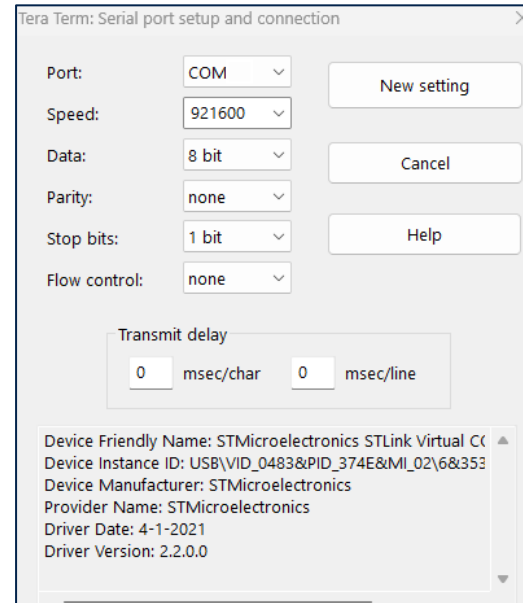
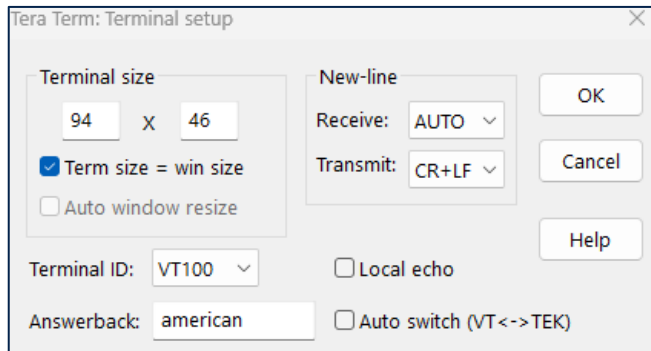
- 

# Flashing ST67W611M1

## Mission Mode (Functional Mode) Verification

Flashing success can be verified by connecting to the Nucleo-U575ZI-Q with a serial terminal.

Configure the serial terminal with the following parameters and connect to the corresponding COM port for the Nucleo-U575ZI-Q.



With the serial terminal program configured, press the reset button the Nucleo-U575ZI-Q and verify application and SDK version.



**\*\*Note:** “Manufacturing Year” date may be incorrect and populated with the year “2000”. This will have no effect on the functionality.

```
w61/>##### Welcome to ST67W6X CLI Application #####
# build: 15:34:46 Sep  5 2025
----- Host info -----
Host FW Version:      1.1.0
----- ST67W6X info -----
ST67W6X MW Version:   1.1.1
AT Version:           1.0.0
SDK Version:          2.0.89
Wi-Fi MAC Version:    1.6.44
BT Controller Version: 1.6.112
BT Stack Version:     1.9.56
Build Date:           Sep  4 2025 11:29:50
Module ID:            Undefined
BOM ID:               0
Manufacturing Year:   2000
Manufacturing Week:   00
Battery Voltage:      3.316 V
Trim Wi-Fi hp:        8,8,8,8,8,7,6,6,7,7,7,8,8,8
Trim Wi-Fi lp:        8,8,8,8,9,9,9,9,10,10,10,11,11,11
Trim BLE:              7,5,5,7,7
Trim XTAL:            37
MAC Address:          40:82:7b:00:0c:2d
Anti-rollback Bootloader: 0
Anti-rollback App:    0
-----
Wi-Fi init is done
Net init is done
MQTT init is done
Starting FOTA task
ready
Application started from bank 1
```





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# How to use X-CUBE-ST67W61 within STM32CubeMX

# Software Development Tools

The image shows two overlapping software interfaces. The background interface is STM32CubeIDE, with a dark blue header and a sidebar. The foreground interface is STM32CubeMX, showing a 'New Project' screen. A pink box highlights a table of software components. Three labels are overlaid on the image: 'IDE' in a dark blue box, 'Code Gen' in a light blue box, and 'Wi-Fi SW' in a pink box.

| Component                         | Version | Selected                 |
|-----------------------------------|---------|--------------------------|
| STMicroelectronics X-CUBE-ST67W61 | 1.0.0   |                          |
| Device Applications               | 1.0.0   |                          |
| Application                       |         | Not selected             |
| Freertos_Tickless                 | 1.0.0   | <input type="checkbox"/> |
| Network ST67W6X_Network_Driver    | 1.0.0   |                          |
| ServiceShell                      | 1.0.0   | <input type="checkbox"/> |
| ServiceAPI                        | 1.0.0   | <input type="checkbox"/> |
| Driver / W61_at_and_bus           | 1.0.0   | <input type="checkbox"/> |
| Utilities                         |         |                          |
| Debug TraceRecorder               | 4.10.2  |                          |
| TraceRecorder                     | 4.10.2  | <input type="checkbox"/> |
| Data Exchange cJSON               | 1.7.18  |                          |
| cJSON                             | 1.7.18  | <input type="checkbox"/> |
| File System LittleFS              | 2.10.1  |                          |
| LittleFS                          | 2.10.1  | <input type="checkbox"/> |
| Utility Utilities                 | 1.4.2   |                          |
| LPM / Tiny LPM                    | 1.4.2   | <input type="checkbox"/> |

**“STM32CubeIDE”**  
development environment

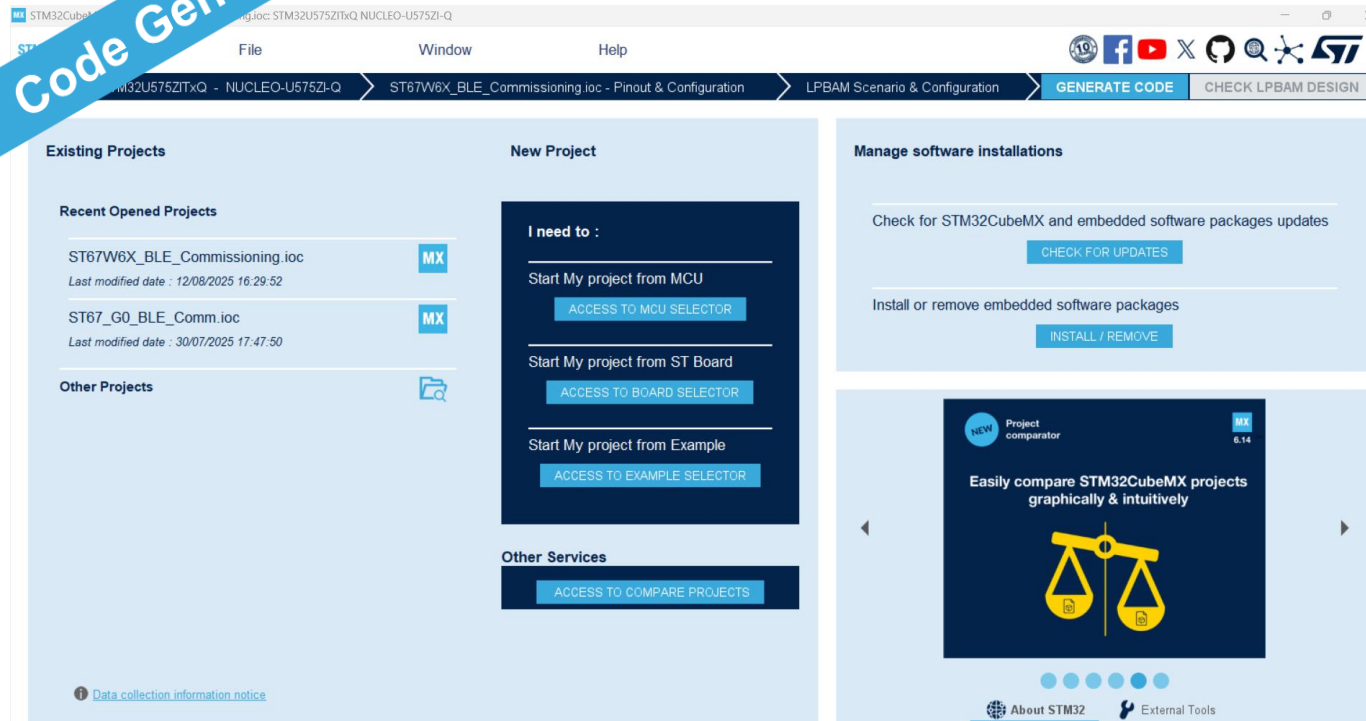
↳ **“STM32CubeMX”**  
initialization code generator

↳ **“X-CUBE-ST67W61”**  
application code and drivers for Wi-Fi

# STM32CubeMX

an All-in-1 Development Tool

Code Gen



Very powerful configuration and code generation

Support all STM32 MCU and MPU, with integrated powerful Finder

Pinouts, clock tree, peripherals and middleware configuration.

Expandable to support wireless connectivity, sensors and much more!



Three possible cases to generate the code from the Cube MX:

- **Case 1:** How to generate a project starting from an existing .ioc
- **Case 2:** How to export a project to a pinout compatible MCU
- **Case 3:** How to generate a project starting from scratch

In this section: We would be having **hands-on** to generate **BLE Commissioning** code using the following:

- Case 1 with NUCLEO-U575ZI-Q as host processor
- Case 3 with NUCLEO-G0B1RE as host processor



# BLE Commissioning App

- This application aims to demonstrate **how to provision Wi-Fi credentials via Bluetooth® LE** to establish a Wi-Fi connection to an access point.





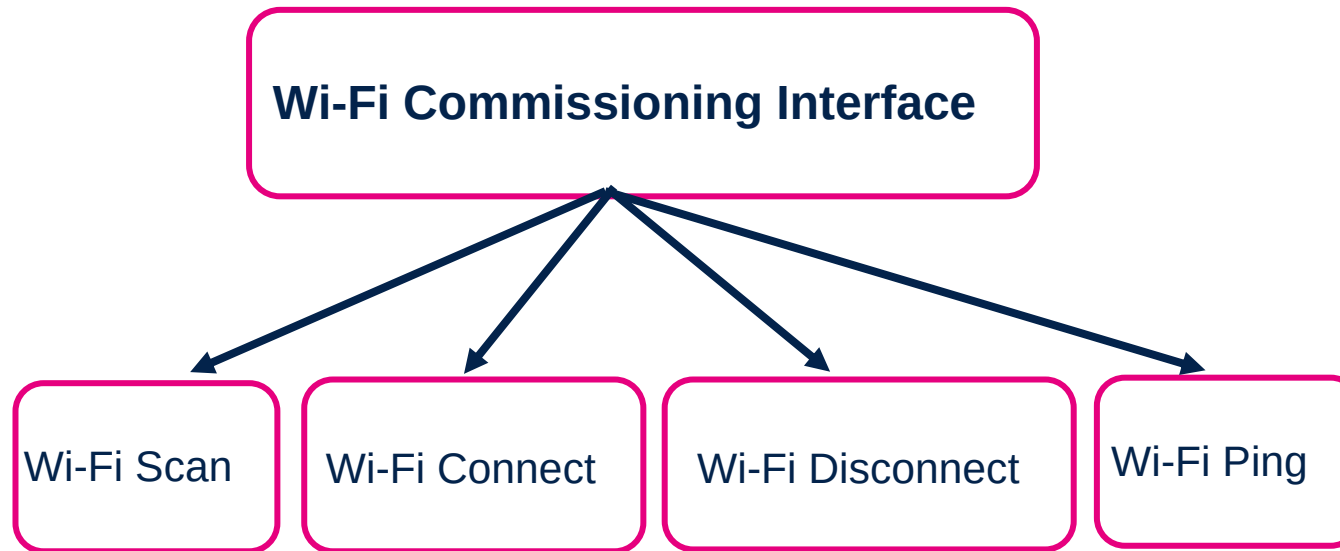
# BLE Commissioning App cont..

The application acts as a peripheral device embedding the commissioning profile and its four characteristics.

At startup, the application starts to advertise.

ST Web Bluetooth® Interface must be used to scan and connect to the commissioning application: (Web-Interface available from the browser) / ST BLE Toolbox (Mobile Application)

Once connected, a Wi-Fi commissioning interface is available on web Bluetooth® page/ ST BLE Toolbox



# Commissioning profile overview

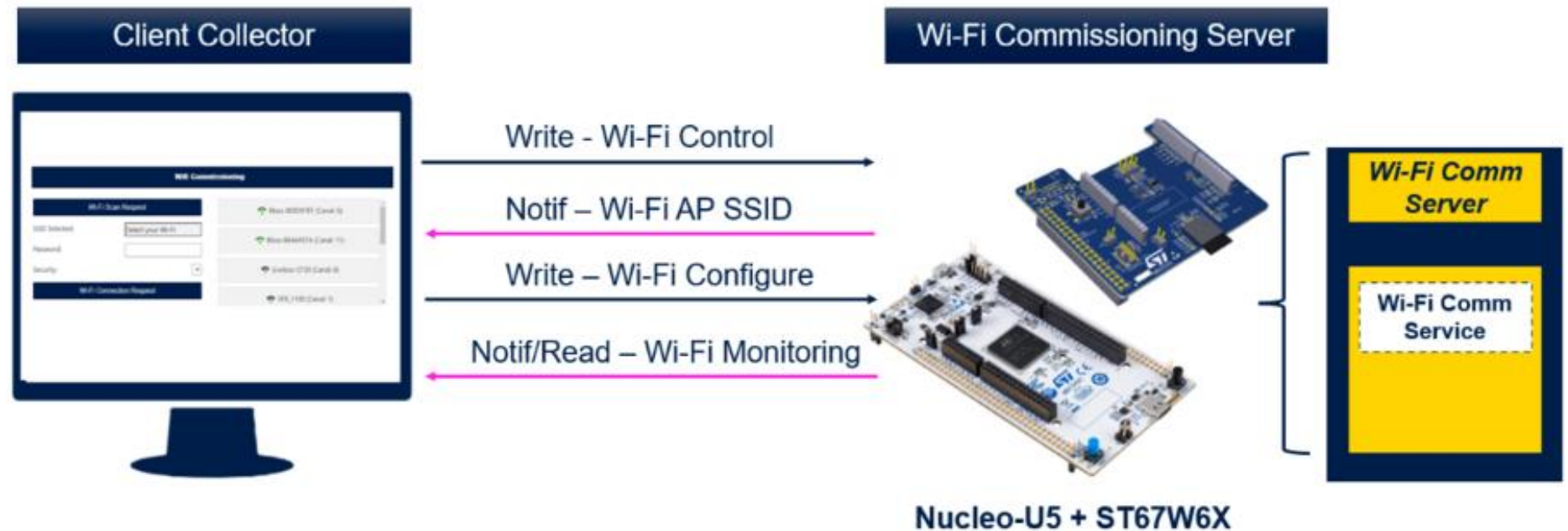
The Wi-Fi commissioning application demonstrates point to point communication using proprietary service and characteristics. It acts as a Peripheral device with one GATT service and four characteristics.

## Wi-Fi commissioning server:

- Contains the Wi-Fi commissioning service, which exposes four characteristics: **Wi-Fi Control (write)**, **Wi-Fi Configure (write)**, **Wi-Fi Access Point List (notification)**, **Monitoring (Notification)** to control and monitor a Wi-Fi connection.
- Is the GATT server

## Client Collector: (STBLE Toolbox/ ST Web Bluetooth page)

- Accesses the information exposed by the Wi-Fi commissioning application, **configures and controls the Wi-Fi connection** with the write characteristics, **monitors the Wi-Fi connection status** by receiving notifications from it
- Is the GATT client





The table below describes the structure of the commissioning service:

| Bluetooth® LE commissioning service specification |                 |                          |
|---------------------------------------------------|-----------------|--------------------------|
| Service                                           | Characteristic  | Mode                     |
| Wi-Fi commissioning                               |                 |                          |
|                                                   | Wi-Fi Control   | Write with Response/Read |
|                                                   | Wi-Fi Configure | Write with Response/Read |
|                                                   | Wi-Fi AP List   | Notify                   |
|                                                   | Monitoring      | Notify                   |

| Wi-Fi commissioning - Wi-Fi control |                                                                                             |
|-------------------------------------|---------------------------------------------------------------------------------------------|
| Byte Index                          | 0                                                                                           |
| Name                                | Action                                                                                      |
| Value                               | 0x01: Wi-Fi Start Scan<br>0x03: Wi-Fi Connect<br>0x04: Wi-Fi Disconnect<br>0x05: Wi-Fi Ping |

**Wi-Fi control characteristic:**  
Used to drive the Wi-Fi by launching scans, connecting or disconnecting from a network



### Wi-Fi commissioning - Wi-Fi configure

| Byte Index | 0                          | 1                  |
|------------|----------------------------|--------------------|
| Name       | Type                       | Data[ ]            |
| Value      | 0x01: AP-SSID<br>0x02: PWD | "ASCII"<br>"ASCII" |

#### Wi-Fi configure characteristic:

Used to setup the Wi-Fi parameters before establishing a connection

### Wi-Fi Commissioning - Wi-Fi AP List

| Byte Index | 0             | 1             | 2 to 3          | 4 to 7           | 8 to 40 |
|------------|---------------|---------------|-----------------|------------------|---------|
| Name       | SSID Length   | Channel       | Signal level    | Security Flag    | SSID    |
| Value      | 0x00 ... 0x20 | 0x00 ... 0xFF | 0x00 ... 0xFFFF | Security_Flag[ ] | Data[ ] |

#### Wi-Fi access point list characteristic:

Used to notify information about scanned access points

### Wi-Fi commissioning - Wi-Fi monitoring

| Byte Index | 0                                                                                      | 1 up to 239                                                          |
|------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Name       | Type                                                                                   | Data[ ]                                                              |
| Value      | 0x03: Connecting<br>0x04: Connection established<br>0x05: Ping response<br>0x06: Error | X<br>SSID<br>Refer to <b>table below</b><br>0x01: Connection Timeout |

#### Monitoring characteristic:

Used to notify information about Wi-Fi network.



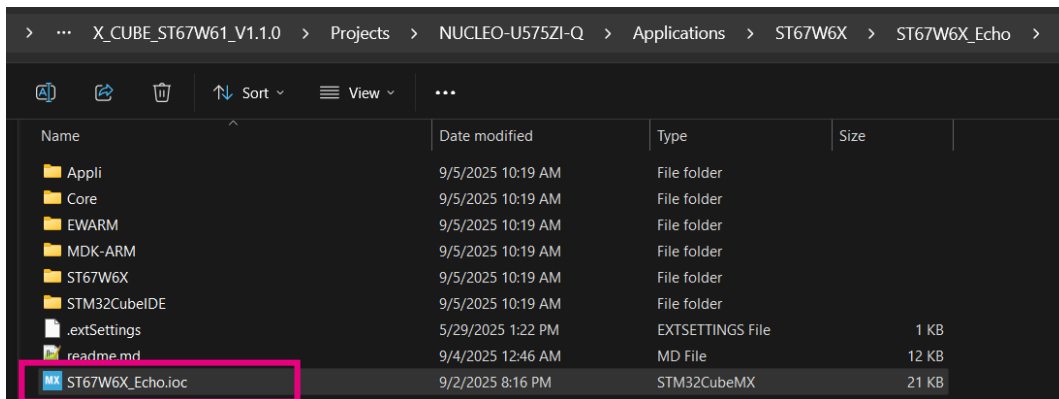
life.augmented

Generating BLE Commissioning  
app using NUCLEO-U575ZI-Q as  
host processor (Hands-on)

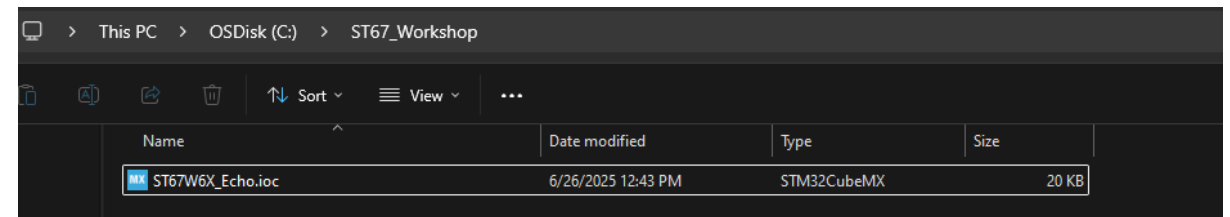
# Case 1: Generating BLE Commissioning app using NUCLEO-U575ZI-Q as host processor

We would start with the ST67W6X\_Echo.ioc file as a starting point and change the configuration to generate the BLE Commissioning application from the STM32CubeMX.

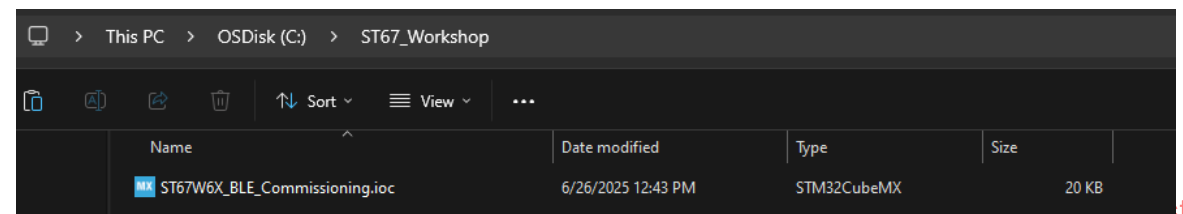
1. Browse through C:\X\_CUBE\_ST67W61\_V1.1.0\Projects\NUCLEO-U575ZI Q\Applications\ST67W6X\ST67W6X\_Echo
2. Copy ST67W6X\_Echo.ioc
3. Create a new directory C:\ST67\_Workshop
4. Paste and rename to ST67W6X\_BLE\_Commissioning.ioc



Copy and Paste



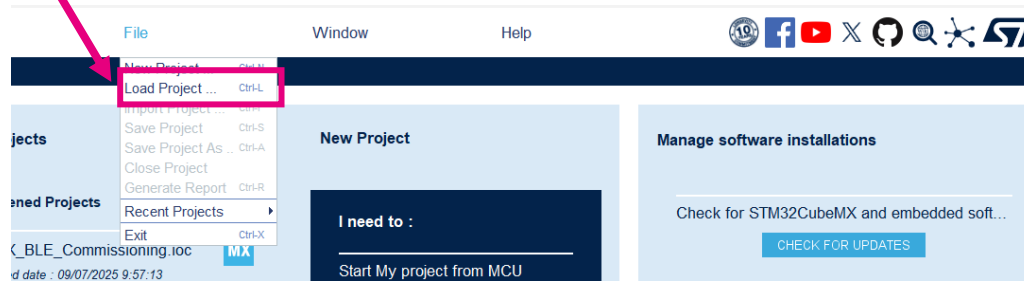
Rename



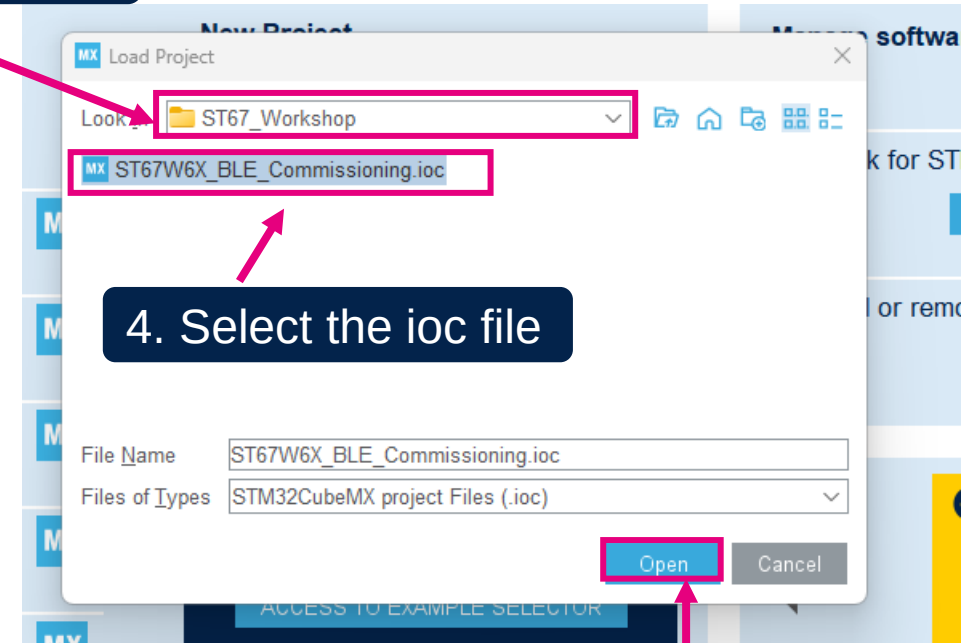
# Loading ioc file in STM32CubeMX

1. Run STM32CubeMX with **admin rights**

2. Load Project



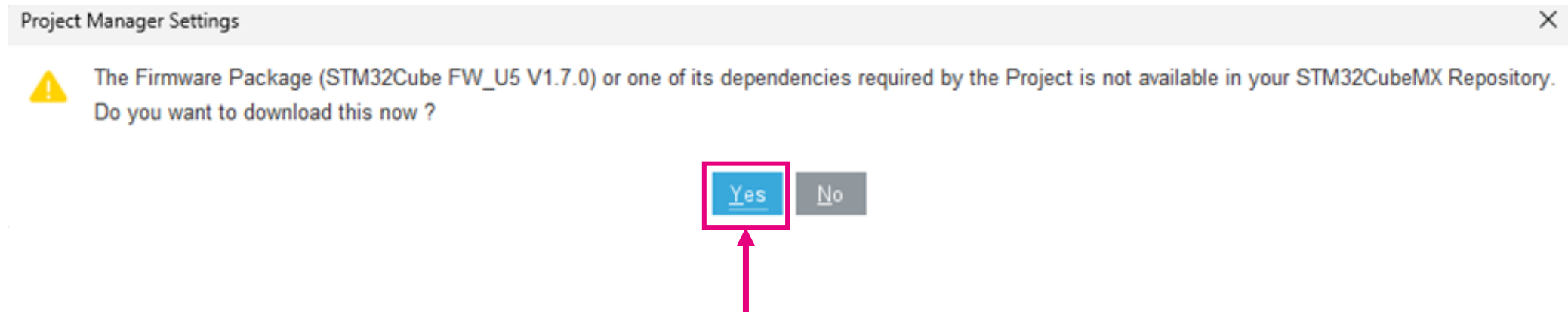
3. Go the ST67\_Workshop



4. Select the ioc file

5. Click open

# If you get the below warnings





# Configuration

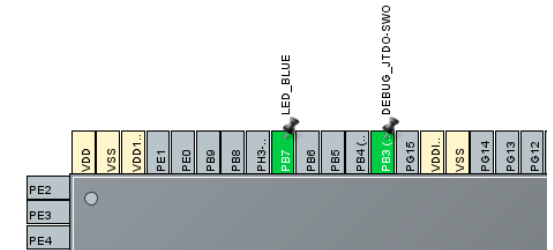
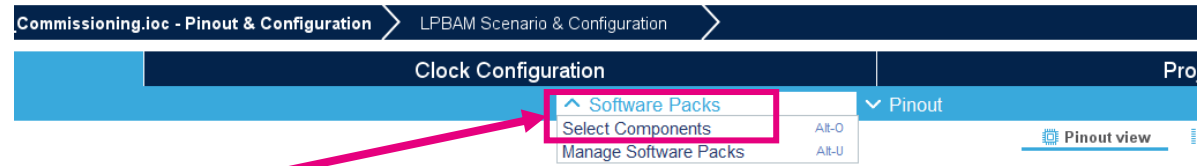
2. Select BLE  
Commissioning app

1. Select Components

3. Select T01(NCP) ST67  
architecture

MX Software Packs Component Selector

| Pack / Bundle / Component           | Status | Version | Selection                           |
|-------------------------------------|--------|---------|-------------------------------------|
| ▼ STMicroelectronics.X-CUBE-ST67W61 | ✓      | 1.1.0   |                                     |
| ▼ Device Applications               | ✓      | 1.1.0   |                                     |
| Application                         | ✓      | 1.1.0   | BLE Commissioning                   |
| FreeRTOS_Tickless                   | ✓      | 1.1.0   | <input checked="" type="checkbox"/> |
| ▼ Network ST67W6X_Network_Driver    | ✓      | 1.1.0   |                                     |
| ST67 Architecture                   | ✓      | 1.1.0   | T01                                 |
| ATDriver                            | ✓      | 1.1.0   | <input checked="" type="checkbox"/> |
| ServiceAPI                          | ✓      | 1.1.0   | <input checked="" type="checkbox"/> |
| ServiceShell                        | ✓      | 1.1.0   | <input type="checkbox"/>            |
| ▼ Utilities                         | ✓      |         |                                     |
| Logging                             | ✓      | 1.1.0   | <input checked="" type="checkbox"/> |
| Shell                               | ✓      | 1.1.0   | <input type="checkbox"/>            |
| Statistics                          | ✓      | 1.1.0   | <input type="checkbox"/>            |
| ▼ Debug TraceRecorder               | ✓      | 4.10.2  |                                     |
| TraceRecorder                       | ✓      | 4.10.2  | <input type="checkbox"/>            |
| ▼ Data Exchange cJSON               | ✓      | 1.7.18  |                                     |
| cJSON                               | ✓      | 1.7.18  | <input type="checkbox"/>            |
| ▼ File System LittleFS              | ✓      | 2.10.1  |                                     |
| LittleFS                            | ✓      | 2.10.1  | <input type="checkbox"/>            |
| ▼ Utility Utilities                 | ✓      | 1.4.2   |                                     |
| LPM / Tiny LPM                      | ✓      | 1.4.2   | <input checked="" type="checkbox"/> |
| ▼ Network LwIP                      | ✓      | 2.2.0   |                                     |
| LwIP                                | ✓      | 2.2.0   | <input type="checkbox"/>            |
| ▼ Data Exchange MQTT-C              | ✓      | 1.1.6   |                                     |
| MQTT-C                              | ✓      | 1.1.6   | <input type="checkbox"/>            |



#### 4. Under Middleware and Software Packs

Categories: A-Z

Middleware and Software Packs

- AIROC-Wi-Fi-Bluetooth-STM32 FILEX
- FP-SNS-FLIGHT1
- FP-SNS-MOTENV1
- FP-SNS-MOTENVWB1
- FP-SNS-SMARTAG2
- FP-SNS-STAIOTCFT
- FP-SNS-STBOX1
- I-CUBE-CANOPEN
- I-CUBE-Cesium
- I-CUBE-FS-RTOS
- I-CUBE-ITTIADB
- I-CUBE-Mongoose
- I-CUBE-embOS
- I-CUBE-wolfMQTT
- I-CUBE-wolfSSH
- I-CUBE-wolfSSL
- I-CUBE-wolfTPM
- I-Cube-SolM-uGOAL
- LEVELX
- NETXDUO
- THREADX
- TOUCHSENSING
- USBPD
- USBX
- X-CUBE-AI
- X-CUBE-ALGOBUILD
- X-CUBE-ALS
- X-CUBE-BLE1
- X-CUBE-BLE2
- X-CUBE-BLEMGR
- X-CUBE-DPower
- X-CUBE-EEPROMA1
- X-CUBE-FREERTOS
- X-CUBE-GNSS1
- X-CUBE-IPS
- X-CUBE-ISPU
- X-CUBE-IoTC-DA16k-PMOD
- X-CUBE-MEMS1
- X-CUBE-NFC4
- X-CUBE-NFC6
- X-CUBE-NFC7
- X-CUBE-NFC9
- X-CUBE-NFC10
- X-CUBE-NFC12
- X-CUBE-PM33A1
- X-CUBE-RT-Thread\_Nano
- X-CUBE-SAFE1
- X-CUBE-SFXS2LP1
- X-CUBE-SMBUS
- X-CUBE-ST60
- X-CUBE-ST67W61
- X-CUBE-ST67W61
- X-CUBE-SUBG2
- X-CUBE-TCPD

STM32CubeMX: STM32CubeMX-STM32W61.1.1.0 Mode and Configuration

Mode

- Device Applications
- Network ST67W6X Network Driver
- Utility Utilities

Configuration

Reset Configuration

Parameter Settings | User Constants | Platform Settings

Configure the below parameters :

Search (Ctrl+F)

W6X Init Modules

- ST67\_ARCH: W6X\_ARCH\_T01
- WIFI\_STATION: Yes
- WIFI\_SAP: No
- BLE: Yes
- NET: Yes
- MQTT: No

Basic Parameters

- DEBUGGER\_ON: ON
- LOG\_OUTPUT\_MODE: LOG\_OUTPUT\_UART
- LOW\_POWER\_MODE: LOW\_POWER\_SLEEP\_ENABLE
- FREERTOS\_TASK\_NOTIF\_ARRAY\_LEN: 8

Logging-Shell

- LOG\_LEVEL: LOG\_DEBUG

W6X Parameters

- W6X\_POWER\_SAVE\_AUTO: No
- W6X\_CLOCK\_MODE: Internal RC oscillator
- W6X\_WIFI\_AUTOCONNECT: No
- W6X\_WIFI\_COUNTRY\_CODE: 00
- W6X\_WIFI\_ADAPTIVE\_COUNTRY\_CODE: No
- W6X\_BLE\_HOSTNAME: ST67W61\_BLE
- W6X\_NET\_HOSTNAME: ST67W61\_WiFi
- W6X\_NET\_DHCP\_MODE: DHCP client STA
- W6X\_NET\_RECV\_TIMEOUT (ms): 10000
- W6X\_NET\_SEND\_TIMEOUT (ms): 10000
- W6X\_NET\_RECV\_BUFFER\_SIZE: 9216
- W6X\_NET\_DNS\_MANUAL: No
- W6X\_NET\_DNS\_IP\_1: {208, 67, 222, 222}
- W6X\_NET\_DNS\_IP\_2: {8, 8, 8, 8}
- W6X\_NET\_DNS\_IP\_3: {0, 0, 0, 0}

W61 drv Parameters

- W61\_WIFI\_MAX\_DETECTED\_AP: 20
- W61\_BLE\_MAX\_DETECTED\_PERIPHERAL: 10

# Configuration

Bluetooth Low Energy connection with NCP in Power save mode requires accurate external low-frequency clock. To setup and use external clock, SW and HW settings must be modified.

Refer to [Wiki Bluetooth LE with low-power setup]([https://wiki.st.com/stm32mcu/wiki/Connectivity:How\\_to\\_measure\\_ST67W611M\\_current\\_consumption#Bluetooth\\_h-C2-AE\\_LE\\_with\\_low-power\\_setup](https://wiki.st.com/stm32mcu/wiki/Connectivity:How_to_measure_ST67W611M_current_consumption#Bluetooth_h-C2-AE_LE_with_low-power_setup)) page to be informed about required changes.

#### 6. Disable W6X\_POWER\_SAVE\_AUTO

#### 7. Rename W6X\_BLE\_HOSTNAME to something unique

#### 5. X-CUBE-ST67W61



STMicroelectronics

Home > STM32U575ZITxQ - NUCLEO-U575ZI-Q > ST67W6X\_BLE\_Commissioning.ioc - Project Manager > LPBAM Scenario & Configuration > GENERATE CODE

Pinout & Configuration | Clock Configuration | Project Manager

Project Settings

Project Name: ST67W6X\_BLE\_Commissioning

Project Location: C:\ST67\_Workshop\ Browse

Application Structure: Advanced ☐ Do not generate the main()

Toolchain Folder Location: C:\ST67\_Workshop\ST67W6X\_BLE\_Commissioning\

Toolchain / IDE: STM32CubeIDE ☒ Generate Under Root

Linker Settings

Minimum Heap Size: 0x200

Minimum Stack Size: 0x400

Thread-safe Settings

CortexM33

☐ Enable multi-threaded support

Thread-safe Locking Strategy: Default - Mapping suitable strategy depending on RTOS selection.

Mcu and Firmware Package

Mcu Reference: STM32U575ZITxQ

Firmware Package Name and Version: STM32Cube\_FW\_U5\_V1.8.0

☒ Use Default Firmware Location

Firmware Relative Path: C:/Users/kulkarni/STM32Cube\_FW\_U5\_V1.8.0/STM32Cube\_FW\_U5\_V1.8.0 Browse

1. Enable "Generate Under Root"

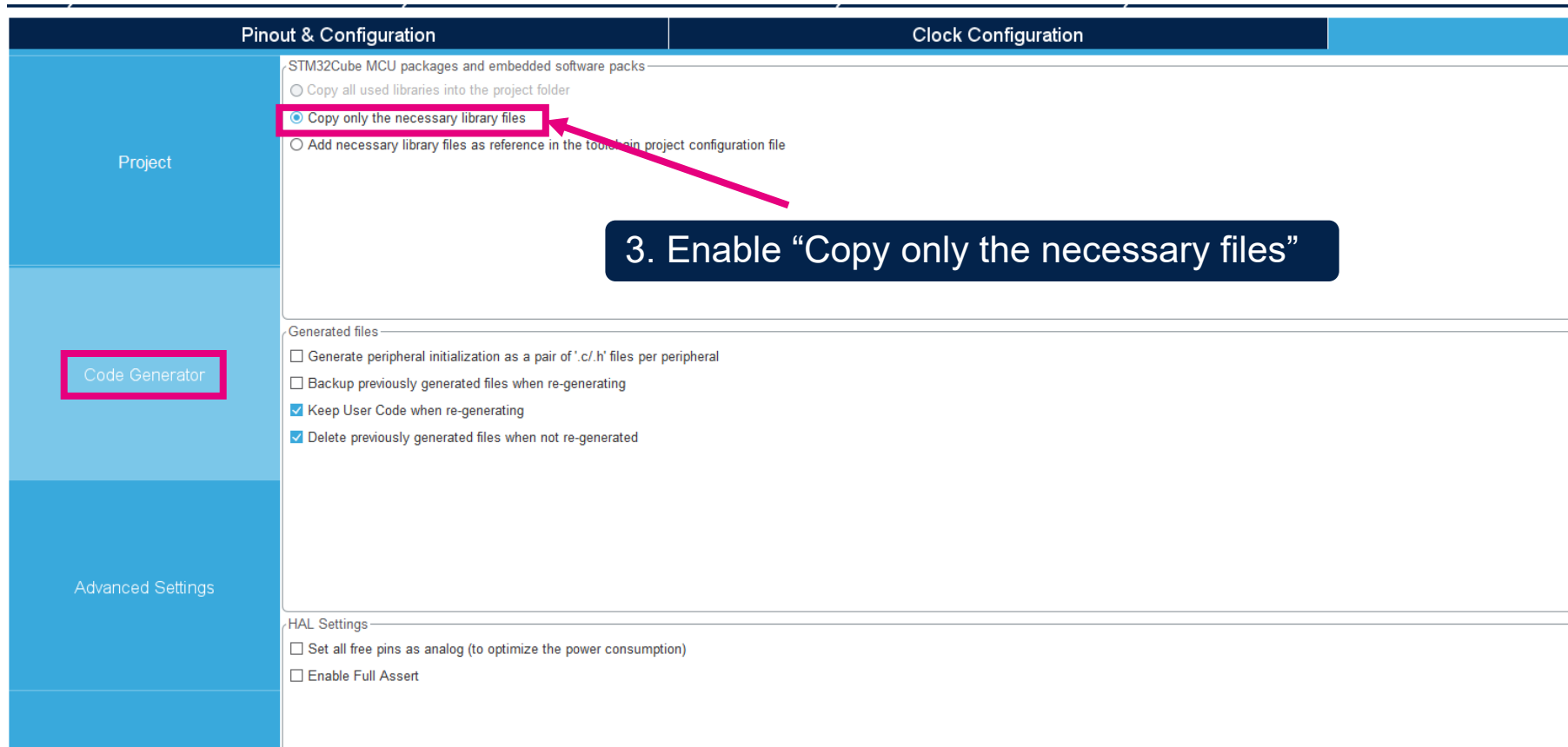
2. Enable "Use Default Firmware Location"

# Project Settings

Here, we are enabling the Flat directory structure: (i.e. Driver and MW directories copied into your project directory)



# Project Settings cont..



Pinout & Configuration

Clock Configuration

Project

Code Generator

Advanced Settings

STM32Cube MCU packages and embedded software packs

- ☐ Copy all used libraries into the project folder
- ☒ Copy only the necessary library files
- ☐ Add necessary library files as reference in the toolchain project configuration file

Generated files

- ☐ Generate peripheral initialization as a pair of '.c/.h' files per peripheral
- ☐ Backup previously generated files when re-generating
- ☒ Keep User Code when re-generating
- ☒ Delete previously generated files when not re-generated

HAL Settings

- ☐ Set all free pins as analog (to optimize the power consumption)
- ☐ Enable Full Assert

3. Enable "Copy only the necessary files"

# Generate Code

Home > STM32U575ZITxQ - NUCLEO-U575ZI-Q > ST67W6X\_BLE\_Commissioning.ioc - Project Manager > LPBAM Scenario & Configuration > **GENERATE CODE** > CHECK LPE

Pinout & Configuration | Clock Configuration | **Project Manager** | T

Project

Code Generator

Advanced Settings

Driver Selector

Search (Ctrl+F)

- CORTEX\_M33\_NS
- PWR
- RCC
- GPIO
- > GPDMA
- ICACHE
- > SPI
- > USART
- > LPTIM
- STMicroelectronics.X-CUBE-ST67W61.1.1.0

HAL  
HAL  
HAL  
HAL  
HAL  
HAL  
HAL  
HAL  
HAL  
HAL

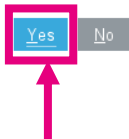
Generated Function Calls

| Generate Code                       | Rank | Function Name       | Peripheral Instance Name          | <input type="checkbox"/> Do Not Generate Function Call | <input type="checkbox"/> Visibility (Static) |
|-------------------------------------|------|---------------------|-----------------------------------|--------------------------------------------------------|----------------------------------------------|
| <input checked="" type="checkbox"/> | 1    | SystemClock_Config  | RCC                               | <input type="checkbox"/>                               | <input type="checkbox"/>                     |
| <input checked="" type="checkbox"/> | 2    | MX_GPIO_Init        | GPIO                              | <input type="checkbox"/>                               | <input checked="" type="checkbox"/>          |
| <input checked="" type="checkbox"/> | 3    | MX_GPDMA1_Init      | GPDMA1                            | <input type="checkbox"/>                               | <input checked="" type="checkbox"/>          |
| <input checked="" type="checkbox"/> | 4    | MX_ICACHE_Init      | ICACHE                            | <input type="checkbox"/>                               | <input checked="" type="checkbox"/>          |
| <input checked="" type="checkbox"/> | 5    | MX_SPI1_Init        | SPI1                              | <input type="checkbox"/>                               | <input checked="" type="checkbox"/>          |
| <input checked="" type="checkbox"/> | 6    | MX_USART1_UART_Init | USART1                            | <input type="checkbox"/>                               | <input checked="" type="checkbox"/>          |
| <input checked="" type="checkbox"/> | 7    | MX_LPTIM1_Init      | LPTIM1                            | <input type="checkbox"/>                               | <input checked="" type="checkbox"/>          |
| <input checked="" type="checkbox"/> | 10   | MX_ST67W6X_Init     | STMicroelectronics.X-CUBE-ST67... | <input type="checkbox"/>                               | <input type="checkbox"/>                     |

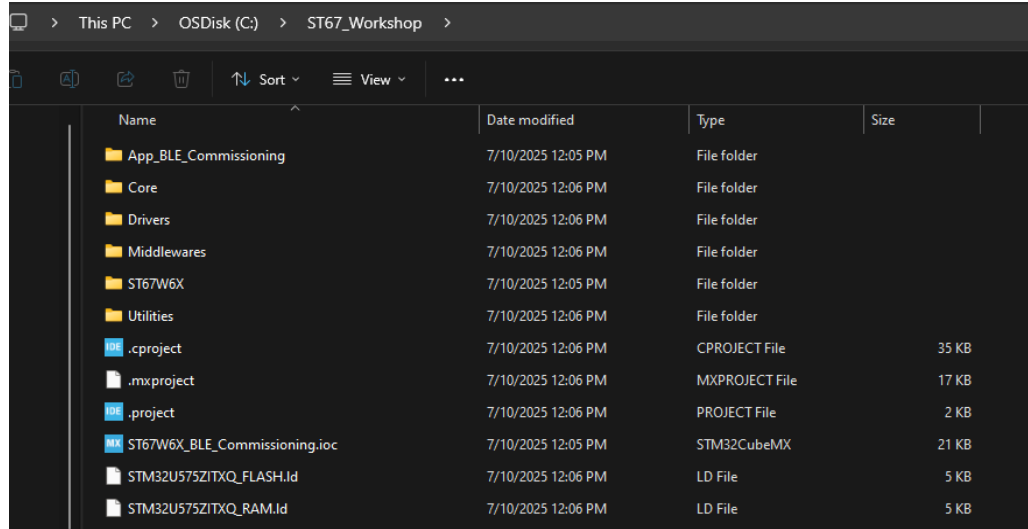
4. Generate Code

Project Manager Settings

⚠ The Firmware Package (STM32Cube FW\_U5 V1.7.0) or one of its dependencies required by the Project is not available in your STM32CubeMX Repository.  
Do you want to download this now ?



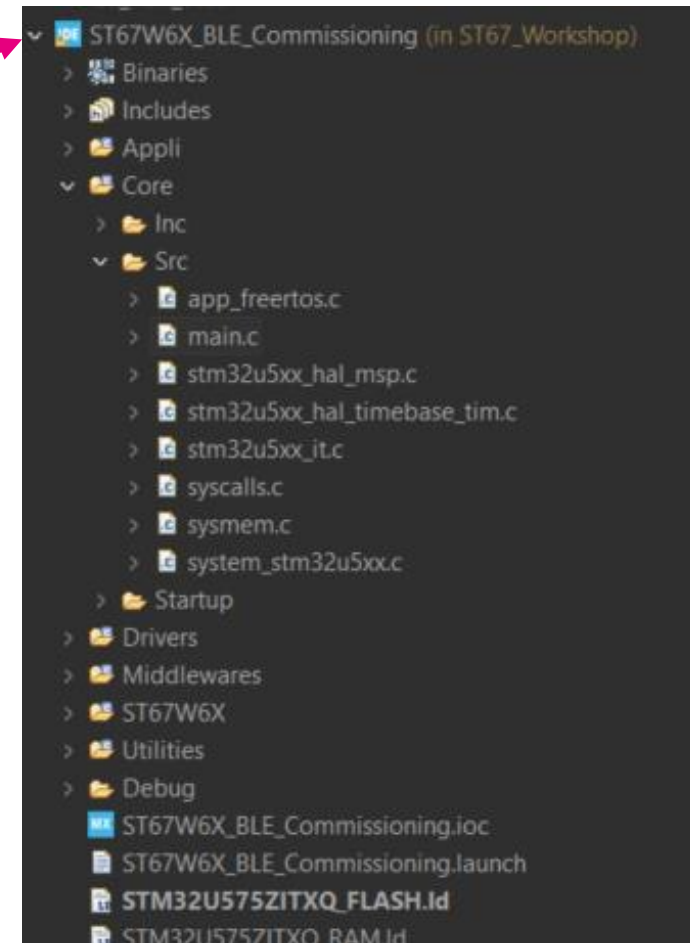
# STM32CubeIDE



| Name                          | Date modified      | Type           | Size  |
|-------------------------------|--------------------|----------------|-------|
| App_BLE_Commissioning         | 7/10/2025 12:05 PM | File folder    |       |
| Core                          | 7/10/2025 12:06 PM | File folder    |       |
| Drivers                       | 7/10/2025 12:06 PM | File folder    |       |
| Middlewares                   | 7/10/2025 12:06 PM | File folder    |       |
| ST67W6X                       | 7/10/2025 12:05 PM | File folder    |       |
| Utilities                     | 7/10/2025 12:06 PM | File folder    |       |
| .cproject                     | 7/10/2025 12:06 PM | CPROJECT File  | 35 KB |
| .mxproject                    | 7/10/2025 12:06 PM | MXPROJECT File | 17 KB |
| .project                      | 7/10/2025 12:06 PM | PROJECT File   | 2 KB  |
| ST67W6X_BLE_Commissioning.ioc | 7/10/2025 12:05 PM | STM32CubeMX    | 21 KB |
| STM32U575ZITXQ_FLASH.ld       | 7/10/2025 12:06 PM | LD File        | 5 KB  |
| STM32U575ZITXQ_RAM.ld         | 7/10/2025 12:06 PM | LD File        | 5 KB  |

After, generating the code, you would view similar folder structure under ST67\_Workshop

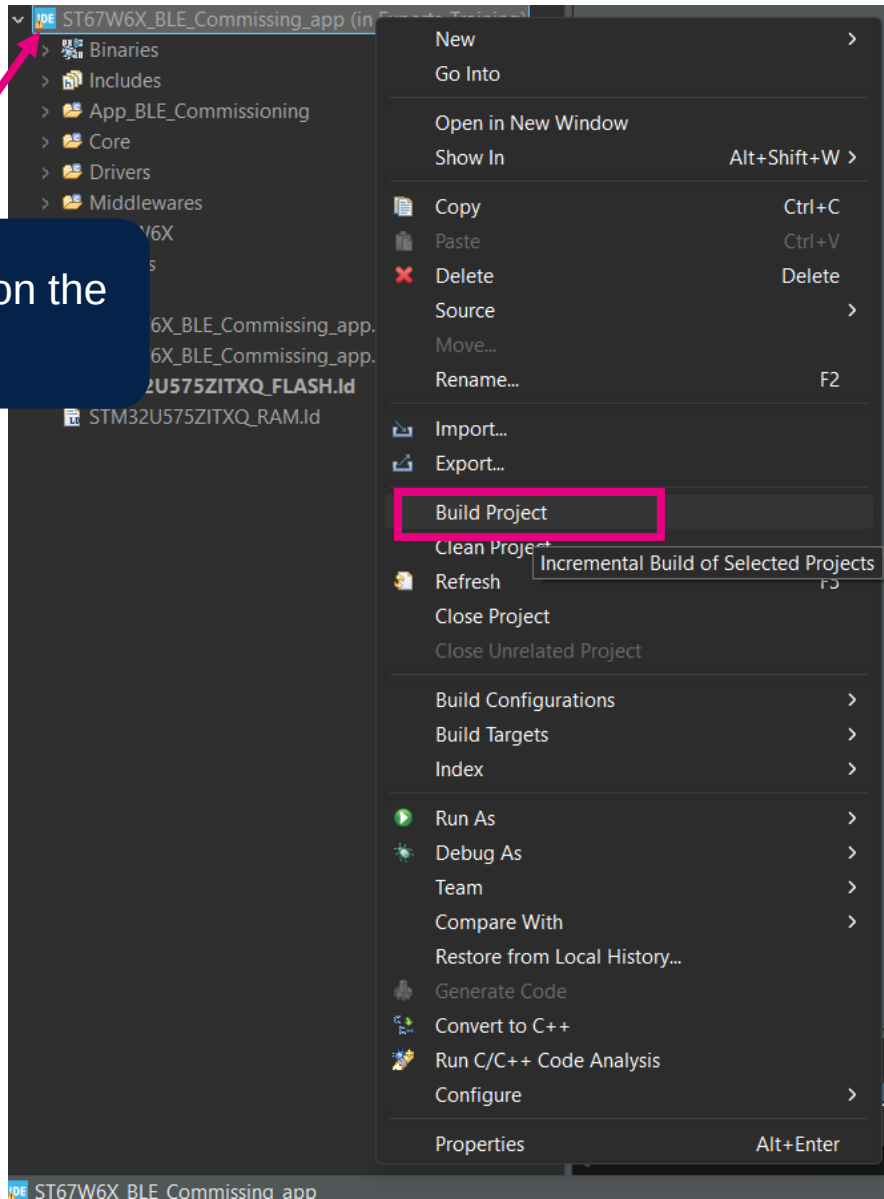
Load the project on the STM32CubeIDE





# Time to build

## Right click on the project



```

CDT Build Console [ST67W6X_BLE_Commissioning]
arm-none-eabi-gcc -mcpu=cortex-m33 -g3 -DDEBUG -DconfigTASK_NOTIFICATION_ARRAY_ENTRIES=8 -C -I../Core/Inc -I../Core/Src/app_freertos.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -DconfigTASK_NOTIFICATION_ARRAY_ENTRIES=8 -c ../Core/Src/main.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -DconfigTASK_NOTIFICATION_ARRAY_ENTRIES=8
arm-none-eabi-gcc -mcpu=cortex-m33 -g3 -DDEBUG -DconfigTASK_NOTIFICATION_ARRAY_ENTRIES=8 -C -I../Core/Inc -I../Core/Src/stm32u5xx_hal_msp.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -DconfigTASK_NOTIFICATION_ARRAY_ENTRIES=8 -c ../Core/Src/stm32u5xx_it.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -DconfigTASK_NOTIFICATION_ARRAY_ENTRIES=8
arm-none-eabi-gcc -mcpu=cortex-m33 -g3 -DDEBUG -DconfigTASK_NOTIFICATION_ARRAY_ENTRIES=8 -C -I../Core/Inc -I../Core/Src/syscalls.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -DconfigTASK_NOTIFICATION_ARRAY_ENTRIES=8 -c ../Core/Src/system_stm32u5xx.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -DconfigTASK_NOTIFICATION_ARRAY_ENTRIES=8
arm-none-eabi-gcc -mcpu=cortex-m33 -g3 -DDEBUG -DconfigTASK_NOTIFICATION_ARRAY_ENTRIES=8 -C -I../Core/Inc -I../App_BLE_Commissioning/Target/freertos_tickless.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -DconfigTASK_NOTIFICATION_ARRAY_ENTRIES=8 -c ../App_BLE_Commissioning/Target/logshell_ctrl.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -DconfigTASK_NOTIFICATION_ARRAY_ENTRIES=8
arm-none-eabi-gcc -mcpu=cortex-m33 -g3 -DDEBUG -DconfigTASK_NOTIFICATION_ARRAY_ENTRIES=8 -C -I../Core/Inc -I../App_BLE_Commissioning/Target/stm32_lpm_if.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -DconfigTASK_NOTIFICATION_ARRAY_ENTRIES=8 -c ../App_BLE_Commissioning/App/main_app.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -DconfigTASK_NOTIFICATION_ARRAY_ENTRIES=8 -o "ST67W6X_BLE_Commissioning.elf" @"objects.list" -mcpu=cortex-m33 -T"C:\ST67_Work\
Finished building target: ST67W6X_BLE_Commissioning.elf

arm-none-eabi-size ST67W6X_BLE_Commissioning.elf
arm-none-eabi-objdump -h -S ST67W6X_BLE_Commissioning.elf > "ST67W6X_BLE_Commissioning.list"
   text    data    bss     dec     hex filename
143952    328    208016  352296  56028 ST67W6X_BLE_Commissioning.elf
Finished building: default.size.stdout

Finished building: ST67W6X_BLE_Commissioning.list

08:12:05 Build Finished. 0 errors, 0 warnings. (took 14s.260ms)

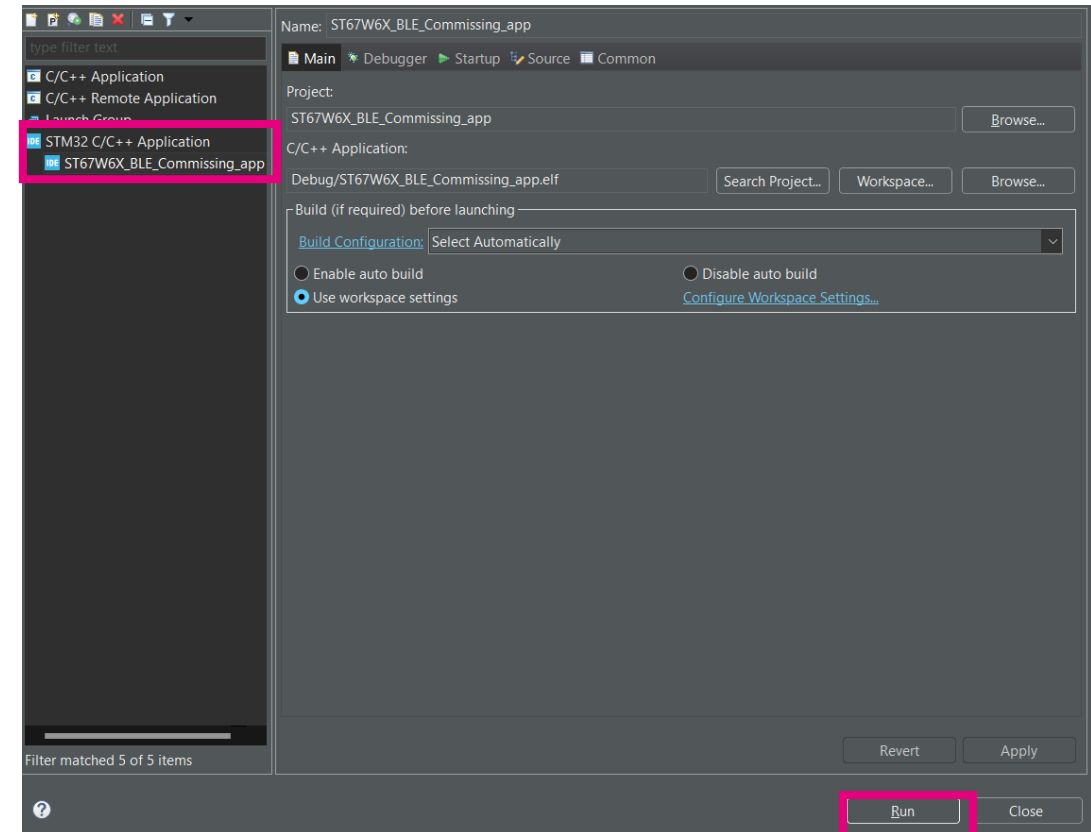
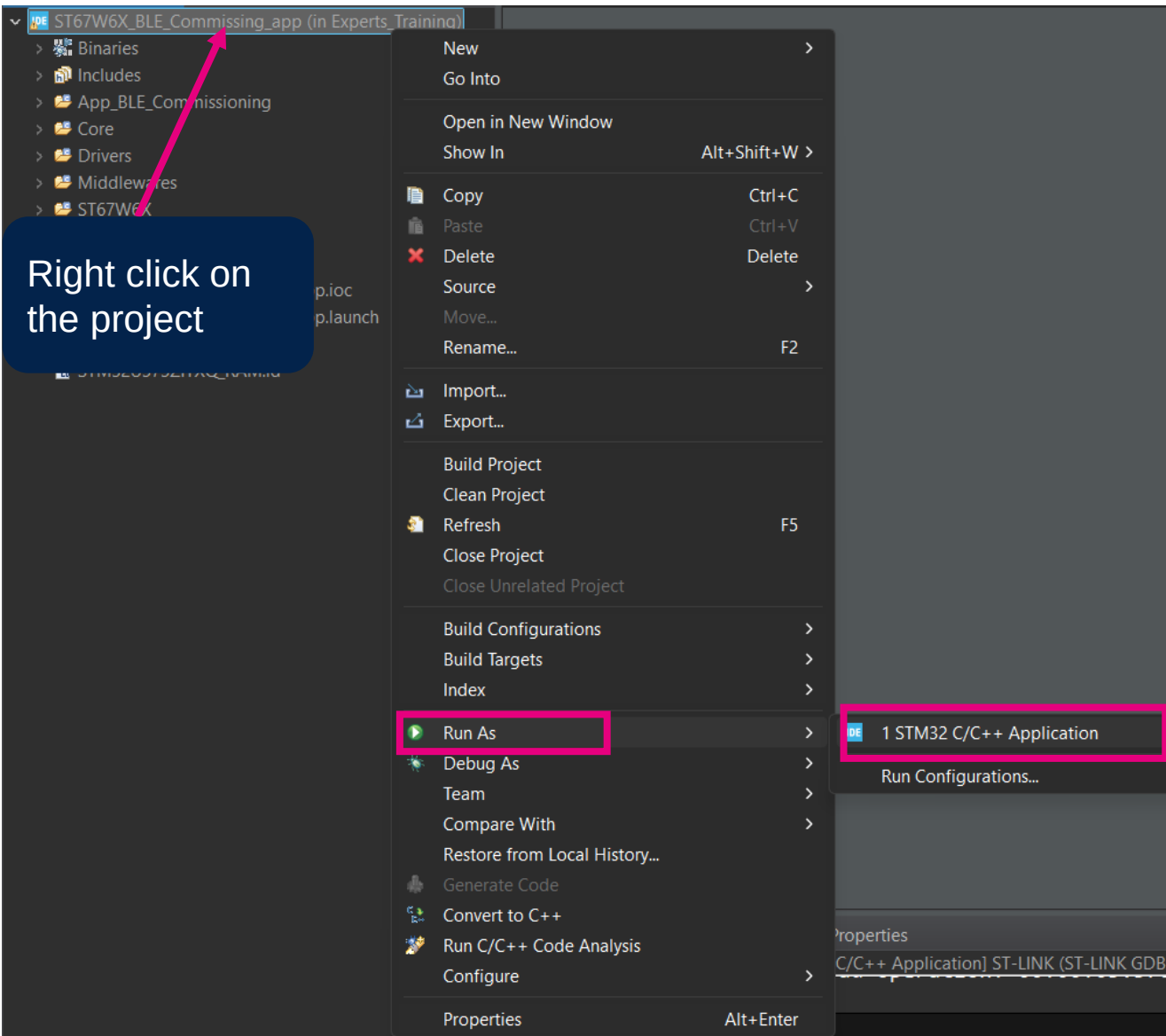
```

## Console after building the application

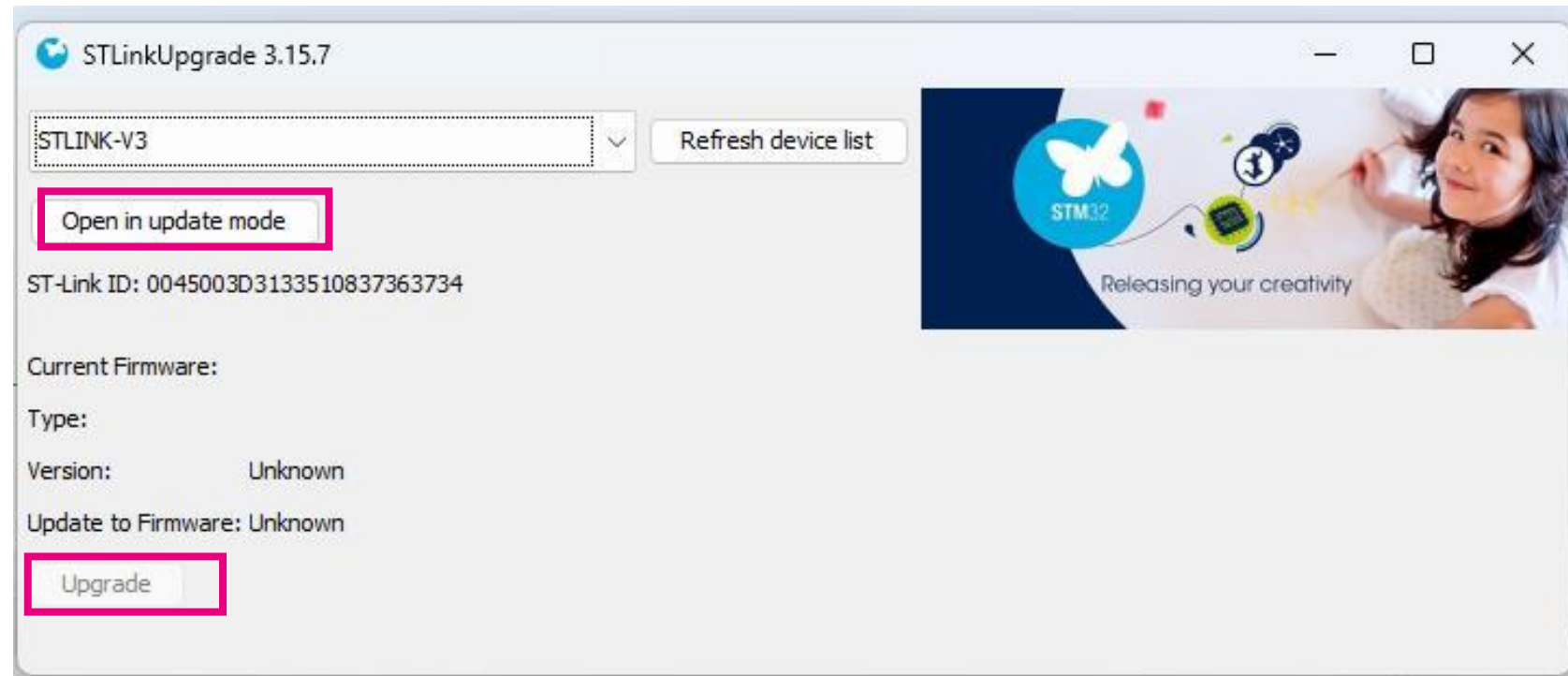
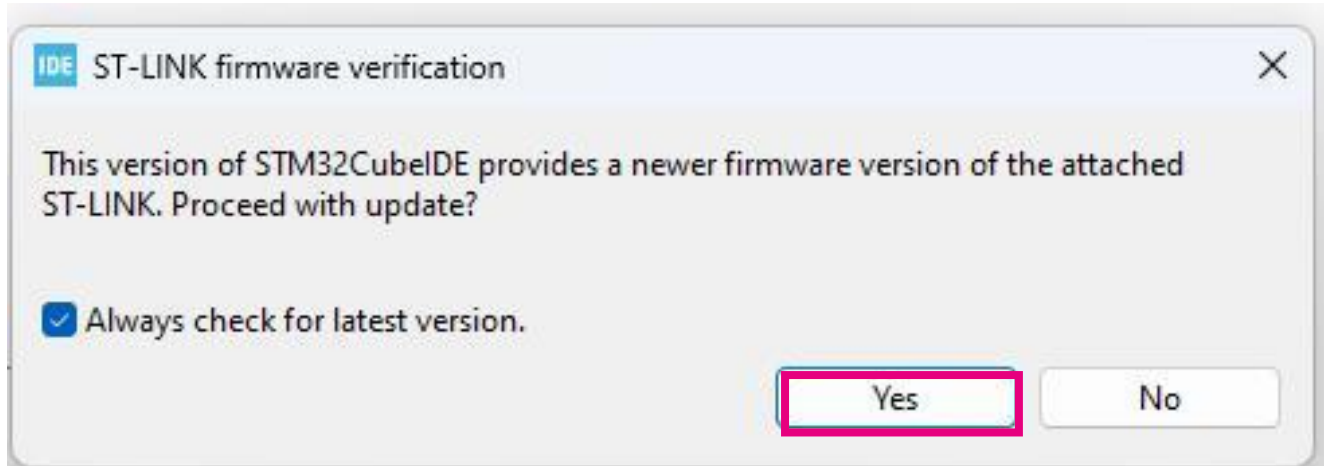


# Flash the project

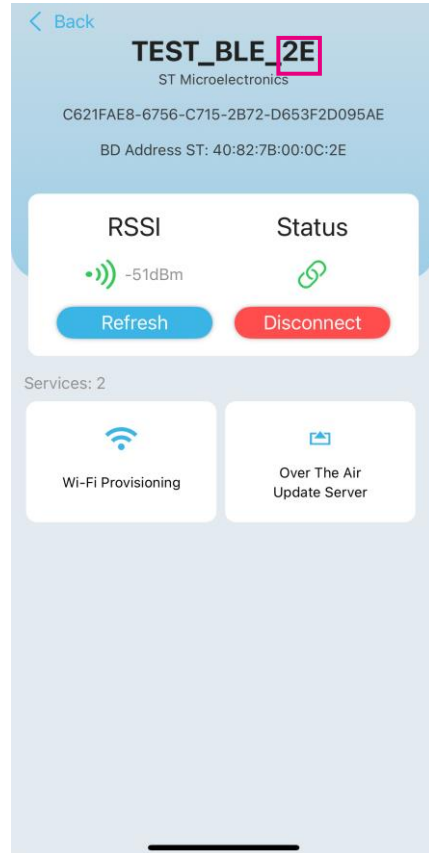
Right click on  
the project



# ST-LINK Update Warning



# Tera Term Logs

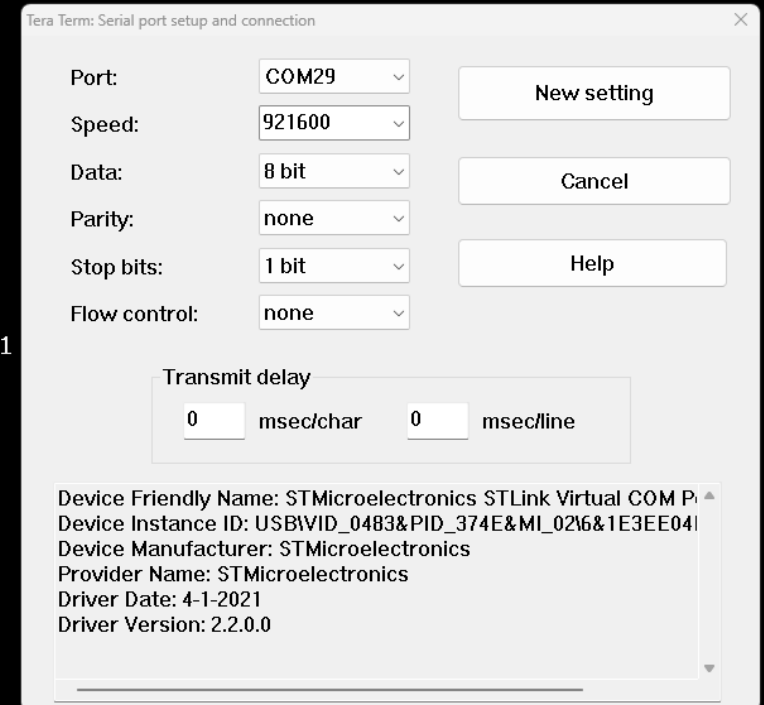


Last 1 byte of BD\_Address would match last byte advertising on the ST\_BLE\_Toolbox

```
COM29 - Tera Term VT
File Edit Setup Control Window Help
##### Welcome to ST67W6X Wi-Fi Commissioning over BLE Application #####
# build: 19:57:12 Sep 7 2025
----- Host info -----
Host FW Version: 1.1.0
----- ST67W6X info -----
ST67W6X MW Version: 1.1.1
AT Version: 1.0.0
SDK Version: 2.0.89
Wi-Fi MAC Version: 1.6.44
BT Controller Version: 1.6.112
BT Stack Version: 1.9.56
Build Date: Sep 4 2025 11:29:50
Module ID: Undefined
BOM ID: 0
Manufacturing Year: 2000
Manufacturing Week: 00
Battery Voltage: 3.324 V
Trim Wi-Fi hp: 8,8,8,8,8,7,6,6,7,7,7,8,8,8
Trim Wi-Fi lp: 8,8,8,8,9,9,9,10,10,10,11,11,11
Trim BLE: 7,5,5,7,7
Trim XTAL: 37
MAC Address: 40:82:7b:00:0c:2d
Anti-rollback Bootloader: 0
Anti-rollback App: 0
-----
Wi-Fi init is done
Net init is done
Ble init is done
BD Address: 40:82:7b:00:0c:2e
Configure BLE
BLE configuration is done

BLE Commissioning Service Creation
- BLE service created
- BLE WIFI Control charac created
- BLE WIFI Configure charac created
- BLE WIFI Monitoring charac created
- BLE FUOTA base address charac created
- BLE FUOTA confirmation charac created
- BLE FUOTA raw data charac created
- BLE services and charac registered
- BLE service and charac creation is done

Start BLE advertising
BLE advertising is started
```



Tera Term logs after flashing the BLE Commissioning application



# ST BLEToolbox/ ST BLE Webpage

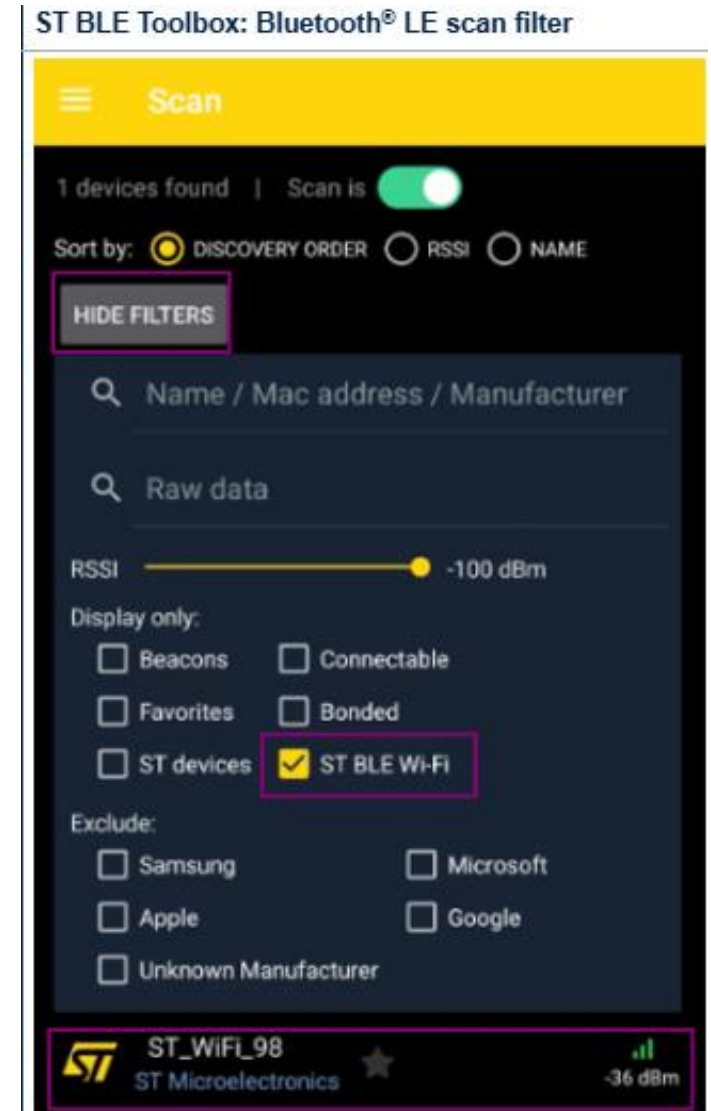
To interface with the Wi-Fi commissioning embedded application, you can use [STBLEToolbox Android/iOS application](#) / [ST Web Bluetooth® Interface](#)

Download [STBLEToolBox](#)

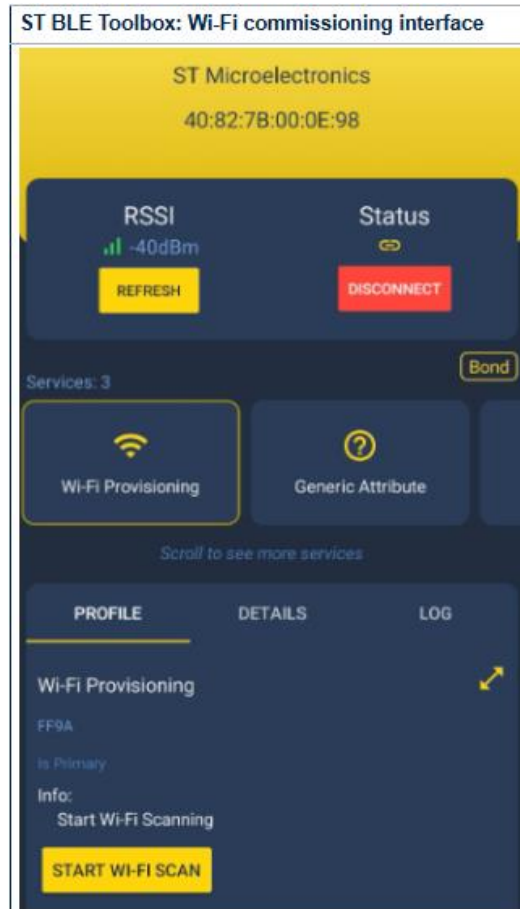
The versions supporting Wi-Fi commissioning service are **v1.4.3 for Android** and **1.2.8 for iOS**

## Steps to launch and connect to Wi-Fi AP:

1. Launch the smartphone application and enable the scan button, if it is not already activated.
2. You can filter scanned devices by clicking on the SHOW FILTERS button and select ST BLE Wi-Fi to show only devices with Wi-Fi commissioning service.
3. Select your device, named ST\_WiFi\_XX where XX represents the last BD address digit, and connect to it.

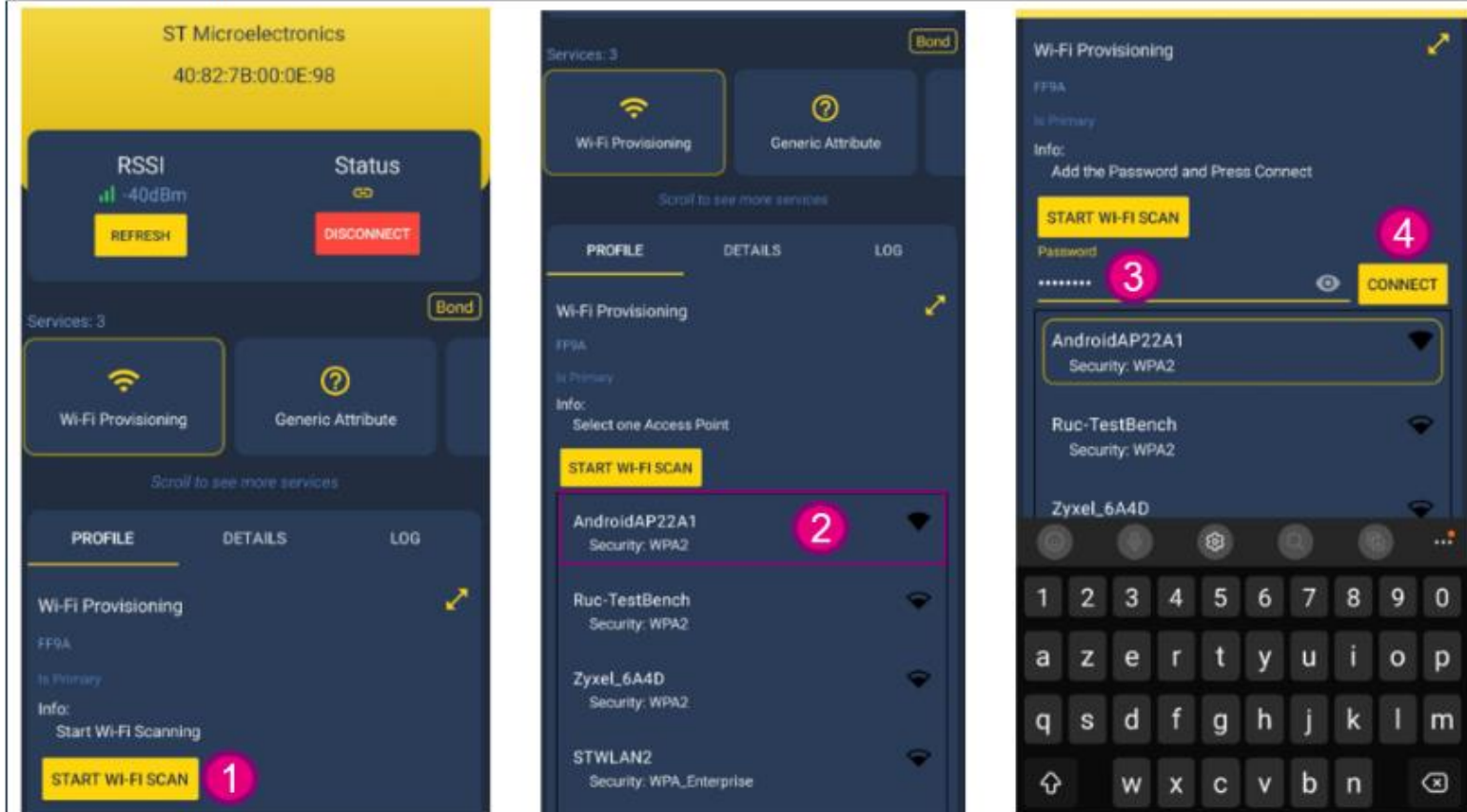


4. Once connected to the Bluetooth® LE device, click on Wi-Fi commissioning button to open the commissioning dedicated interface.



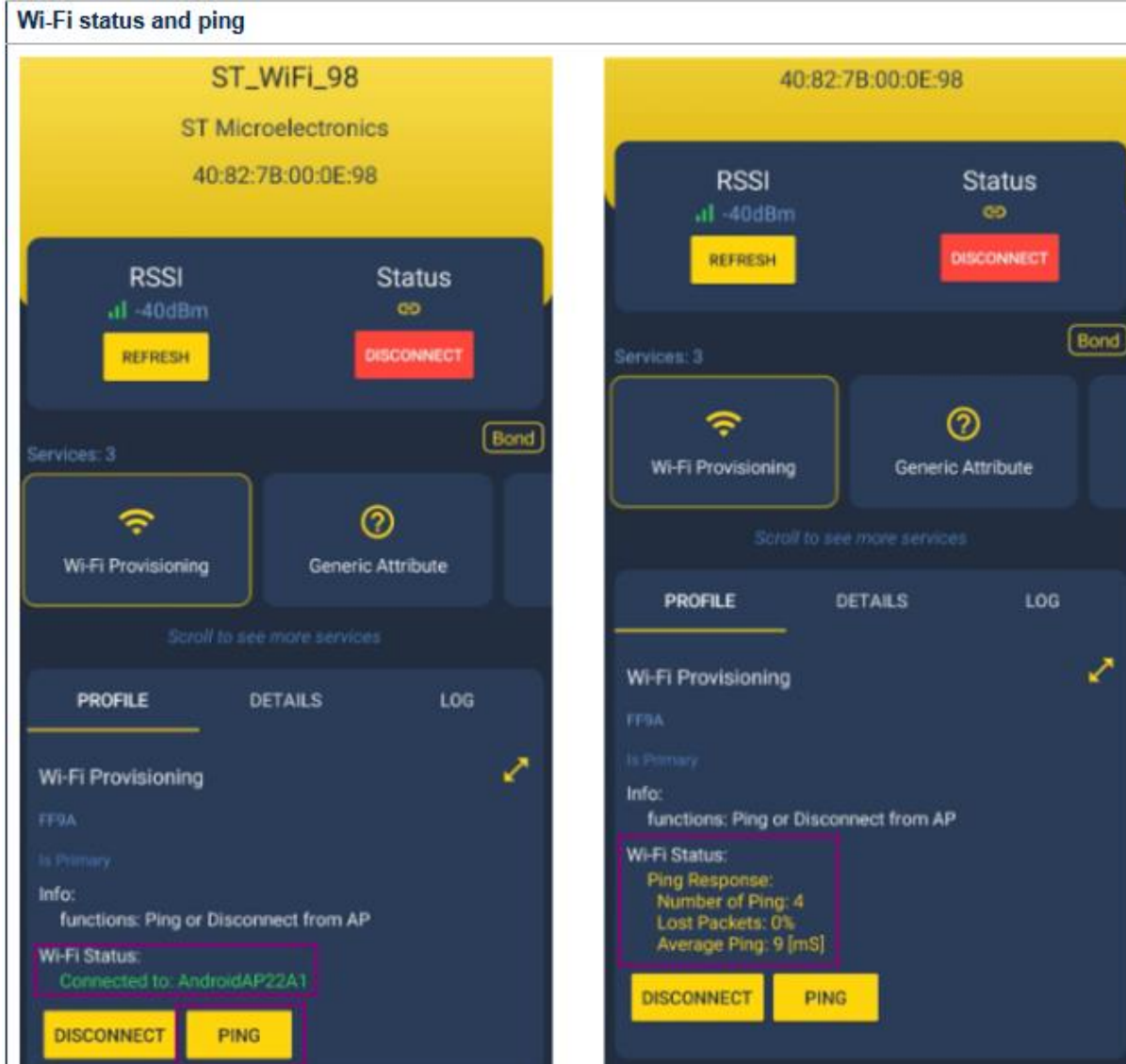
From this interface,  
Launch a Wi-Fi scan (1),  
To detect and list all available networks(2).  
Once the available networks are displayed, select the one you want to connect to (2)  
Type a password if needed (3)  
Click on CONNECT button (4).

Wi-Fi connection interface





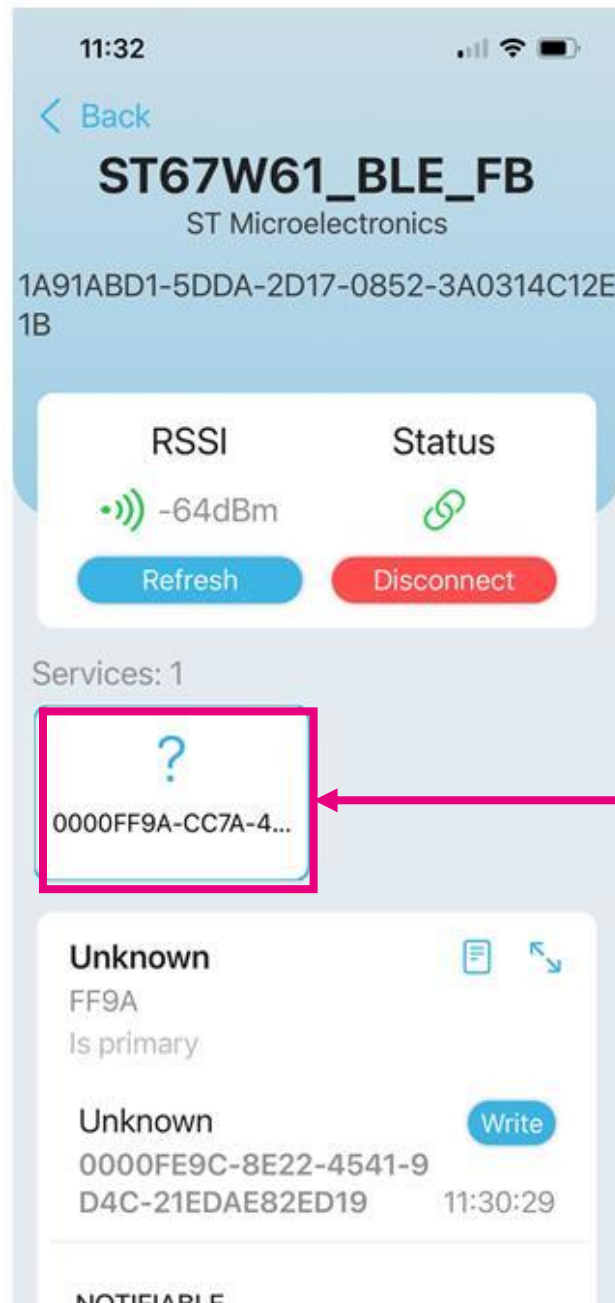
Clicking on the Ping button start a ping test with the Wi-Fi access point and returns results in the interface.

[illegible]

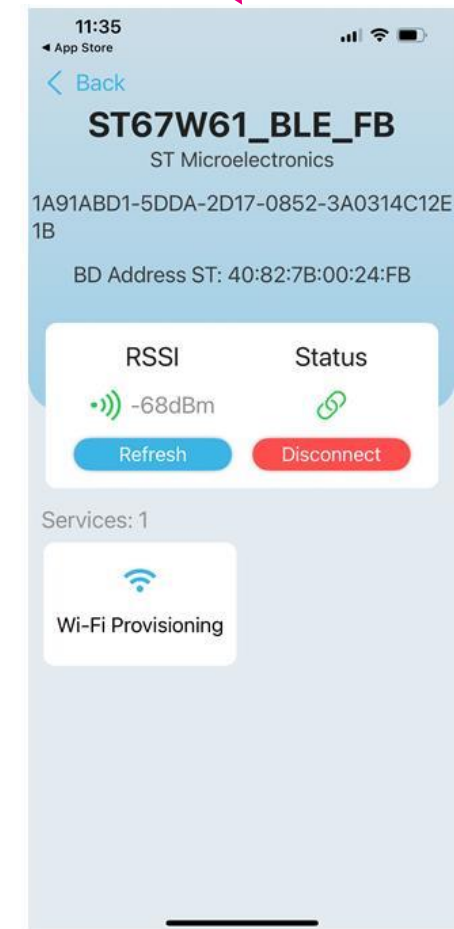


# ST BLE Toolbox

Solution: Uninstall the app and re-install the app back



Cannot see the Wi-Fi Provisioning button





life.augmented

## Porting to a new host mcu using STM32CubeMX

# Porting over new host mcu(1/3)

## 1. Select the Host MCU

Choose the target STM32 part number from the STM32CubeMX

## 2. Check schematics of Host MCU and ST67

Review pin mappings for SPI, GPIO, USART

## 3. Configure USART for Debugging

Used for debugging logs via STLink virtual port

## 4. Configure the SPI Interface for Host MCU and ST67 communication

Set SPI pins, SPI mode, SPI Priority

## 5. Configure DMA Channels for SPI

Fast and reliable SPI data transfer, reducing CPU load during transmission



# Porting over new host mcu(2/3)

## 6. **Configure GPIO Pins** ( CHIP\_EN, SPI\_RDY, SPI\_CS, BOOT, User\_Button)

Set GPIO Mode, GPIO output level, GPIO Maximum Output speed

## 7. **Configure SYS** (Time base Source)

## 8. **Configure Clock**

## 9. **Enable FreeRTOS Middleware**

Setup default task, heap size, stack size and required priorities

## 10. **Enable X-CUBE-ST67W61**

Setup Application, W61 Drivers, Utilities

## 11. **Configure NVIC Interrupt Settings**

SPI, DMA, USART, EXTI



# Porting over new host mcu(3/3)

## 12. Project Creation and Code Generation

Generate project files (STM32CubeIDE/IAR)

Open in IDE for application development

## 13. Application Entry

This initializes ST67W6X Driver, ST67W6X Wi-Fi module, ST67W6X Network module, Application specific functions

## 14. Build and Flash the application







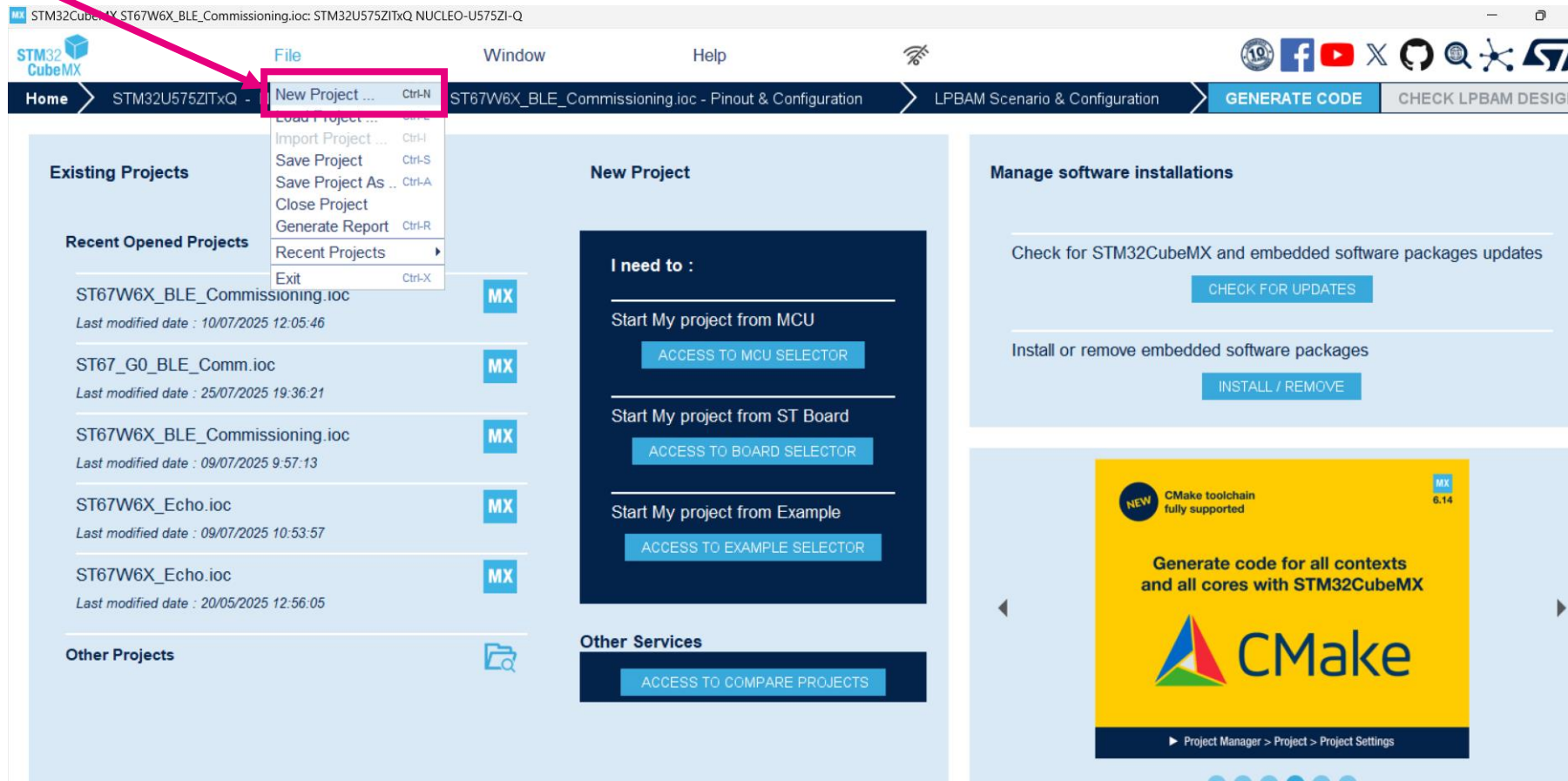
life.augmented

Generating BLE Commissioning  
app using NUCLEO-G0B1RE as  
host processor (Hands-On)



# Case 3: Generating BLE Commissioning app using NUCLEO-G0B1RE as host processor

Open STM32CubeMX and create the new project.



Select the board selector

Type STM32G0

Select STM32G0

Select NUCLEO-G0B1RE

Start Project



ST Restricted

Board Filters

Commercial Part Number

STM32G0

PRODUCT INFO

Type

Supplier

MCU / MPU Series

Check/Uncheck All (1)

STM32C0

STM32F0

STM32F1

STM32F2

STM32F3

STM32F4

STM32F7

☒ STM32G0

STM32G4

STM32H5

STM32H7

STM32L0

STM32L1

STM32L4

STM32L4+

STM32L5

Features Large Picture Docs & Resources Datasheet Buy Start Project

STM32G0 Series

**NUCLEO-G0B1RE**

**STM32 Nucleo-64 development board with STM32G0B1RE MCU, supports Arduino and ST morpho connectivity**

**ACTIVE**  
Product is in mass production

Part Number : NUCLEO-G0B1RE  
Commercial Part Number : NUCLEO-G0B1RE

Unit Price (US\$) : 10.32  
Mounted Device : [STM32G0B1RET6](#)

The STM32 Nucleo-64 board provides an affordable and flexible way for users to try out new concepts and build prototypes by choosing from the various combinations of performance and power consumption features provided by the STM32 microcontroller. For the compatible boards, the internal or external SMPS significantly reduces power consumption in Run mode.

Arduino® Uno V3 connectivity support and the ST morpho headers allow the easy integration of the functionality of the STM32 Nucleo open development platform with a wide choice of specialized shields.

STM32 Nucleo-64 board does not require any separate probe as it integrates the

Board Project Options: NUCLEO-G0B1RE

Initialize all peripherals with their default Mode ?

Yes No

Boards List: 8 items

|   | Overview | Commercial Part No | Type          | Marketing Status | Unit Price (US\$) | Mounted Device                |
|---|----------|--------------------|---------------|------------------|-------------------|-------------------------------|
| ☆ |          | NUCLEO-G070RB      | Nucleo-64     | Active           | 10.32             | <a href="#">STM32G070RBT6</a> |
| ☆ |          | NUCLEO-G071RB      | Nucleo-64     | Active           | 10.32             | <a href="#">STM32G071RBT6</a> |
| ☆ |          | NUCLEO-G0B1RE      | Nucleo-64     | Active           | 10.32             | <a href="#">STM32G0B1RET6</a> |
| ☆ |          | STM32G0316-DISCO   | Discovery Kit | Active           | 9.89              | <a href="#">STM32G0316M6</a>  |

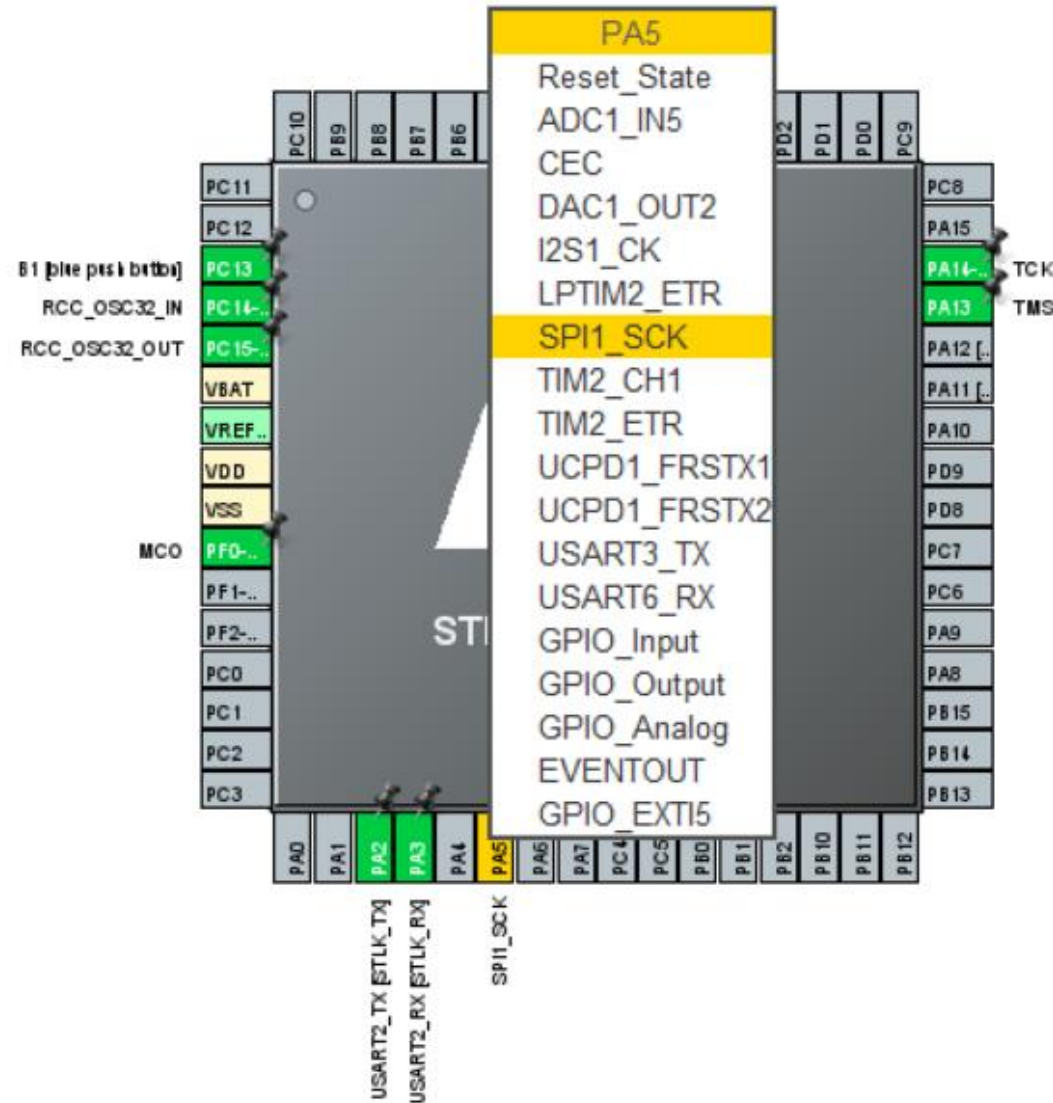
# Pinout

When generating an application from scratch and if X-Nucleo-67W61M1 is used, the pinout must be set in accordance with the ST67W61M1 Arduino interfaces.

| Pin function | Arduino Connector | STM32G0 Pin | GPIO mode           | GPIO pull mode | GPIO speed |
|--------------|-------------------|-------------|---------------------|----------------|------------|
| SPI_CLK      | CN5.D13           | PA5         | AF PP               | No Pull        | Very high  |
| SPI_MISO     | CN5.D12           | PA6         | AF PP               | No Pull        | Very high  |
| SPI_MOSI     | CN5.D11           | PA7         | AF PP               | No Pull        | Very high  |
| SPI_CS       | CN5.D10           | PB0         | Output PP           | No Pull        | high       |
| USER_BUTTON  | -                 | PC13        | EXTI Falling        | Pull-up        | N/A        |
| CHIP_EN      | CN9.D5            | PB4         | Output PP           | No Pull        | High       |
| BOOT         | CN9.D6            | PB14        | Output PP           | No Pull        | Low        |
| SPI_RDY      | CN9.D3            | PB3         | EXTI Falling/Rising | No Pull        | N/A        |

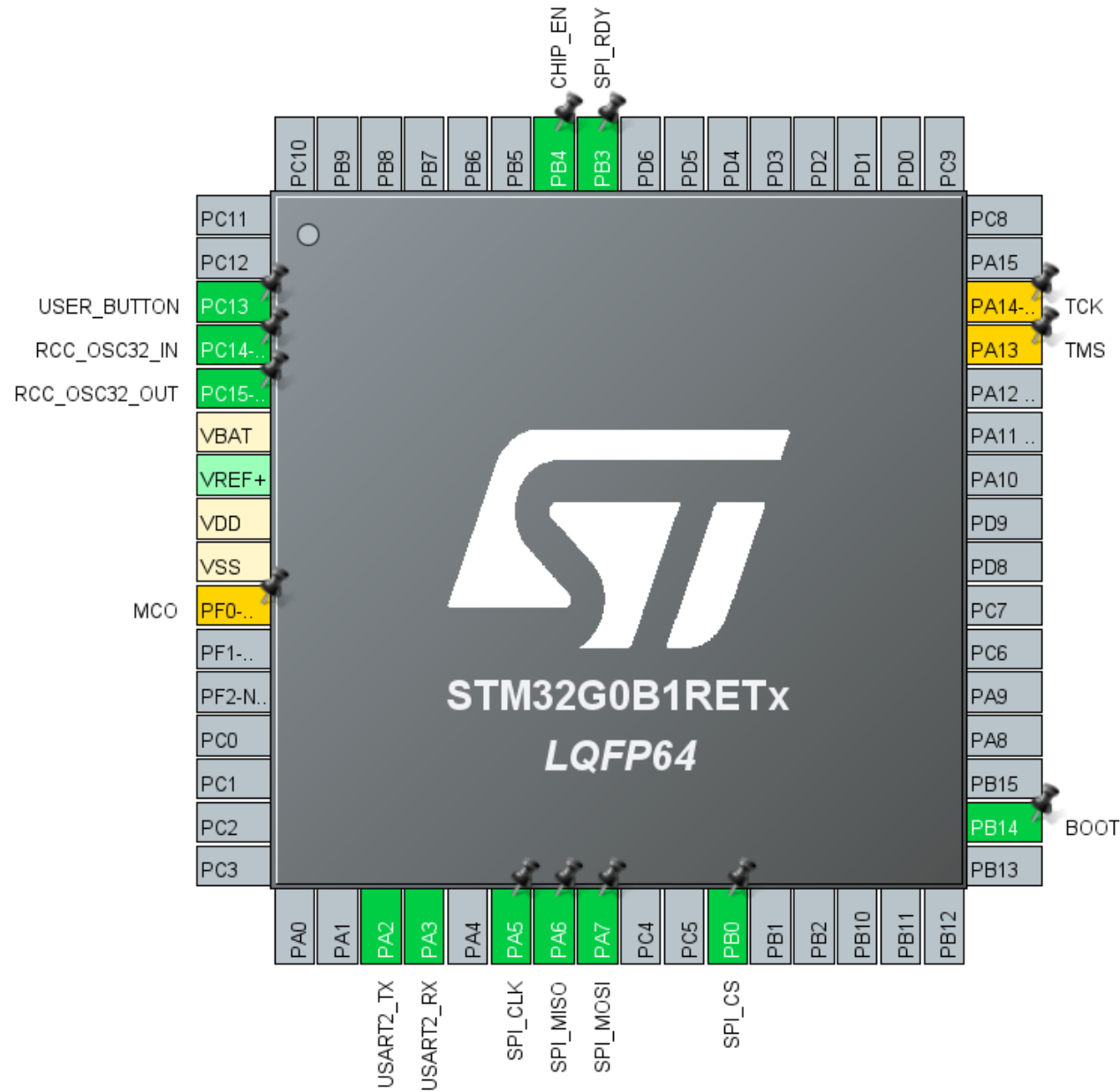
# Assigning functionality

For assigning, click on the pin and select functionality.  
(e.g) PA5->SPI1\_SCK





# Pinout View



After, assigning all the pins mentioned from the pin out table, this would be overall pinout view



# UART GPIO(USART 2)

PA2: USART2\_TX  
PA3: USART2\_RX

**Parameter Settings**

Mode: Asynchronous

Hardware Flow Control (RS232): Disable

Hardware Flow Control (RS485): ☐

Slave Select(NSS) Management: Disable

Configuration

Reset Configuration

Parameter Settings | User Constants | NVIC Settings | DMA Settings | GPIO Settings

Configure the below parameters :

Search (Ctrl+F)

**Basic Parameters**

Baud Rate: 115200 Bits/s

Word Length: 8 Bits (including Parity)

Parity: None

Stop Bits: 1

**Advanced Parameters**

Data Direction: Receive and Transmit

Over Sampling: 16 Samples

Single Sample: Disable

ClockPrescaler: 1

Fifo Mode: Disable

Txfifo Threshold: 1 eighth full configuration

Rxfifo Threshold: 1 eighth full configuration

**Advanced Features**

Auto Baudrate: Disable

TX Pin Active Level Inversion: Disable

RX Pin Active Level Inversion: Disable

Data Inversion: Disable

TX and RX Pins Swapping: Disable

Overrun: Enable

DMA on RX Error: Enable

MSB First: Disable

**GPIO Settings**

Configuration

Reset Configuration

Parameter Settings | User Constants | NVIC Settings | DMA Settings | GPIO Settings

Search Signals

Search (Ctrl+F)

☐ Show only Modified Pins

| Pin Name | Signal on Pin | GPIO output level | GPIO mode            | GPIO Pull-up/Pull-down      | Maximum out... | Fast Mode | User Label | Modified                 |
|----------|---------------|-------------------|----------------------|-----------------------------|----------------|-----------|------------|--------------------------|
| PA2      | USART2_TX     | Low               | Alternate Functio... | No pull-up and no pull-down | Low            | n/a       |            | <input type="checkbox"/> |
| PA3      | USART2_RX     | Low               | Alternate Functio... | No pull-up and no pull-down | Low            | n/a       |            | <input type="checkbox"/> |

# SPI GPIO(SPI1)

PA5: SPI\_SCK  
PA6: SPI\_MISO  
PA7: SPI\_MOSI

Home

STM32G0B1RETx - NUCLEO-G0B1RE

ST67\_G0\_BLE\_Comm.ioc - Pinout & Configuration

Pinout & Configuration

Clock Configuration

Project Manager

Software Packs

Pinout

SPI1 Mode and Configuration

Mode

Full-Duplex Master

Hardware NSS Signal

Disable

Configuration

Reset Configuration

Parameter Settings

User Constants

NVIC Settings

DMA Settings

GPIO Settings

Search Signals

Search (Ctrl+F)

Show only Modified Pins

| Pin Name | Signal on Pin | GPIO output level | GPIO mode          | GPIO Pull-up/Pull-down      | Maximum output current | Fast Mode | User Label | Modified |
|----------|---------------|-------------------|--------------------|-----------------------------|------------------------|-----------|------------|----------|
| PA5      | SPI1_SCK      | Low               | Alternate Function | No pull-up and no pull-down | Very High              | n/a       | SPI_SCK    | ✓        |
| PA6      | SPI1_MISO     | Low               | Alternate Function | No pull-up and no pull-down | Very High              | n/a       | SPI_MISO   | ✓        |
| PA7      | SPI1_MOSI     | Low               | Alternate Function | No pull-up and no pull-down | Very High              | n/a       | SPI_MOSI   | ✓        |

Select SPI1

Set the Maximum output to Very High

Set the User Labels



# SPI GPIO(SPI1)

Parameter  
Settings

SPI1 Mode and Configuration

Mode

Mode Full-Duplex Master

Hardware NSS Signal Disable

Configuration

Reset Configuration

Parameter Settings User Constants NVIC Settings DMA Settings GPIO Settings

Configure the below parameters :

Search (Ctrl+F)

Basic Parameters

Frame Format

Data Size 8 Bits

First Bit MSB First

Clock Parameters

Prescaler (for Baud Rate) 2

Baud Rate 8.0 MBits/s

Clock Polarity (CPOL) Low

Clock Phase (CPHA) 1 Edge

Advanced Parameters

CRC Calculation Disabled

NSSP Mode Enabled

NSS Signal Type Software

Set the Data size to 8

Set the Prescaler to 2

# DMA Channels

Select DMA

The screenshot shows the STM32CubeMX Pinout & Configuration tab. On the left, the 'System Core' section has 'DMA' selected and highlighted with a red box. A red arrow points from the 'Select DMA' text to this box. The main area shows the 'DMA Mode and Configuration' section. Under 'Mode', 'DMA1\_DMA2' and 'MemToMem' are selected. Below this is a table with the following data:

| DMA Request | Channel        | Direction            | Priority  |
|-------------|----------------|----------------------|-----------|
| SPI1_TX     | DMA1 Channel 1 | Memory To Peripheral | High      |
| SPI1_RX     | DMA1 Channel 2 | Peripheral To Memory | Low       |
|             |                |                      | Medium    |
|             |                |                      | High      |
|             |                |                      | Very High |

The 'High' priority for the second row is highlighted with a red box. Below the table are 'Add' and 'Delete' buttons. The 'DMA Request Settings' section shows 'Mode' set to 'Normal', 'Increment Address' unchecked, and 'Data Width' set to 'Byte'. The 'DMA Request Synchronization Settings' section has 'Enable synchronization' unchecked, 'Synchronization signal' and 'Signal polarity' dropdowns, 'Enable event' unchecked, and a 'Request number' input field.

**SPI1\_TX -> DMA1 CHANNEL 1->High**  
**SPI1\_RX -> DMA1 CHANNEL 2-> High**

# GPIO

Select GPIO

Categories A->Z

System Core

- ✓ DMA
- ✓ **GPIO**
- ✓ IWDG
- ✓ NVIC
- ⚠ RCC
- ⚠ SYS
- WWDG

Analog >

Timers >

Connectivity >

Multimedia >

### GPIO Mode and Configuration

Mode

Configuration

Group By Peripherals

✓ GPIO   ✓ Single Mapped Signals   ✓ RCC   ✓ SPI   ✓ USART   ✓ NVIC

Search Signals  
Search (Ctrl+F)

☐ Show only Modified Pins

| Pin Name | Signal on Pin | GPIO output level | GPIO mode                                                          | GPIO Pul...  | Maximu... | Fast Mode | User Label | Modified |
|----------|---------------|-------------------|--------------------------------------------------------------------|--------------|-----------|-----------|------------|----------|
| PB0      | n/a           | Low               | Output Push Pull                                                   | No pull-u... | High      | n/a       | SPI_CS     | ✓        |
| PB3      | n/a           | n/a               | External Interrupt Mode with Rising/Falling edge trigger detection | No pull-u... | n/a       | n/a       | SPI_RDY    | ✓        |
| PB4      | n/a           | Low               | Output Push Pull                                                   | No pull-u... | High      | n/a       | CHIP_EN    | ✓        |
| PB14     | n/a           | Low               | Output Push Pull                                                   | No pull-u... | Low       | n/a       | BOOT       | ✓        |
| PC13     | n/a           | n/a               | External Interrupt Mode with Falling edge trigger detection        | No pull-u... | n/a       | n/a       | USER_BU... | ✓        |

GPIO pin names

GPIO output level

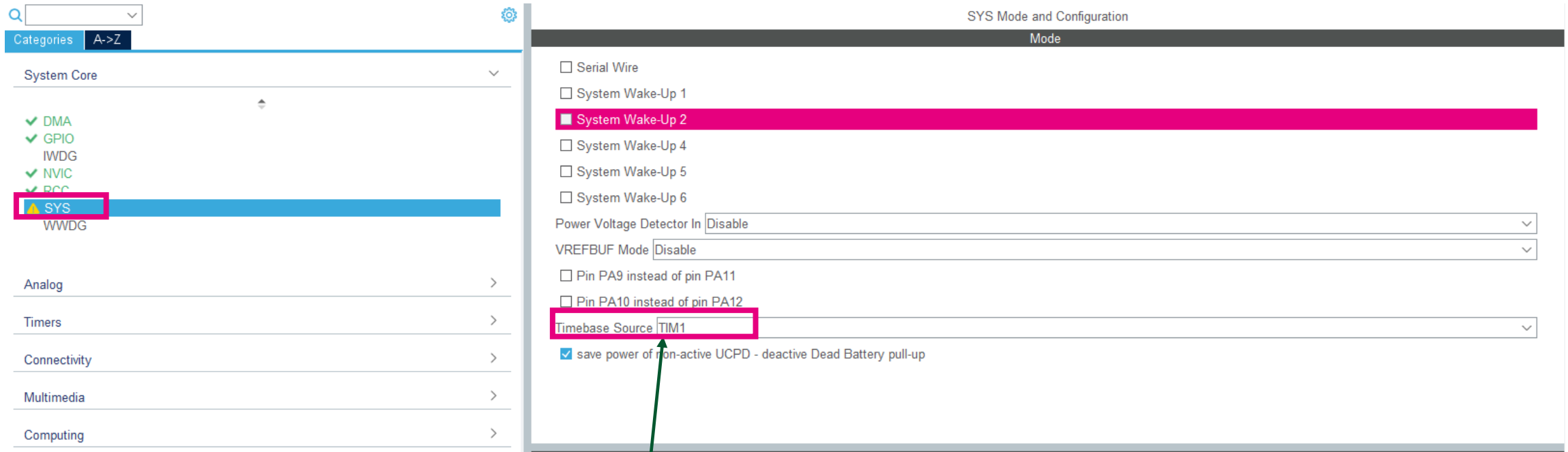
GPIO mode

GPIO maximum speed

GPIO user label



# Timer



Sys mode configuration: TIM1 in place of SysTick: to avoid RTOS conflicts, enable low power operation.



# RCC

Home > STM32G0B1RETx - NUCLEO-G0B1RE > Untitled - Pinout & Configuration

Pinout & Configuration | Clock Configuration | Project Manager

Software Packs | Pinout

RCC Mode and Configuration

Mode

High Speed Clock (HSE) **Disable**

☐ OSC enable

Low Speed Clock (LSE) Crystal/Ceramic Resonator

☐ OSC32 enable

☐ Master Clock Output

☐ Master Clock Output 2

☒ LSCO Clock Output

☐ Audio Clock Input (I2S\_CKIN)

CRS SYNC **Disable**

HSE disabled

Configuration

Reset Configuration

Parameter Settings | User Constants | NVIC Settings | GPIO Settings

Configure the below parameters :

Search (Ctrl+F)

System Parameters

VDD voltage (V) 3.3 V

Instruction Cache Enabled

Prefetch Buffer Enabled

Data Cache Enabled

Flash Latency(WS) 2 WS (3 CPU cycle)

RCC Parameters

HSI Calibration Value 64

HSE Startup Timeout Value (ms) 100

LSE Startup Timeout Value (ms) 5000

Power Parameters

Power Regulator Voltage Scale Power Regulator Voltage Scale 1

Peripherals Clock Configuration

Generate the peripherals clock configuration TRUE

|                     |                 |
|---------------------|-----------------|
| Clock Configuration | Project Manager |
|---------------------|-----------------|

- 
- The diagram illustrates the clock architecture of the STM32H743, showing the following components and their connections:
- Input Frequencies:** 32.768 KHz (LSE), 8 MHz (HSI), 48 MHz (HSI48), and 12.288 MHz (MCO2).
  - Dividers and Multipliers:** Various dividers (e.g., /32, /1, /2, /4, /8, /16, /64) and multipliers (e.g., X8, X16, X48) are used to generate different clock rates.
  - System Clock Mux:** A central multiplexer that selects the system clock source. The diagram shows PLLCLK selected, which is then divided by 1 to produce a 64 MHz SYSCLK.
  - Other Clock Muxes:** Various other multiplexers are shown for different peripherals, such as APB, AHB, LPTIM, I2S, and USB.
  - Red Callout:** A red box with the text "Set System Clock MUX to PLLCLK (64MHz)" points to the System Clock Mux, indicating the required configuration for the system clock.

# Middleware & Software Packs

## FreeRTOS

- FreeRTOS component can be either be selected from Middleware (FREERTOS) or from the Software Pack (X-CUBE-FREERTOS) depending on the chosen STM32.
- By default, the stack size of the default task defined by STM32CubeMx tool is equal to 128 words. The application default task required **512 words**.

The screenshot displays the STM32CubeMX software interface for configuring FreeRTOS. On the left, the 'Middleware and Software Packs' list shows 'FREERTOS' selected. The main window, titled 'FREERTOS Mode and Configuration', has the 'Interface' set to 'CMSIS\_V2'. The 'Configuration' section shows various options checked, including 'Tasks and Queues'. A table lists the tasks, with 'defaultTask' having a stack size of 512 words. An 'Edit Task' dialog box is open, showing the configuration for 'defaultTask' with a stack size of 512 words.

| Task Name   | Priority         | Stack Size (Words) | Entry Function   | Code Generation | Parameter | Allocation | Buffer Name | Control Block Na... |
|-------------|------------------|--------------------|------------------|-----------------|-----------|------------|-------------|---------------------|
| defaultTask | osPriorityNormal | 512                | StartDefaultTask | Default         | NULL      | Dynamic    | NULL        | NULL                |

| Queue Name | Queue Size | Buffer Name | Control Block Name |
|------------|------------|-------------|--------------------|
|------------|------------|-------------|--------------------|

# Middleware & Software Packs

## FreeRTOS

- Adjust heap :The size heap value should be set at least to 40Kbytes. To optimize the memory, this value could be adjusted during integration and validation tests of the application

The screenshot displays the STM32CubeIDE interface for configuring FreeRTOS. On the left, the 'Middleware and Software Packs' list includes various packs, with 'FREERTOS' highlighted. On the right, the 'FREERTOS Mode and Configuration' dialog is open, showing the 'Config parameters' tab. The 'TOTAL\_HEAP\_SIZE' is set to '40000 Bytes'.

| Parameter                         | Value            |
|-----------------------------------|------------------|
| FreeRTOS version                  | 10.3.1           |
| CMSIS-RTOS version                | 2.00             |
| MPU/FPU                           |                  |
| ENABLE_MPU                        | Disabled         |
| ENABLE_FPU                        | Disabled         |
| Kernel settings                   |                  |
| USE_PREEMPTION                    | Enabled          |
| CPU_CLOCK_HZ                      | SystemCoreClock  |
| TICK_RATE_HZ                      | 1000             |
| MAX_PRIORITIES                    | 56               |
| MINIMAL_STACK_SIZE                | 128 Words        |
| MAX_TASK_NAME_LEN                 | 16               |
| USE_16_BIT_TICKS                  | Disabled         |
| IDLE_SHOULD_YIELD                 | Enabled          |
| USE_MUTEXES                       | Enabled          |
| USE_RECURSIVE_MUTEXES             | Enabled          |
| USE_COUNTING_SEMAPHORES           | Enabled          |
| QUEUE_REGISTRY_SIZE               | 8                |
| USE_APPLICATION_TASK_TAG          | Disabled         |
| ENABLE_BACKWARD_COMPATIBILITY     | Enabled          |
| USE_PORT_OPTIMISED_TASK_SELECTION | Disabled         |
| USE_TICKLESS_IDLE                 | Disabled         |
| USE_TASK_NOTIFICATIONS            | Enabled          |
| RECORD_STACK_HIGH_ADDRESS         | Disabled         |
| Memory management settings        |                  |
| Memory Allocation                 | Dynamic / Static |
| TOTAL_HEAP_SIZE                   | 40000 Bytes      |
| Memory management scheme          | heap_4           |
| Hook function related definitions |                  |
| USE_IDLE_HOOK                     | Disabled         |
| USE_TICK_HOOK                     | Disabled         |

# Middleware & Software Packs

## FreeRTOS

Pinout & Configuration

Clock Configuration

Project Manager

Software Packs

Pinout

FREERTOS Mode and Configuration

Mode

Interface: CMSIS\_V2

Configuration

Reset Configuration

Config parameters

Include parameters

Advanced settings

User Constants

Tasks and Queues

Timers and Semaphores

Mutexes

Events

FreeRTOS Heap Usage

Configure the below parameters :

Search (Ctrl+F)

Newlib settings (see parameter description first)

USE\_NEWLIB\_REENTRANT

Enabled

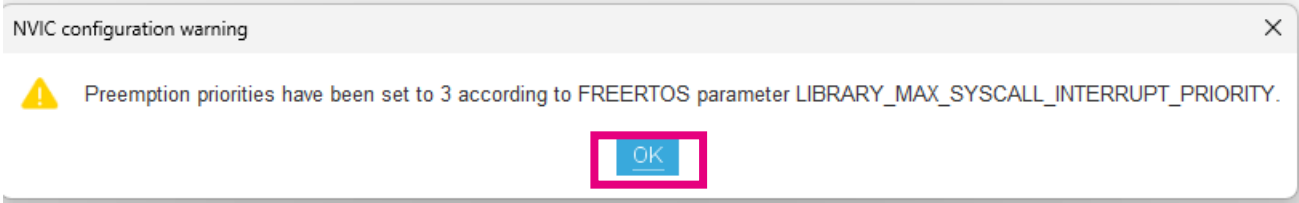
Project settings (see parameter description first)

Use FW pack heap file

Enabled



Enable  
USE\_NEWLIB\_REENTRANT



Categories A-Z

System Core

- ✓ DMA
- ✓ GPIO
- IWDG
- ✓ **NVIC**
- ✓ RCC
- ⚠ SYS
- WWDG

Analog >

Timers >

Connectivity >

Multimedia >

Computing >

Middleware and Software Packs >

Utilities >

NVIC Mode and Configuration

Mode

Configuration

✓ NVIC

Code generation

☐ Sort by Preemption Priority and Sub Priority

☐ Sort by interrupts names

Search

Show  available interrupts

☒ Force DMA channels Interrupts

| NVIC Interrupt Table                                  | Enabled                  | Preemption Priority | Uses FreeRTOS functions  |
|-------------------------------------------------------|--------------------------|---------------------|--------------------------|
| Non maskable interrupt                                | ✓                        | 0                   | <input type="checkbox"/> |
| Hard fault interrupt                                  | ✓                        | 0                   | <input type="checkbox"/> |
| System service call via SWI instruction               | ✓                        | 0                   | <input type="checkbox"/> |
| Pendable request for system service                   | ✓                        | 3                   | ✓                        |
| System tick timer                                     | ✓                        | 3                   | ✓                        |
| PVD through EXTI line 16, PVM (monit. VDDIO2) thro... | <input type="checkbox"/> | 3                   | ✓                        |
| Flash global interrupt                                | <input type="checkbox"/> | 3                   | ✓                        |
| RCC global interrupt                                  | <input type="checkbox"/> | 3                   | ✓                        |
| EXTI line 2 and line 3 interrupts                     | ✓                        | 3                   | ✓                        |
| EXTI line 4 to 15 interrupts                          | ✓                        | 3                   | ✓                        |
| DMA1 channel 1 interrupt                              | ✓                        | 3                   | ✓                        |
| DMA1 channel 2 and channel 3 interrupts               | ✓                        | 3                   | ✓                        |
| Time base: TIM1 break, update, trigger and commuta... | ✓                        | 3                   | <input type="checkbox"/> |
| SPI1/I2S1 Interrupt                                   | ✓                        | 3                   | ✓                        |
| USART2 + LPUART2 Interrupt                            | ✓                        | 3                   | ✓                        |



# Middleware & Software Packs

## X-CUBE-ST67W61 Software package

| Pack / Bundle / Component           | Status | Version | Selection                           |
|-------------------------------------|--------|---------|-------------------------------------|
| ▼ STMicroelectronics.X-CUBE-ST67W61 | ✓      | 1.1.0   | ▼                                   |
| ▼ Device Applications               | ✓      | 1.1.0   |                                     |
| Application                         | ✓      | 1.1.0   | BLE_Commissioning ▼                 |
| FreeRTOS_Tickless                   |        | 1.1.0   | <input type="checkbox"/>            |
| ▼ Network ST67W6X_Network_Driver    | ✓      | 1.1.0   |                                     |
| ST67 Architecture                   | ✓      | 1.1.0   | T01 ▼                               |
| ATDriver                            | ✓      | 1.1.0   | <input checked="" type="checkbox"/> |
| ServiceAPI                          | ✓      | 1.1.0   | <input checked="" type="checkbox"/> |
| ServiceShell                        |        | 1.1.0   | <input type="checkbox"/>            |
| ▼ Utilities                         | ✓      |         |                                     |
| Logging                             | ✓      | 1.1.0   | <input checked="" type="checkbox"/> |
| Shell                               |        | 1.1.0   | <input type="checkbox"/>            |
| Statistics                          |        | 1.1.0   | <input type="checkbox"/>            |
| ▼ Debug TraceRecorder               |        | 4.10.2  |                                     |
| TraceRecorder                       |        | 4.10.2  | <input type="checkbox"/>            |
| ▼ Data Exchange cJSON               |        | 1.7.18  |                                     |
| cJSON                               |        | 1.7.18  | <input type="checkbox"/>            |
| ▼ File System LittleFS              |        | 2.10.1  |                                     |
| LittleFS                            |        | 2.10.1  | <input type="checkbox"/>            |
| ▼ Utility Utilities                 |        | 1.4.2   |                                     |
| LPM / Tiny LPM                      |        | 1.4.2   | <input type="checkbox"/>            |
| ▼ Network LwIP                      |        | 2.2.0   |                                     |
| LwIP                                |        | 2.2.0   | <input type="checkbox"/>            |
| ▼ Data Exchange MQTT-C              |        | 1.1.6   |                                     |
| MQTT-C                              |        | 1.1.6   | <input type="checkbox"/>            |

Select BLE\_Commissioning

Select T01 ST67  
Architecture

Enable AT Driver

Enable  
ServiceAPI

Enable Logging

# Middleware & Software Packs

## X-CUBE-ST67W61 Software package

The screenshot displays the STM32CubeIDE interface with the 'Pinout & Configuration' tab selected. The left sidebar shows a list of software packs, with 'X-CUBE-ST67W61' highlighted. The main area shows the configuration for 'STMicroelectronics.X-CUBE-ST67W61.1.0.0 Mode and Configuration'. The 'Mode' section has 'Device Applications' and 'Network ST67W6X Network Driver' checked. The 'Configuration' section has a 'Reset Configuration' button and a tabbed interface with 'W6X Modules', 'Parameter Settings', 'User Constants', and 'Platform Settings' (highlighted by a pink arrow and a blue box labeled 'Platform settings'). The 'Platform Settings' tab shows a 'Platform proposal' table with two rows: 'LogSh\_UART' and 'NCP\_SPI'. The 'LogSh\_UART' row has 'USART:Asynchronous' in the 'IPs or Components' column, 'USART2' in the 'Found Solutions' column, and 'Unknown' in the 'BSP API' column. The 'NCP\_SPI' row has 'SPI:Full-Duplex Master' in the 'IPs or Components' column, 'SPI1' in the 'Found Solutions' column, and 'Unknown' in the 'BSP API' column. The 'Found Solutions' column for both rows is highlighted with a pink box.

Pinout & Configuration

Clock Configuration

Project Manager

Software Packs

Pinout

STMicroelectronics.X-CUBE-ST67W61.1.0.0 Mode and Configuration

Mode

☒ Device Applications

☒ Network ST67W6X Network Driver

Configuration

Reset Configuration

W6X Modules

Parameter Settings

User Constants

Platform Settings

Platform settings

Platform proposal

BSP

| Name       | IPs or Components      | Found Solutions | BSP API |
|------------|------------------------|-----------------|---------|
| LogSh_UART | USART:Asynchronous     | USART2          | Unknown |
| NCP_SPI    | SPI:Full-Duplex Master | SPI1            | Unknown |



# Middleware & Software Packs

## X-CUBE-ST67W61 Software package

STMicrollectronics.X-CUBE-ST67W61.1.1.0 Mode and Configuration

Mode

- ☒ Device Applications
- ☒ Network ST67W6X Network Driver

Configuration

Reset Configuration

Parameter Settings | User Constants | Platform Settings

Configure the below parameters :

Search (Ctrl+F)

W6X Init Modules

|              |              |
|--------------|--------------|
| ST67_ARCH    | W6X_ARCH_T01 |
| WIFI_STATION | Yes          |
| WIFI_SAP     | No           |
| BLE          | Yes          |
| NET          | Yes          |
| MQTT         | No           |

Basic Parameters

|                               |                 |
|-------------------------------|-----------------|
| LOG_OUTPUT_MODE               | LOG_OUTPUT_UART |
| FREERTOS_TASK_NOTIF_ARRAY_LEN | 8               |

Logging-Shell

|           |           |
|-----------|-----------|
| LOG_LEVEL | LOG_DEBUG |
|-----------|-----------|

W6X Parameters

|                                |                        |
|--------------------------------|------------------------|
| W6X_POWER_SAVE_AUTO            | No                     |
| W6X_CLOCK_MODE                 | Internal RC oscillator |
| W6X_WIFI_AUTOCONNECT           | No                     |
| W6X_WIFI_COUNTRY_CODE          | 00                     |
| W6X_WIFI_ADAPTIVE_COUNTRY_CODE | No                     |
| W6X_BLE_HOSTNAME               | ST67W61_BLE            |
| W6X_NET_HOSTNAME               | ST67W61_WIFI           |
| W6X_NET_DHCP_MODE              | DHCP STA+SAP           |
| W6X_NET_RECV_TIMEOUT (ms)      | 10000                  |
| W6X_NET_SEND_TIMEOUT (ms)      | 10000                  |
| W6X_NET_RECV_BUFFER_SIZE       | 9216                   |
| W6X_NET_DNS_MANUAL             | No                     |
| W6X_NET_DNS_IP_1               | {208, 67, 222, 222}    |
| W6X_NET_DNS_IP_2               | {8, 8, 8, 8}           |
| W6X_NET_DNS_IP_3               | {0, 0, 0, 0}           |

Disable W6X\_POWER\_SAVE\_AUTO

Rename W6X\_BLE\_HOSTNAME to something unique



# Project Creation

Home > STM32G0B1RETx - NUCLEO-G0B1RE > ST67\_G0\_BLE\_Comm.ioc - Project Manager

Pinout & Configuration | Clock Configuration | **Project Manager**

**Project**

Code Generator

Advanced Settings

Project Settings

Project Name: ST67\_G0\_BLE\_Comm

Project Location: C:\ST67\_Test Browse

Application Structure: Advanced ☐ Do not generate the main()

Toolchain Folder Location: C:\ST67\_Test\ST67\_G0\_BLE\_Comm\

Toolchain / IDE: STM32CubeIDE ☒ **Generate Under Root**

Linker Settings

Minimum Heap Size: 0x200

Minimum Stack Size: 0x400

Thread-safe Settings

Cortex-M0+NS

☐ Enable multi-threaded support

Thread-safe Locking Strategy: Default - Mapping suitable strategy depending on RTOS selection.

Mcu and Firmware Package

Mcu Reference: STM32G0B1RETx

Firmware Package Name and Version: STM32Cube\_FW\_G0\_V1.6.2 ☒ Use latest available version

☒ **Use Default Firmware Location**

Firmware Relative Path: C:/Users/kulkams/STM32Cube/Repository/STM32Cube\_FW\_G0\_V1.6.2 Browse

1. Enable “Generate Under Root”

Here, we are enabling the Flat directory structure: (i.e. Driver and MW directories copied into your project directory)

2. Enable “ Use Default Firmware Location”

# Code Generator

Home > STM32G0B1RETx - NUCLEO-G0B1RE > ST67\_G0\_BLE\_Comm.ioc - Project Manager

Pinout & Configuration | Clock Configuration | Project Manager

Project

Code Generator

Advanced Settings

STM32Cube MCU packages and embedded software packs

- ☐ Copy all used libraries into the project folder
- ☒ Copy only the necessary library files
- ☐ Add necessary library files as reference in the toolchain project configuration file

Generated files

- ☐ Generate peripheral initialization as a pair of '.c/.h' files per peripheral
- ☐ Backup previously generated files when re-generating
- ☒ Keep User Code when re-generating
- ☒ Delete previously generated files when not re-generated

HAL Settings

- ☐ Set all free pins as analog (to optimize the power consumption)
- ☐ Enable Full Assert

3. Enable "Copy only the necessary files"

# Advanced settings

Home > STM32G0B1RETx - NUCLEO-G0B1RE > Experts\_Training\_G0.ioc - Project Manager > GENE

Pinout & Configuration | Clock Configuration | Project Manager

Project

Code Generator

Advanced Settings

Driver Selector

Search (Ctrl+F)

RCC HAL  
GPIO HAL  
DMA HAL  
> SPI HAL  
> USART HAL  
STMicroelectronics.X-CUBE-ST67W61.1.1.0 HAL

Generated Function Calls

| Generate Code                       | Rank | Function Name       | Peripheral Instance Name  | <input type="checkbox"/> Do Not Generate Function Call | <input type="checkbox"/> Visibility (Static) |
|-------------------------------------|------|---------------------|---------------------------|--------------------------------------------------------|----------------------------------------------|
| <input checked="" type="checkbox"/> | 1    | SystemClock_Config  | RCC                       | <input type="checkbox"/>                               | <input type="checkbox"/>                     |
| <input checked="" type="checkbox"/> | 2    | MX_GPIO_Init        | GPIO                      | <input type="checkbox"/>                               | <input checked="" type="checkbox"/>          |
| <input checked="" type="checkbox"/> | 3    | MX_DMA_Init         | DMA                       | <input type="checkbox"/>                               | <input checked="" type="checkbox"/>          |
| <input checked="" type="checkbox"/> | 4    | MX_SPI1_Init        | SPI1                      | <input type="checkbox"/>                               | <input checked="" type="checkbox"/>          |
| <input checked="" type="checkbox"/> | 5    | MX_USART2_UART_Init | USART2                    | <input type="checkbox"/>                               | <input checked="" type="checkbox"/>          |
| <input checked="" type="checkbox"/> | 6    | MX_ST67W6X_Init     | STMicroelectronics.X-C... | <input type="checkbox"/>                               | <input type="checkbox"/>                     |

4. Uncheck this option for SPI1 and USART2

5. Check Visibility for SPI1 and USART2



# Generate Code

## 5. Generate Code

Home > STM32G0B1RETx - NUCLEO-G0B1RE > Experts\_Training\_G0.ioc - Project Manager > **GENERATE CODE**

**Pinout & Configuration** | **Clock Configuration** | **Project Manager** | **Tools**

**Project**

**Code Generator**

**Advanced Settings**

**Driver Selector**

Search (Ctrl+F)

|                                         |     |
|-----------------------------------------|-----|
| RCC                                     | HAL |
| GPIO                                    | HAL |
| DMA                                     | HAL |
| > SPI                                   | HAL |
| > USART                                 | HAL |
| STMicroelectronics.X-CUBE-ST67W61.1.1.0 | HAL |

**Generated Function Calls**

| Generate Code                       | Rank | Function Name       | Peripheral Instance Name  | <input type="checkbox"/> Do Not Generate Function Call | <input type="checkbox"/> Visibility (Static) |
|-------------------------------------|------|---------------------|---------------------------|--------------------------------------------------------|----------------------------------------------|
| <input checked="" type="checkbox"/> | 1    | SystemClock_Config  | RCC                       | <input type="checkbox"/>                               | <input type="checkbox"/>                     |
| <input checked="" type="checkbox"/> | 2    | MX_GPIO_Init        | GPIO                      | <input type="checkbox"/>                               | <input checked="" type="checkbox"/>          |
| <input checked="" type="checkbox"/> | 3    | MX_DMA_Init         | DMA                       | <input type="checkbox"/>                               | <input checked="" type="checkbox"/>          |
| <input checked="" type="checkbox"/> | 4    | MX_SPI1_Init        | SPI1                      | <input type="checkbox"/>                               | <input checked="" type="checkbox"/>          |
| <input checked="" type="checkbox"/> | 5    | MX_USART2_UART_Init | USART2                    | <input type="checkbox"/>                               | <input checked="" type="checkbox"/>          |
| <input checked="" type="checkbox"/> | 6    | MX_ST67W6X_Init     | STMicroelectronics.X-C... | <input type="checkbox"/>                               | <input type="checkbox"/>                     |

**Register CallBack**

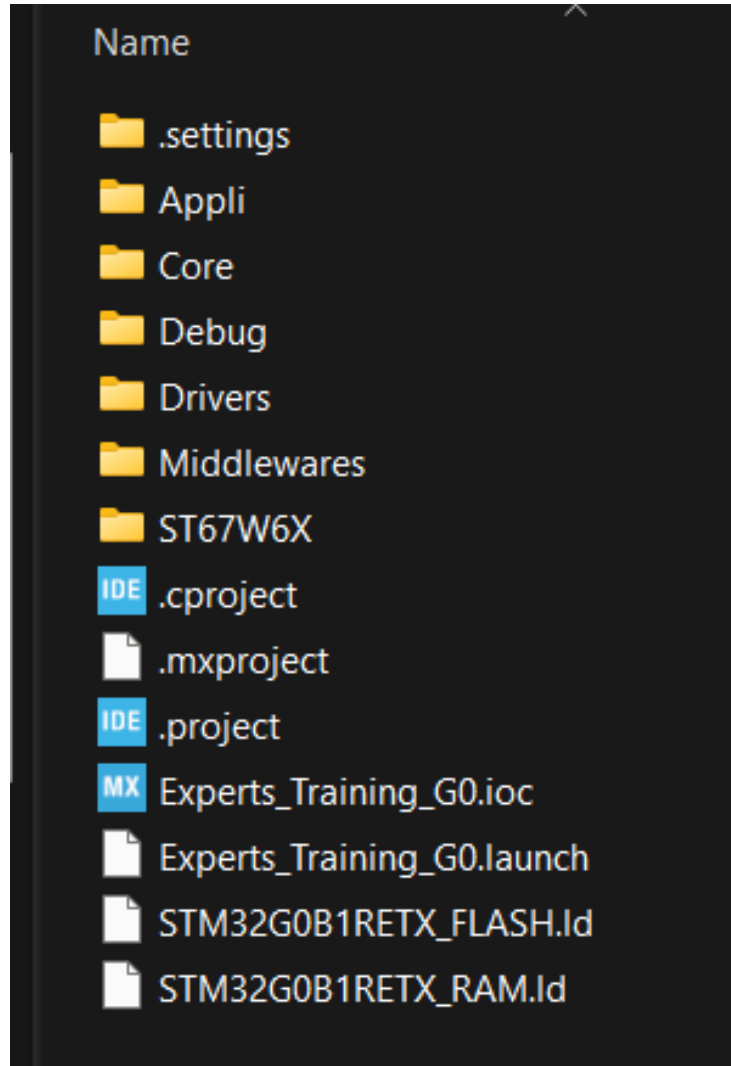
Search (Ctrl+F)

|       |         |
|-------|---------|
| ADC   | DISABLE |
| CEC   | DISABLE |
| COMP  | DISABLE |
| CRYP  | DISABLE |
| DAC   | DISABLE |
| FDCAN | DISABLE |
| HCD   | DISABLE |
| I2C   | DISABLE |
| I2S   | DISABLE |
| IRDA  | DISABLE |
| LPTIM | DISABLE |
| PCD   | DISABLE |
| RNG   | DISABLE |
| RTC   | DISABLE |
| SMBUS | DISABLE |
| SPI   | DISABLE |
| TIM   | DISABLE |
| UART  | DISABLE |
| USART | DISABLE |
| WWDG  | DISABLE |

Note: While generating the project, make sure the MX\_SPI1\_Init and MX\_USART2\_UART\_Init are unchecked (Do Not Generate Function Call) or if they are getting enabled when you hit the Generate Code: try to close CubeMX and try again



# Folder Structure

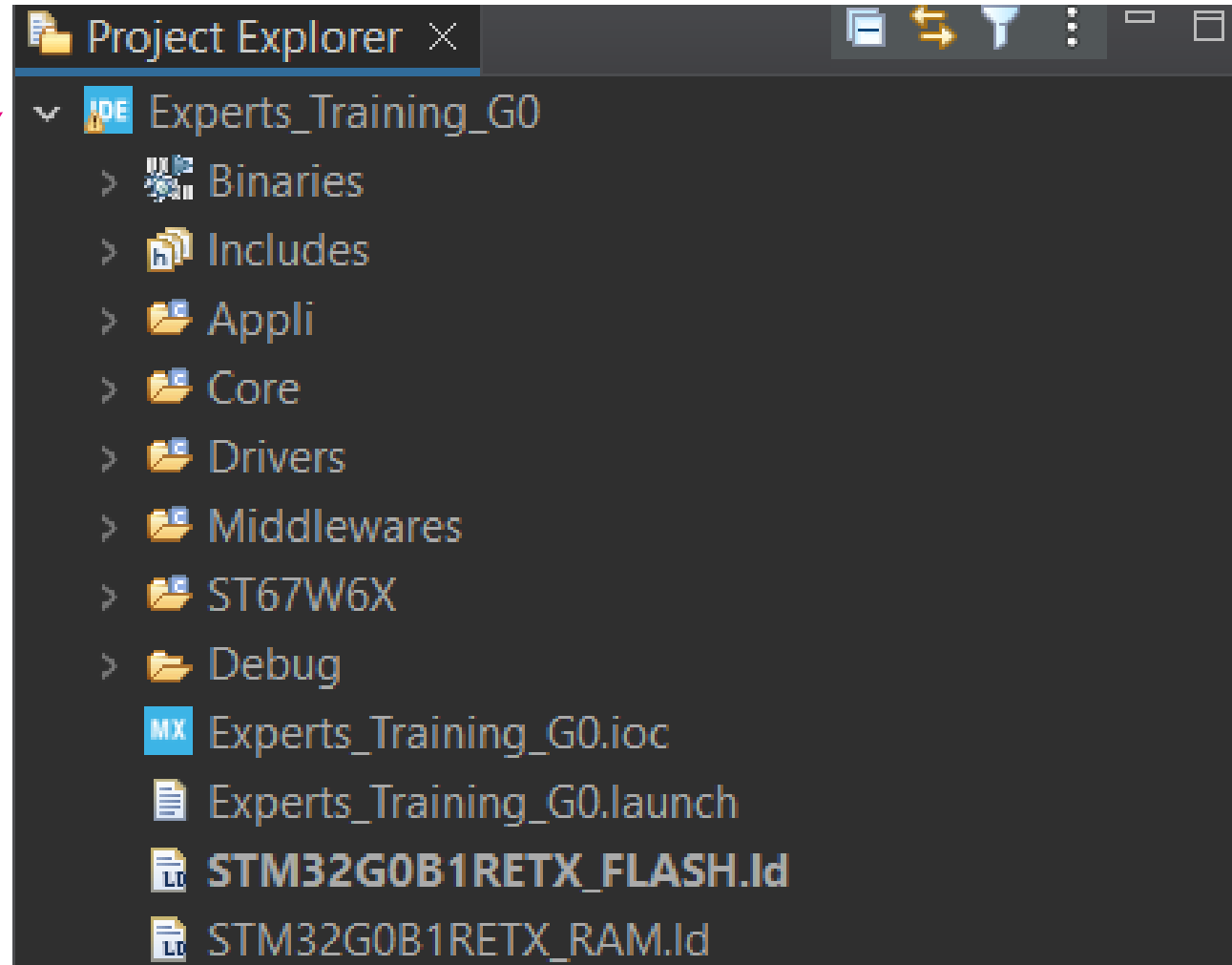


Folder structure view after generating the code from CubeMX

- App code
- Core
- Drivers
- Middlewares
- ST67W6X

# STM32CubeIDE

Load the project on  
the STM32CubeIDE



# xPortIsInsideInterrupt()

- If the used device embeds an Arm® Cortex®M0+ (e.g. STM32G0) and its FreeRTOS version (v10.3.1) which is older than of V10.6, definition below **must be added manually after project has been generated**.
- The below function definition must be added in the **portmacro.h** file (Project/Middlewares/Third\_Party/FreeRTOS/Source/portable/GCC/ARM\_CM0)

```
#define portINLINE __inline
#ifndef portFORCE_INLINE
#define portFORCE_INLINE inline __attribute__((always_inline))
#endif

/*-----*/

portFORCE_INLINE static BaseType_t xPortIsInsideInterrupt( void )
{
    uint32_t ulCurrentInterrupt;
    BaseType_t xReturn;

    /* Obtain the number of the currently executing interrupt. */
    __asm volatile ( "mrs %0, ipsr" : "=r" ( ulCurrentInterrupt )::"memory" );

    if( ulCurrentInterrupt == 0 )
    {
        xReturn = pdFALSE;
    }
    else
    {
        xReturn = pdTRUE;
    }

    return xReturn;
}
```



# STM32CubeIDE

Call `xPortIsInsideInterrupt()` function under `portmacro.h` file

IMP: As we are using G0 (CortexM0) -> older version of FreeRTOS, so we need to add this function definition manually.

```
#define portINLINE __inline
#ifndef portFORCE_INLINE
#define portFORCE_INLINE inline
__attribute__((always_inline))
#endif

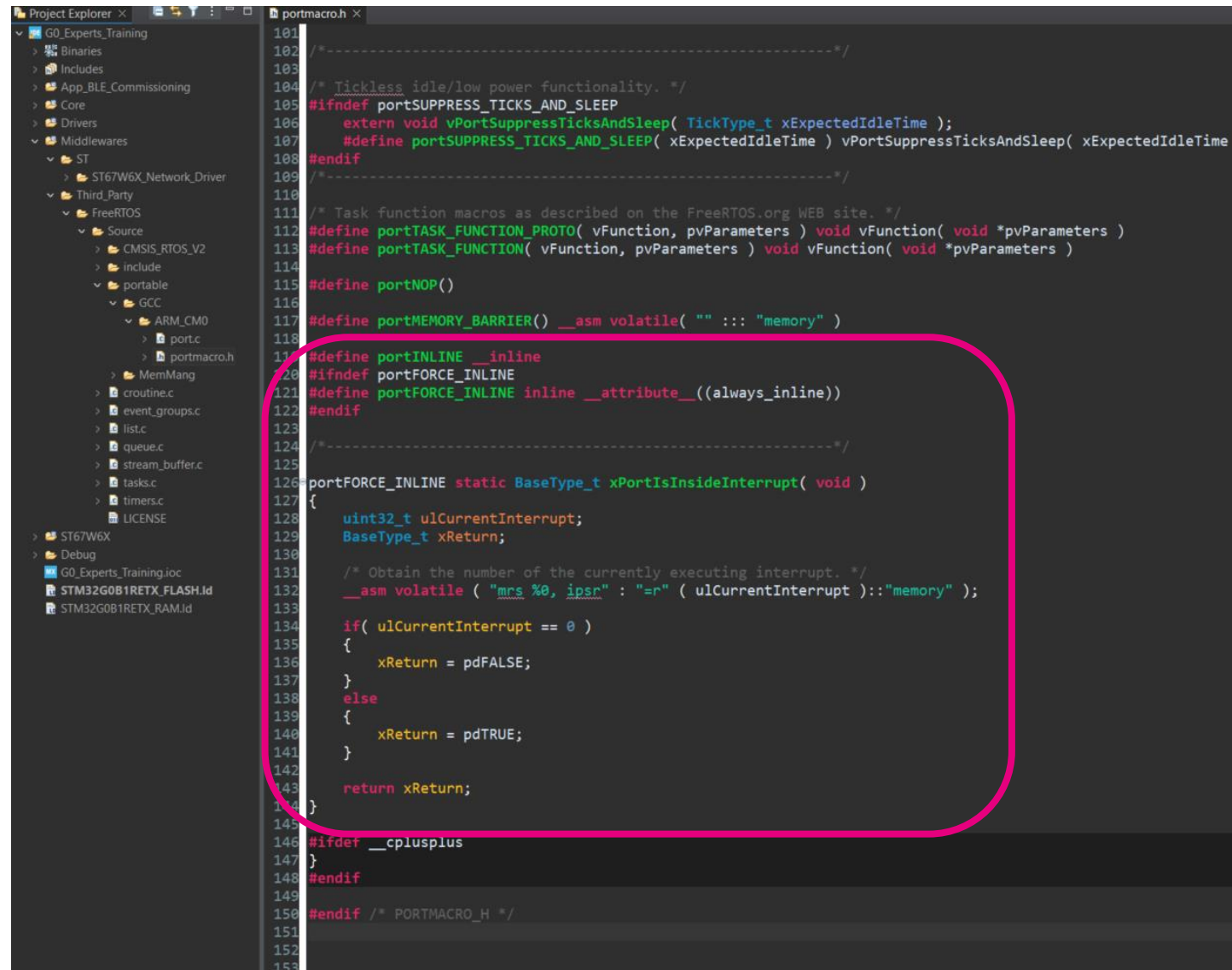
/*-----*/

portFORCE_INLINE static BaseType_t xPortIsInsideInterrupt(
void )
{
    uint32_t ulCurrentInterrupt;
    BaseType_t xReturn;

    /* Obtain the number of the currently executing interrupt. */
    __asm volatile ( "mrs %0, ipsr" : "=r" ( ulCurrentInterrupt
)::"memory" );

    if( ulCurrentInterrupt == 0 )
    {
        xReturn = pdFALSE;
    }
    else
    {
        xReturn = pdTRUE;
    }

    return xReturn;
}
```



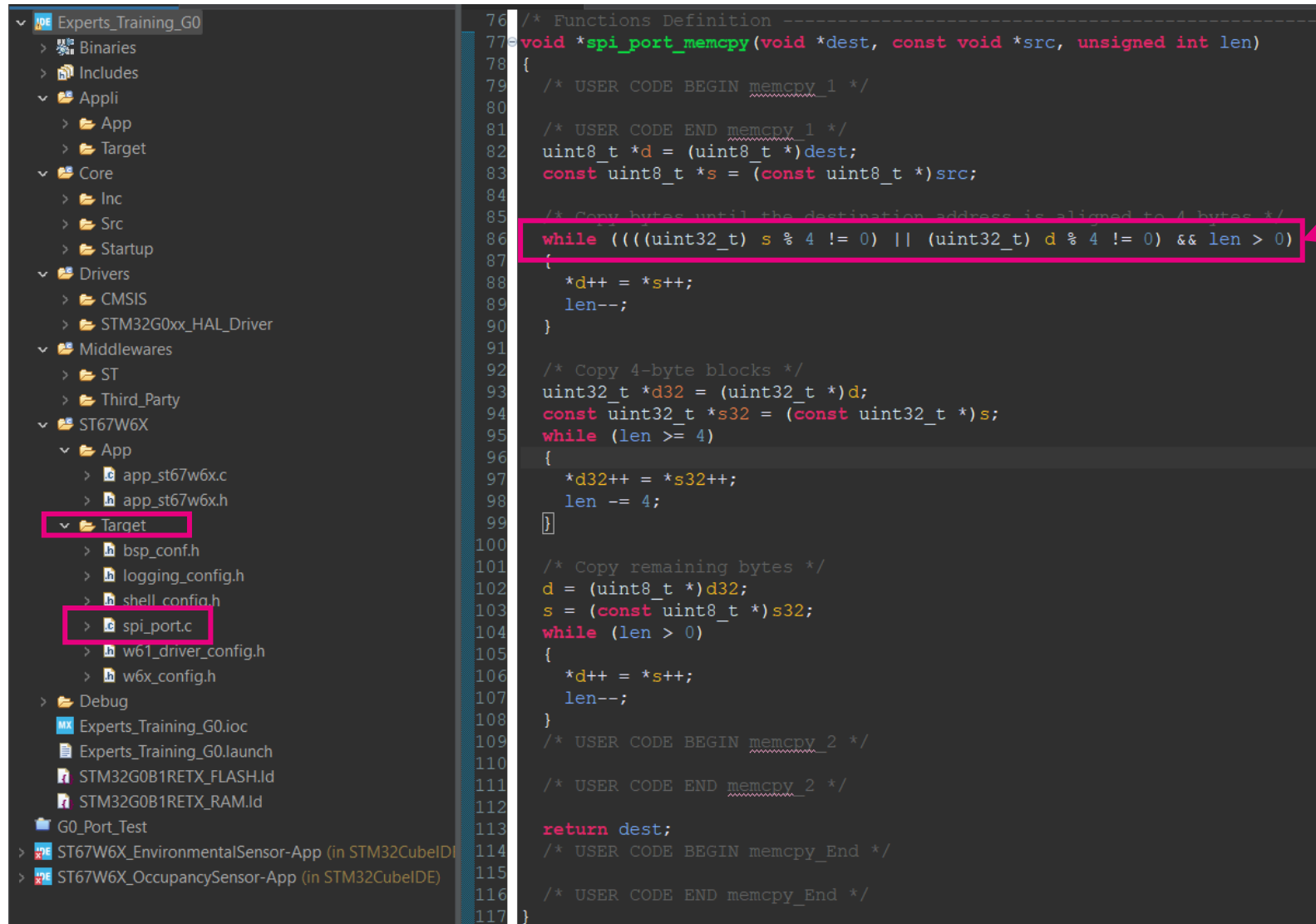
Note: For the Cortex M0+, there is a memory alignment to be followed. Introducing a 4-byte alignment check section to see if source or destination are not aligned.

If not, it handles copy byte by byte, until alignment is reached.

# STM32CubeIDE

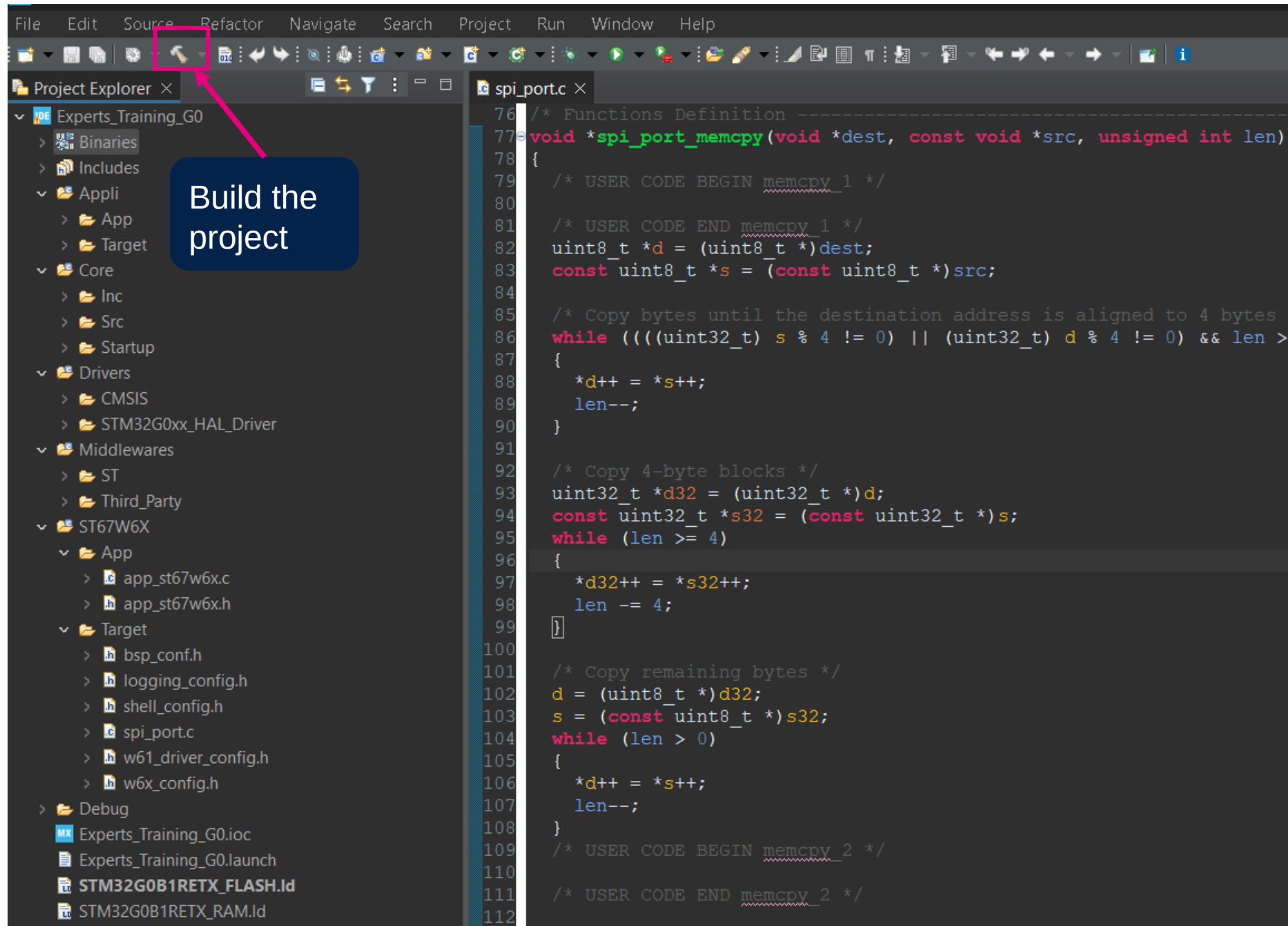
In Project/ST67W6X/spi\_port.c  
Add the below line of code to  
check for the source as well.

while (((uint32\_t) s % 4 != 0) || (uint32\_t) d % 4 != 0) && len > 0)



```
76 /* Functions Definition -----
77 void *spi_port_memcpy(void *dest, const void *src, unsigned int len)
78 {
79     /* USER CODE BEGIN memcpy_1 */
80
81     /* USER CODE END memcpy_1 */
82     uint8_t *d = (uint8_t *)dest;
83     const uint8_t *s = (const uint8_t *)src;
84
85     /* Copy bytes until the destination address is aligned to 4 bytes */
86     while (((uint32_t) s % 4 != 0) || (uint32_t) d % 4 != 0) && len > 0)
87     {
88         *d++ = *s++;
89         len--;
90     }
91
92     /* Copy 4-byte blocks */
93     uint32_t *d32 = (uint32_t *)d;
94     const uint32_t *s32 = (const uint32_t *)s;
95     while (len >= 4)
96     {
97         *d32++ = *s32++;
98         len -= 4;
99     }
100
101     /* Copy remaining bytes */
102     d = (uint8_t *)d32;
103     s = (const uint8_t *)s32;
104     while (len > 0)
105     {
106         *d++ = *s++;
107         len--;
108     }
109     /* USER CODE BEGIN memcpy_2 */
110
111     /* USER CODE END memcpy_2 */
112
113     return dest;
114     /* USER CODE BEGIN memcpy_End */
115
116     /* USER CODE END memcpy_End */
117 }
```

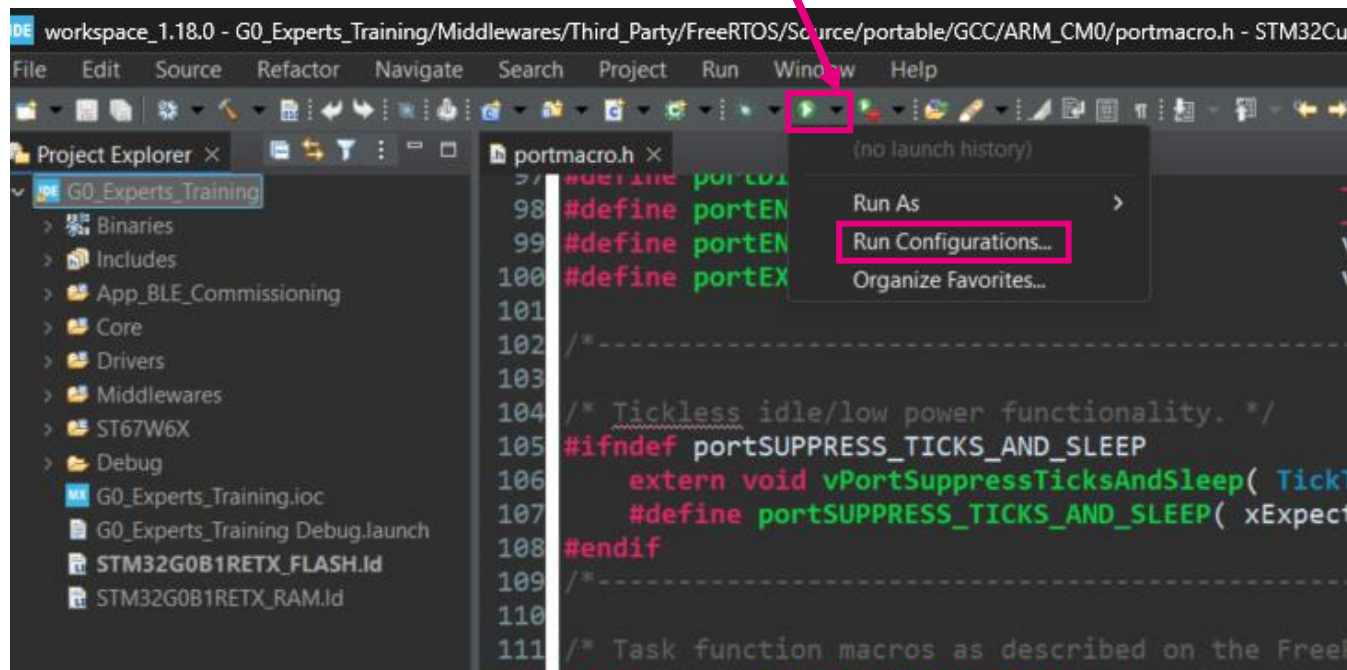
# Build!



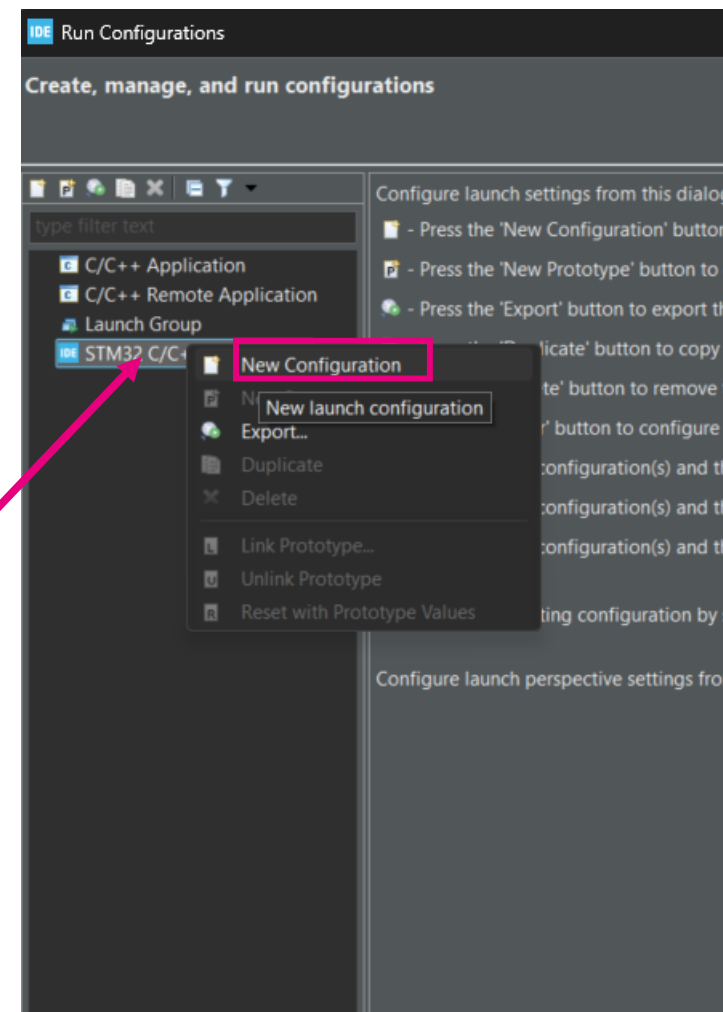


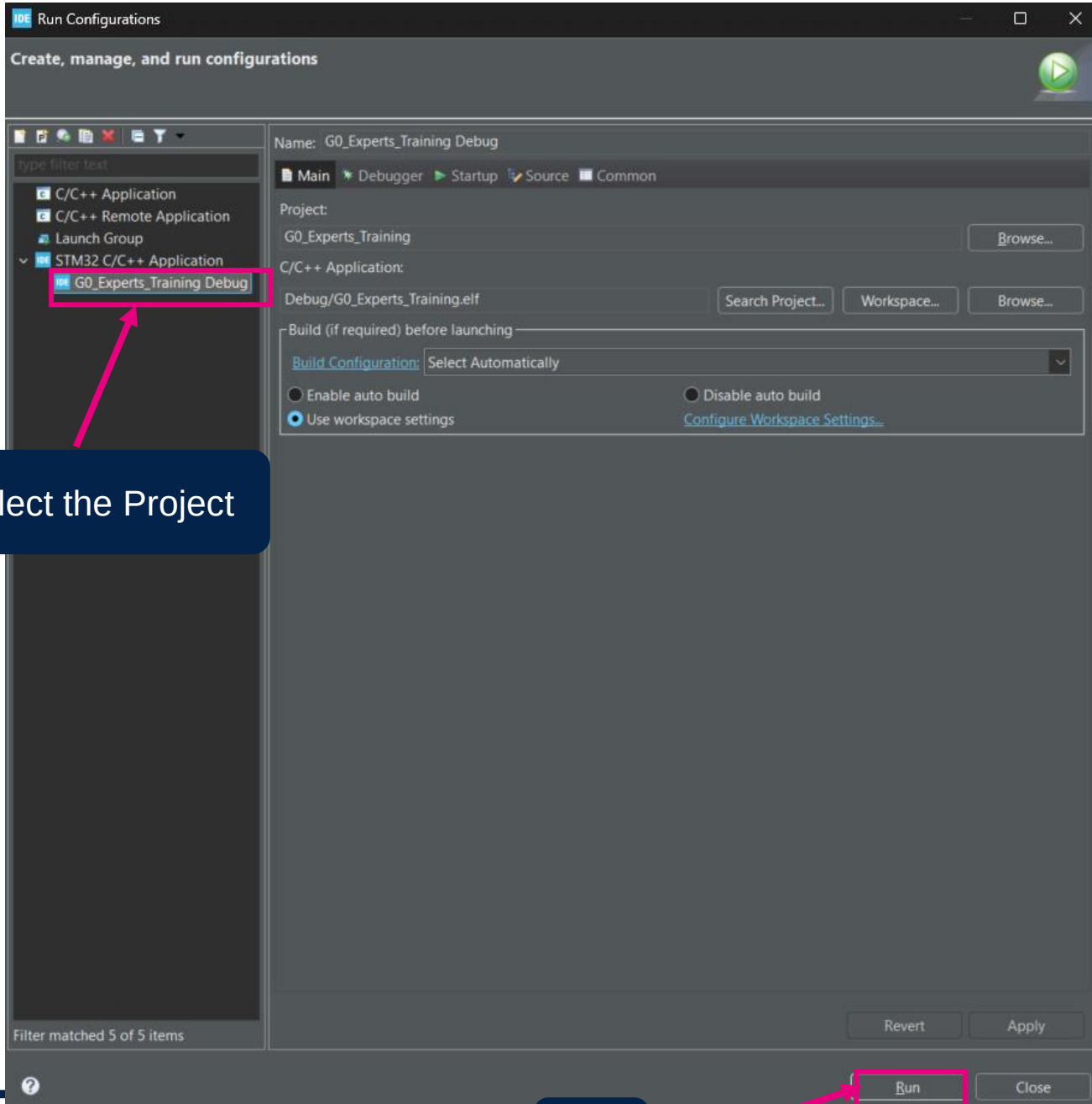
# Flash!

Flash the project



Right-Click on STM32 C/C++ application





Select the Project

Run

Console logs after  
flashing the project

Flash!

```
Problems Tasks Console Properties
<terminated> G0_Experts_Training Debug [STM32 C/C++ Application] ST-LINK (ST-LINK GDB server) (Terminated Aug 20, 2025, 3:14:43 PM) [pid: ...]

Waiting for debugger connection...
Debugger connected
Waiting for debugger connection...
Debugger connected
Waiting for debugger connection...
-----
STM32CubeProgrammer v2.19.0
-----

Log output file: C:\Users\kulkarns\AppData\Local\Temp\STM32CubeProgrammer_a15436.log
ST-LINK SN : 0670FF485570854967113128
ST-LINK FW : V2J46M31
Board : NUCLEO-G0B1RE
Voltage : 3.23V
SWD freq : 4000 KHz
Connect mode: Under Reset
Reset mode : Hardware reset
Device ID : 0x467
Revision ID : Rev Z
Device name : STM32G0B0xx/B1xx/C1xx
Flash size : 512 KBytes
Device type : MCU
Device CPU : Cortex-M0+
BL Version : 0x92

Opening and parsing file: ST-LINK_GDB_server_a15436.srec

Memory Programming ...
File : ST-LINK_GDB_server_a15436.srec
Size : 122.02 KB
Address : 0x08000000

Erasing memory corresponding to sector 0:
Erasing internal memory sectors [0 61]
Download in Progress:

File download complete
Time elapsed during download operation: 00:00:03.410

Verifying ...

Download verified successfully

Shutting down...
Exit.
```



## Tera Term Logs after flashing

# Tera Term Logs

```
#### Welcome to ST67W6X Wi-Fi Commissioning over BLE Application ####
# build: 23:14:47 Sep  7 2025
----- Host info -----
Host FW Version:      1.1.0
----- ST67W6X info -----
ST67W6X MW Version:   1.1.1
AT Version:           1.0.0
SDK Version:          2.0.89
Wi-Fi MAC Version:    1.6.44
BT Controller Version: 1.6.112
BT Stack Version:     1.9.56
Build Date:           Sep  4 2025 11:29:50
Module ID:            Undefined
BOM ID:               0
Manufacturing Year:   2000
Manufacturing Week:   00
Battery Voltage:      3.316 V
Trim Wi-Fi hp:        8,8,8,8,8,7,6,6,7,7,7,8,8,8
Trim Wi-Fi lp:        8,8,8,8,9,9,9,10,10,10,11,11,11
Trim BLE:              7,5,5,7,7
Trim XTAL:            37
MAC Address:          40:82:7b:00:0c:2d
Anti-rollback Bootloader: 0
Anti-rollback App:    0

-----
Wi-Fi init is done
Net init is done
Ble init is done
BD Address: 40:82:7b:00:0c:2e
Configure BLE
BLE configuration is done

BLE Commissioning Service Creation
- BLE service created
- BLE WIFI Control charac created
- BLE WIFI Configure charac created
- BLE WIFI Monitoring charac created
- BLE FUOTA base address charac created
- BLE FUOTA confirmation charac created
- BLE FUOTA raw data charac created
- BLE services and charac registered
BLE service and charac creation is done

Start BLE advertising
BLE advertising is started
█
```

Tera Term: Serial port setup and connection

|               |        |             |
|---------------|--------|-------------|
| Port:         | COM21  | New setting |
| Speed:        | 115200 |             |
| Data:         | 8 bit  | Cancel      |
| Parity:       | none   |             |
| Stop bits:    | 1 bit  | Help        |
| Flow control: | none   |             |

Transmit delay

|   |           |   |           |
|---|-----------|---|-----------|
| 0 | msec/char | 0 | msec/line |
|---|-----------|---|-----------|

Device Friendly Name: STMicroelectronics STLink Virtual COM P  
Device Instance ID: USB\VID\_0483&PID\_374B&MI\_02\6&38AF34B  
Device Manufacturer: STMicroelectronics  
Provider Name: STMicroelectronics  
Driver Date: 4-1-2021  
Driver Version: 2.2.0.0





life.augmented

# Documentation



## Access:

The wifi wiki is opened to all



## Wiki address of the main page

[https://wiki.st.com/stm32mcu/wiki/Connectivity:Introduction\\_to\\_Wi-Fi](https://wiki.st.com/stm32mcu/wiki/Connectivity:Introduction_to_Wi-Fi)



**We will be happy to get your feedback!**

# Wiki Wi-Fi®: pages breakdown

About Wi-Fi

## 1 Wi-Fi® overview

1.1 What is Wi-Fi®

1.2 Wi-Fi® network architecture

1.3 Wi-Fi® features details

Details about  
ST67W611M1

## 2 ST67W611M1

2.1 ST67W611M1 overview

2.2 X-CUBE-ST67W61

2.3 ST67W611M1: Hardware and board description

2.4 ST67W611M1 Security overview

2.5 ST67W611M1-based customer end-Product

2.6 ST67W611M1 evaluation

X-CUBE-ST67W61

## 3 X-CUBE-ST67W61 software application notes and user manuals

3.1 Getting started with ST67W611M1

Host boards  
supported

## 4 ST67W611M1 associated hosts boards

## 5 Specific tools

Tools & References

## 6 Terms and definitions

## 7 References



# Other documentation

- Getting started with X-CUBE-ST67W61 [here](#)

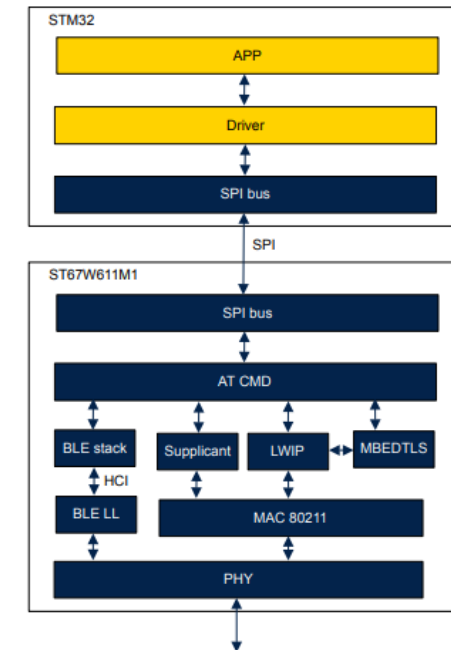
## 2

### X-CUBE-ST67W61 architecture overview

The X-CUBE-ST67W61 is a set of software components implementing host applications driving a Wi-Fi® and Bluetooth® LE coprocessor. The coprocessor is controlled via AT command over SPI interface.

The figure below illustrates the architecture of the solution:

Figure 1. X-CUBE-ST67W61 architecture view



- MW API documentation is available:
  - X-CUBE-ST67W61\_V1.0.0\Middlewares\ST\ST67W6X\_Network\_Driver\Doc\ST67W6X\_Network\_Driver.chm



# ST Customer Support

- [ST Support Home](#)
  - Tickets can be submitted under the part number X-CUBE-ST67W61 or ST67W611M1



# Our technology starts with You



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